

TM Forum Best Practice

End-to-End Business Flows

GB921E

Team Approved Date: 29-May-2020

Release Status: Production	Approval Status: TM Forum Approved Suitable for Conformance
Version 20.0.1	IPR Mode: RAND

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Executive Summary

Due to increasing competition in the global telecommunication markets, it is essential for telecom operators to continuously improve their cost and quality position.

Within this context, the definition of efficient and customer-oriented business processes is a crucial success factor. The Business Process Framework eTOM comprises of functional process structures, i.e. Market/Sales, Product, Customer, Service, Resource and Engaged Party processes. The Process Model discussed in this paper complements eTOM by providing an end-to-end process flow driven view to the different functional groupings. Within the TM Forum Business Process Framework eTOM Levels 1–3 already exists as a standard. However, the implementation of this standard can be facilitated through the further enhancement in terms of process flows at Level 3 as demonstrated in Figure 1. Therefore, this document gives clear recommendations on how to develop the core Customer Centric, Network and Product process flows which are compliant with eTOM Level 1–3.

This document is an Application Note, aiming to document an approach based on industry experience that sets out a structure for considering how these flows can be organized against the overall business needs, and provides a number of process flows that address these needs. Note that, as an Application Note, this material should not be read as normative – i.e. a single standardized approach – but rather as a representative mechanism that provides a useful base for others to build on. Other approaches, with different organizational structures which may lead to other process flows, are also possible. It is the goal of work like this to assist convergence for the industry, but not to impose a single approach, where there are other variations and alternatives that make sense.

The original process flows were first published in April 2011 (Release 9.1). Since then the original process flows have been redesigned and updated as part of the development of the Releases 13.5, 15.0 and 16.0.

Introduction

Due to increasing competition in the global telecommunication markets, it is essential for telecom operators to continuously improve their cost and quality position. One of the most renowned and discussed approaches for improving competitiveness in the last decade is Business Process Management. Business Process Management is about improving organizational performance through the definition, implementation and management of operational, as well as enterprise (management) processes.

Over the last years, customer satisfaction has become one of the most critical factors to success in an environment which is characterized by growing market transparency, increasing demands from the customers, and reduced customer loyalty. By improving customer loyalty, existing turnover can be secured or even increased – either through growth in terms of turnover per existing customer, or through new customers. The key to success is, however, not purely a case of clever customer loyalty programs, but also the consistent orientation of the core processes towards customers' requirements. The benefit of marketing measures will be short term if customer orientation is not taken seriously in terms of the company's interaction with the customers.

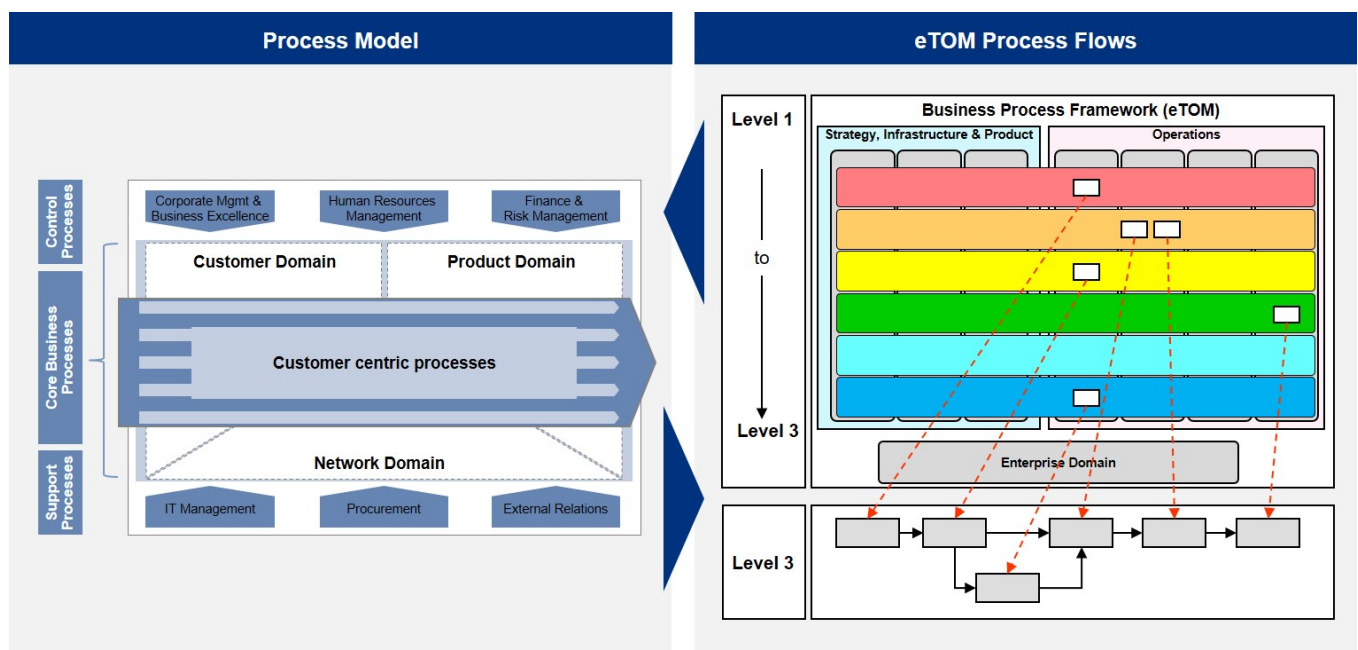


Figure 1. Process Model and eTOM Flows

Within this context, the definition of efficient and customer-oriented business processes is a crucial success factor. The Business Process Framework eTOM comprises of functional process structures, i.e. Market/Sales, Product, Customer, Service, Resource and Engaged Party processes. The Process Model discussed in this paper complements eTOM by providing an end-to-end process flow driven view to the different functional groupings.

Note that this can be considered as a customized extension to the standardized Business Process Framework structure, which may be useful for some areas of application. This does not preclude other such extensions and customized views, where these are found useful, too. In particular, note that the structure shown in Figure 1 and subsequent figures is not a “different” process model,

since this is not proposed as a different process hierarchy. The process hierarchy is still that of the Business Process Framework, and no changes to the decomposition structure in the Business Process Framework are being introduced here. Instead, the structure shown can be thought of as a view of what happens inside a company's organization, when the process elements from the Business Process Framework are allocated to departments and business units. The view on the left in Figure 1, which is then shown also in Figure 2, can be considered as such an organizational view, and not in any sense in conflict with the decomposition hierarchy of the Business Process Framework, since this organizational view makes direct use of the process elements from the Business Process Framework. The concepts here are discussed at some length in Addendum G "Guide to Applying the Business Process Framework", and the approach above is in line with this.

Within the TM Forum Business Process Framework, eTOM Level 1–3 already exists as a standard. However, the implementation of this standard can be facilitated through further enhancement in terms of process flows at Level 3 as demonstrated in Figure 1. Therefore, this document gives clear guidance on how to develop the identified core Customer Centric, Network and Product process flows, which are designed to be compliant with eTOM Level 1–3.

1. End-to-End Business Streams

Generally, a service provider can be structured into core business processes, as well as control and support processes. One perspective that can be useful for the core business processes is to view these as consisting of the Customer Centric processes and three additional process areas: Customer, Product and Network. These can be seen to represent process domains arranged as end-to-end business streams which show a focus across the Service Provider for the area of business activity concerned. Note that the use of the terms “process domains” and “end-to-end business streams” here is just intended to explain this broad business context and is not a new element of structure or terminology within the core Business Process Framework.

In this model, the Customer Centric processes are characterized as being initiated by the customer and ending at the customer. Additionally, the processes involved include all other core processes, which are not directly initiated by a customer (e.g. marketing campaign, product lifecycle management, network operations).

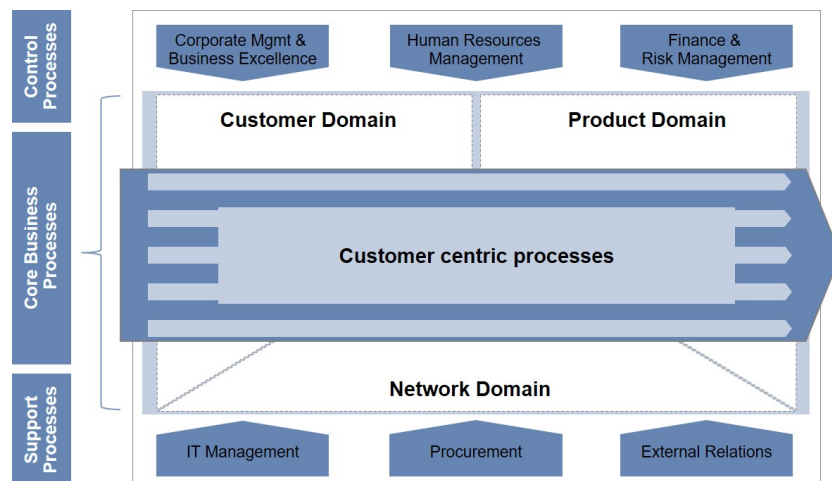


Figure 2. A process view focused on customer-centric processes

The Customer Centric processes have several interactions with the other three process areas/domains. For instance, they might be initiated by relevant activities in the sense of CRM at the Customer domain, and interactions may include trouble solving in the Network domain and the Product domain.

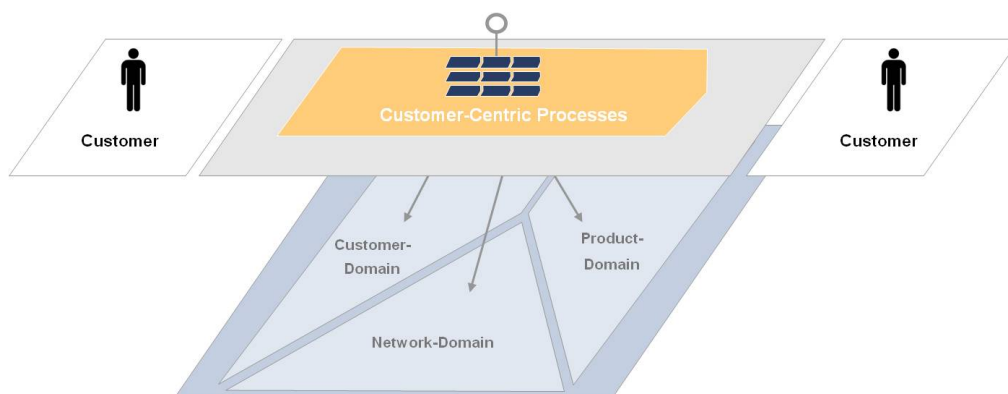


Figure 3. Process domain view – example

1.1. Customer-Centric End-to-End Business Streams

A total of seven Customer Centric Processes represent the customer view and interaction with the telecommunication company. These processes start with the customer initiating the contact. They end with the fulfilment of his/her request. Customer Centric processes include activities such as handling information requests, new sale, billing and invoice generation or problem and complaint handling.

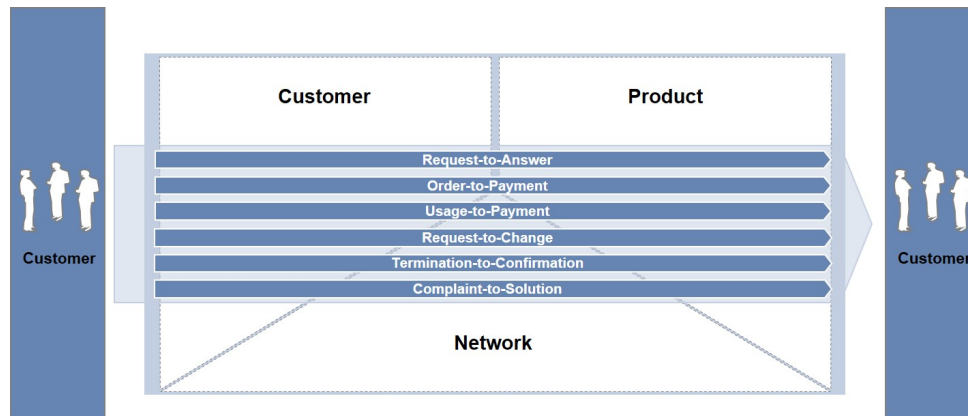


Figure 4. Customer-Centric processes

1.2. Network End-to-End Business Streams

This domain consists of six Network processes, which represent the network operations' view and interaction within the telecommunications company. Network processes include activities such as order handling, trouble ticket management, billing, capacity management, and service lifecycle management.

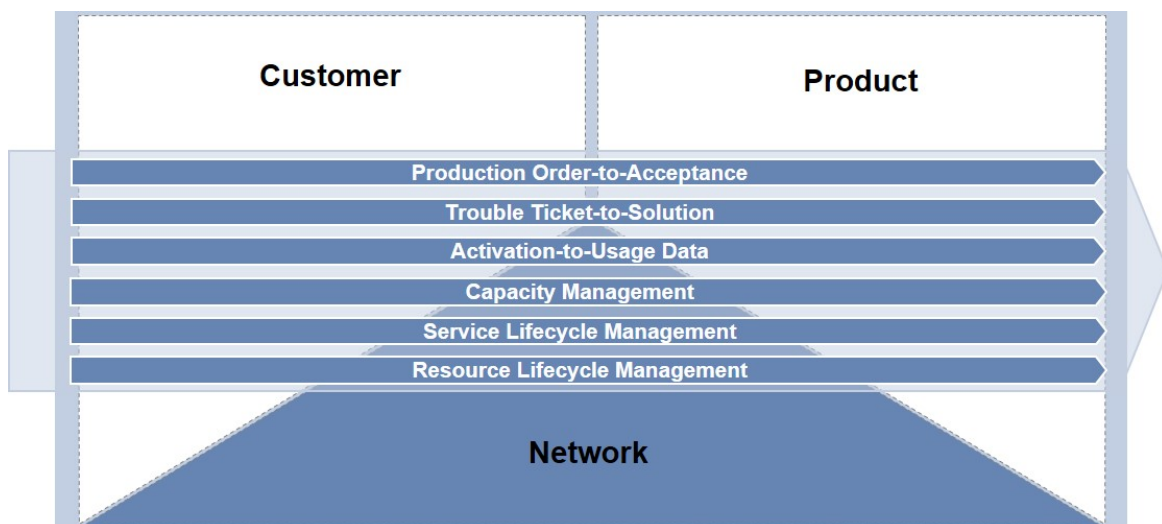


Figure 5. Network processes

1.3. Product End-to-End Business Streams

In this domain there are six Product related core workflows for an operator and build the basis for the generation of revenues. Product processes include activities such as product development, launch, retirement etc.

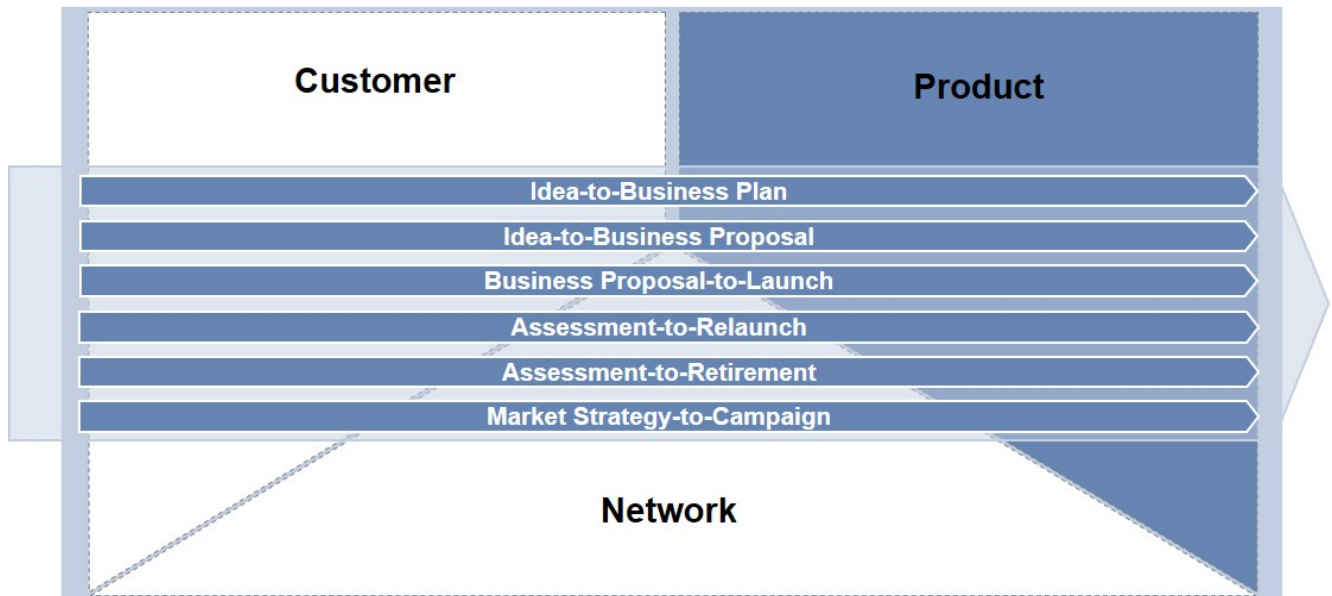


Figure 6. Product processes

1.4. Engaged Party Flows

In this section we explore Engaged Party Domain of Business Process Framework and using engaged party processes in real world scenarios. This consists of worked examples of using eTOM processes for various most common processes of the organization. The framework discussed can be used as best practice or guideline to arrive at a good understanding of using Engaged Party Domain.

2. Customer Centric Process Flows

This section includes sample process-flows (eTOM level 3) relevant to the Customer Centric focus. The process flows for the seven Customer Centric end-to-end processes recommend how to apply the Operations part of eTOM to a service provider.

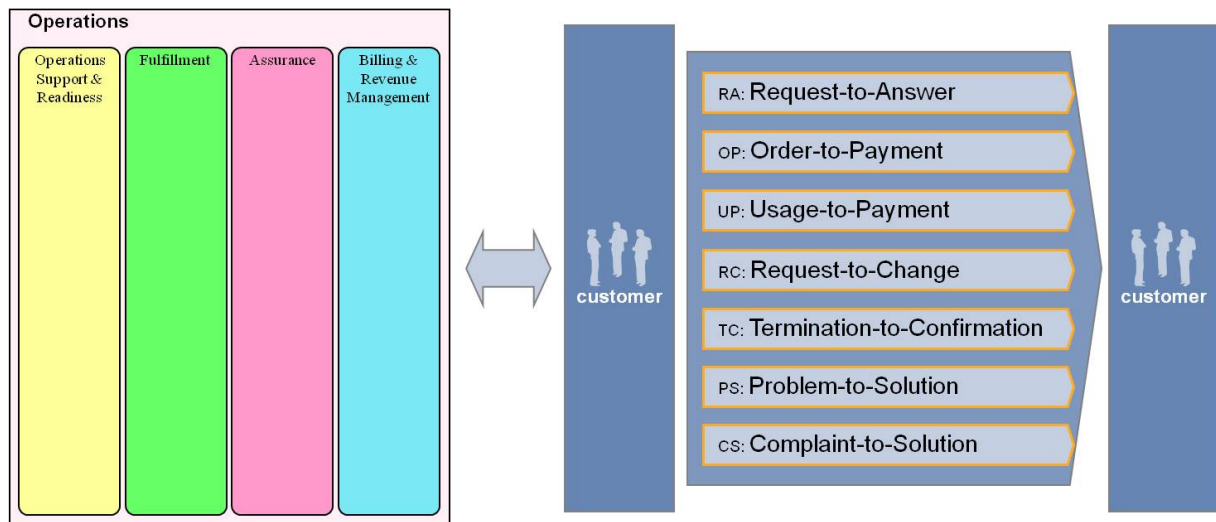


Figure 7. eTOM and Customer-Centric processes

Remark: The flow illustration uses level 3 process elements of the Business Process Framework eTOM. In some diagrams you will find databases which are added to explain the flow. The Databases: Customer Subscription Inventory, Service Inventory and Resource Inventory are shown to provide a clearer understanding of the relevant process element. They have the following relation to the Information Framework (SID): Subscription Inventory to the ABE "Customer", Service Inventory to "Service Configuration" and Resource Inventory to "Resource Configuration".

2.1. Process Flows for Request-to-Answer

2.1.1. Overview

- This process comprises of activities relevant to managing customer requests across all communication channels (customer interfaces).
- Specific information requests or product requests from the customer are qualified and addressed.
- This could lead to the preparation of a pre-sales offer if the customer shows interest in a particular product.

2.1.2. Assumptions:

- The offer preparation may or may not lead to a reservation of resources in advance.
- Resources might be reserved for some special products or customers depending on the operator's policies and procedures.

- If the offer is accepted by the customer, the next process 'Order to Payment' is triggered.

2.1.3. eTOM Mapping

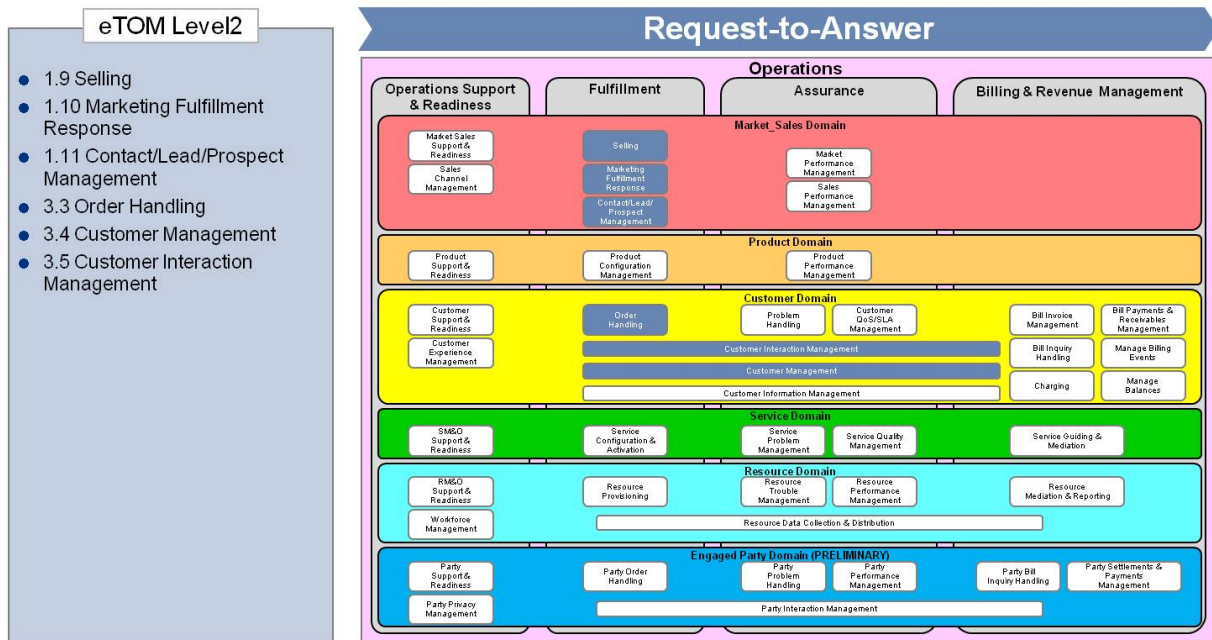


Figure 8. Request-to-Answer - Mapping on eTOM Level 2

2.1.4. Detailed Process-Flow (Level 3)

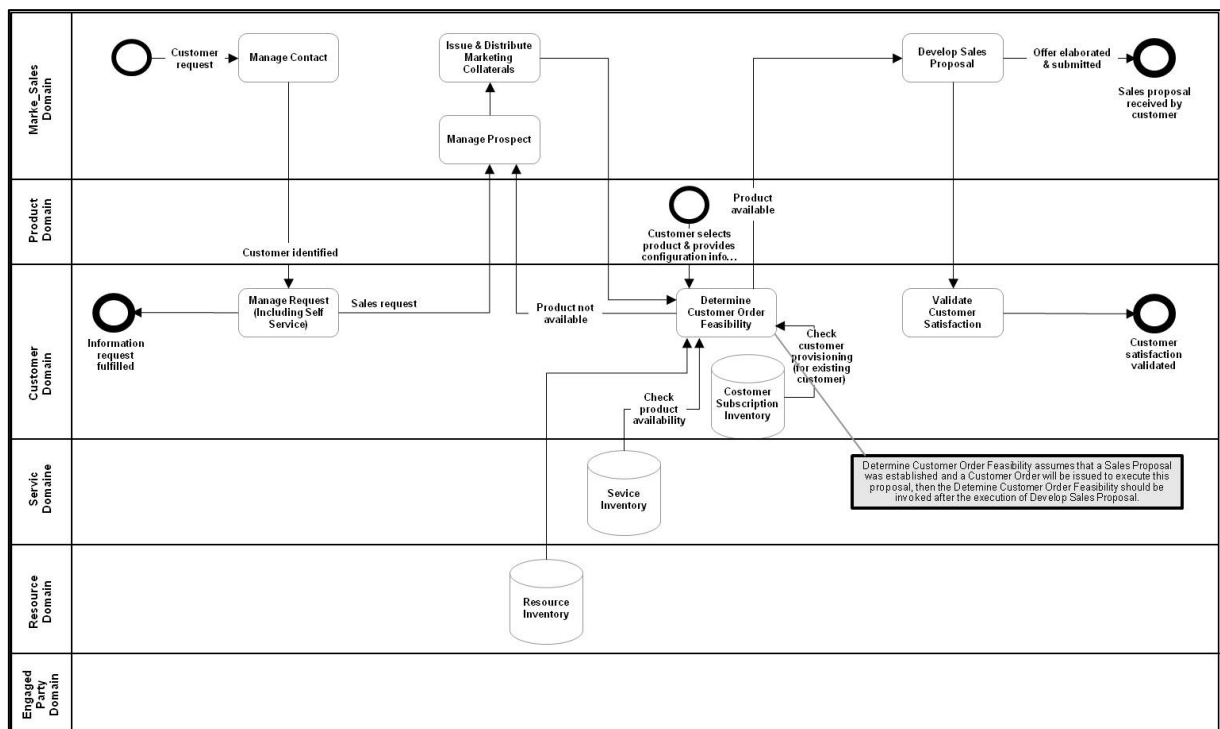


Figure 9. Request-to-Answer – Process-Flow (Level 3) BPNM

2.1.5. Detailed Information

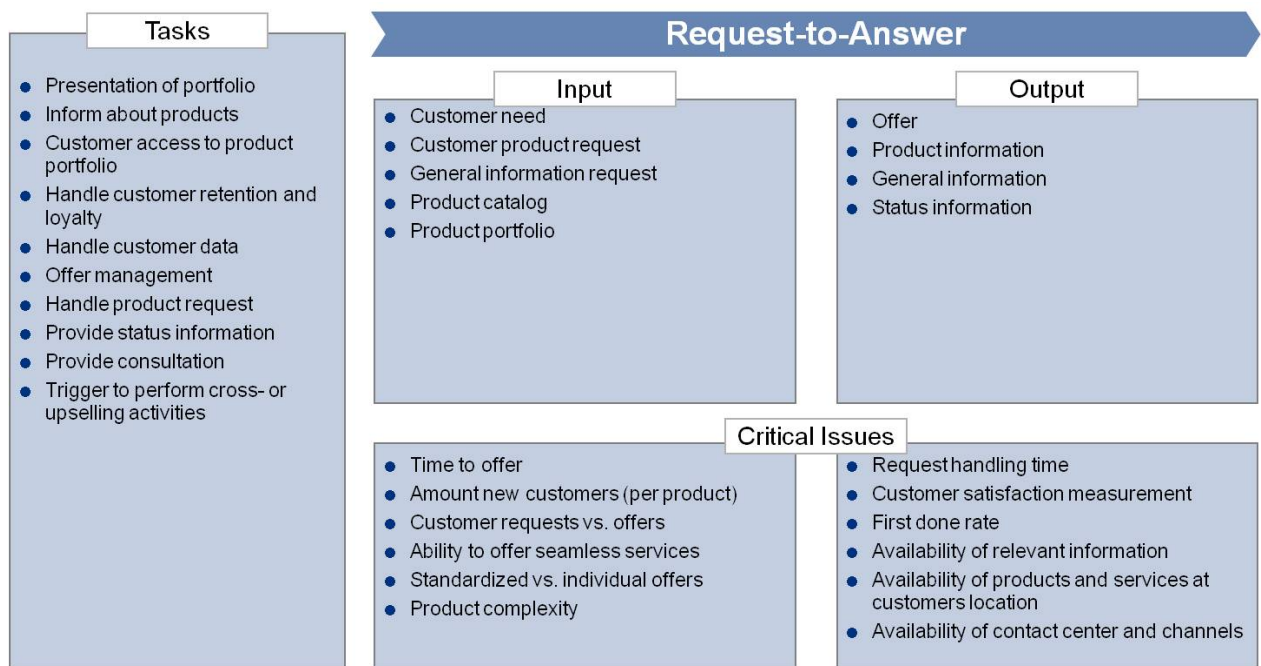


Figure 10. Request-to-Answer – Detailed Information

2.2. Process Flows for Order-to-Payment

2.2.1. Overview

- This process deals with all activities which convert the customer request or an accepted offer into a 'Ready for use' product.
- This process involves capturing customer order information, triggering the relevant provisioning process and handing over the order to the Service layer.
- Once the product is successfully provisioned, the customer order is closed, and the customer satisfaction is validated.

2.2.2. Assumptions:

- This scenario is relevant for products offered to the mass customer base.
- Pre-order feasibility check verifies whether the requested product can be offered to the customer based on the service / resource availability, the product portfolio, and the customer's configuration.
- Pre-order feasibility check would not generally involve the reservation of resources prior to issuance of customer order.
- However, there might be a reservation of some critical resources, depending on the operator's policies and procedures.

2.2.3. eTOM Mapping

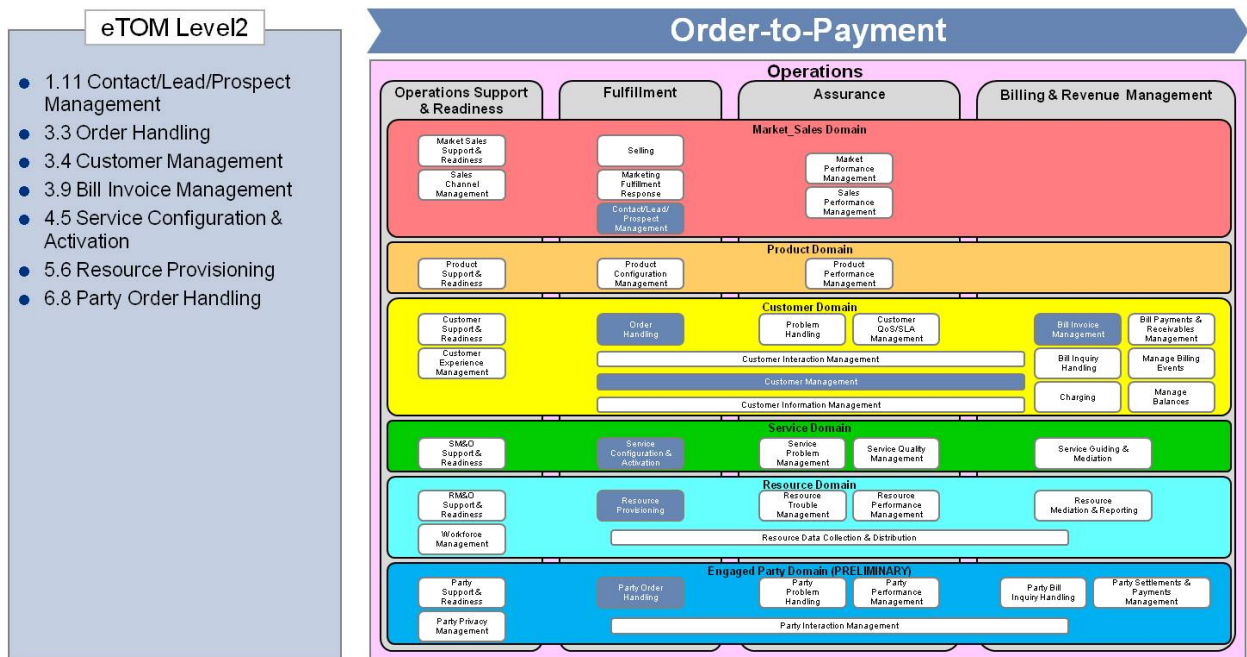


Figure 11. Order-to-Payment - Mapping on eTOM Level 2

2.2.4. Detailed Process-Flow (Level 3)

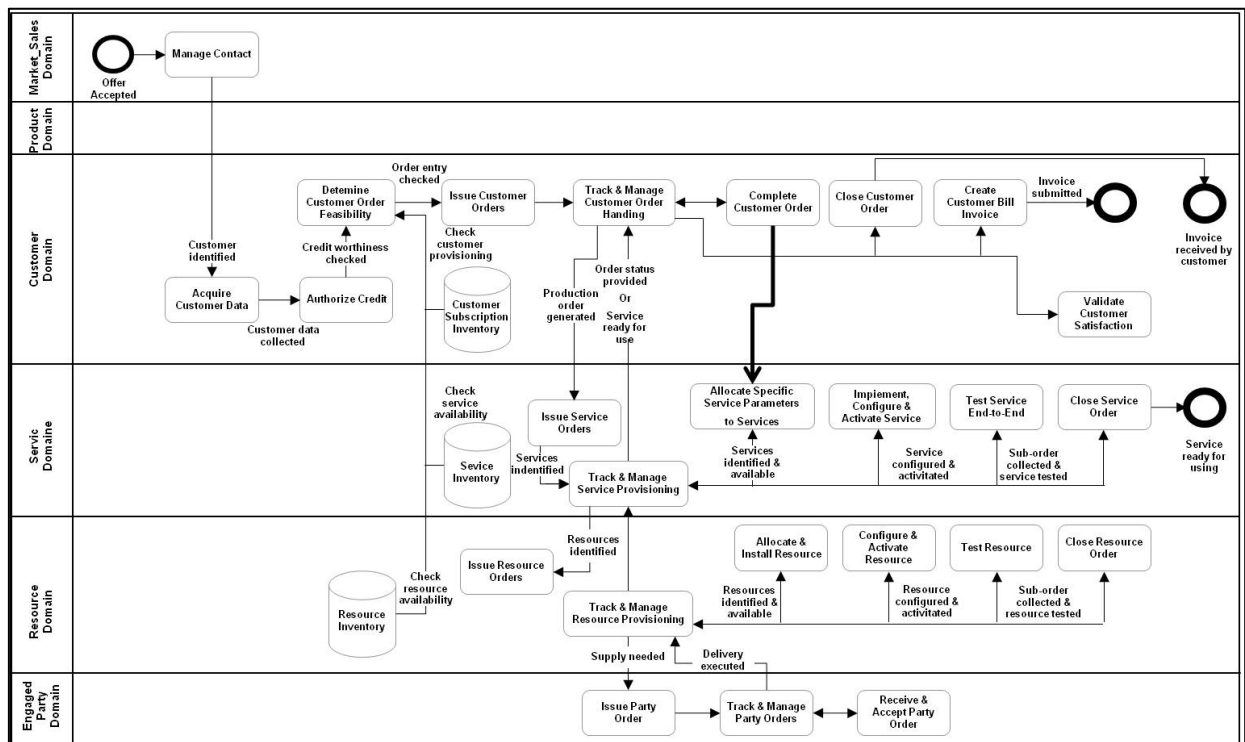


Figure 12. Order-to-Payment – Process-Flow (Level 3) BPNM

2.2.5. Detailed Information

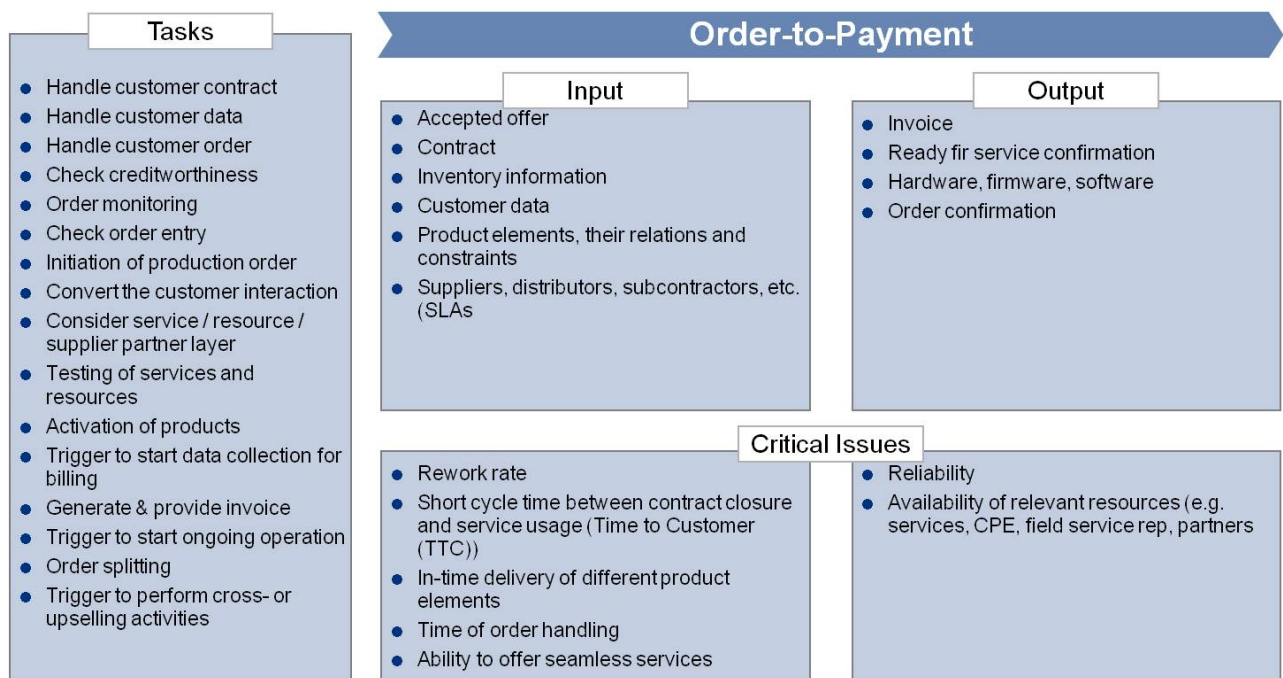


Figure 13. Order-to-Payment – Detailed Information

2.3. Process Flows for Usage-to-Payment

2.3.1. Overview

- This process deals with all activities related to the handling of the product/service usage.
- The accuracy of pricing is ensured, and all usage data is captured and duly processed for billing information requests and bill generation.

2.3.2. Assumptions:

- This scenario is relevant for products offered to the mass customer base.
- The process elements 'Mediate Resource Usage Records' and 'Guide Resource Usage Records' are not always applicable.
- 'Perform Rating' element is usually under the Customer layer for customized products, and under the Service layer for standardized products.

2.3.3. eTOM Mapping

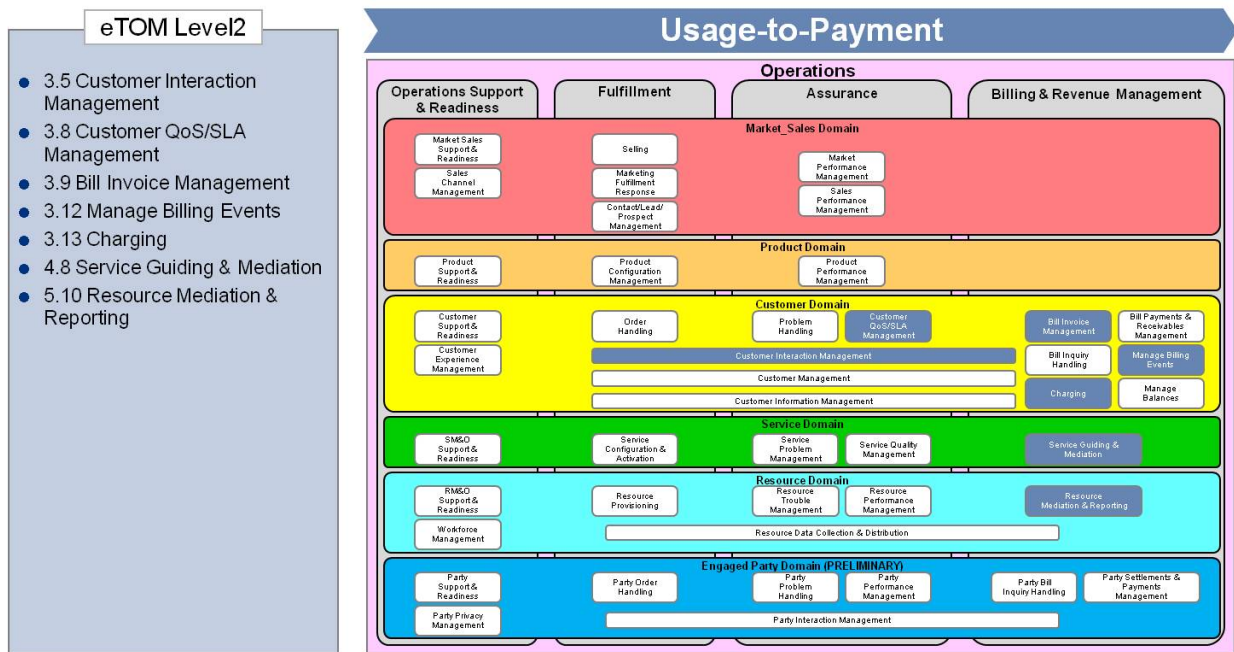


Figure 14. Usage-to-Payment - Mapping on eTOM Level 2

2.3.4. Detailed Process-Flow (Level 3)

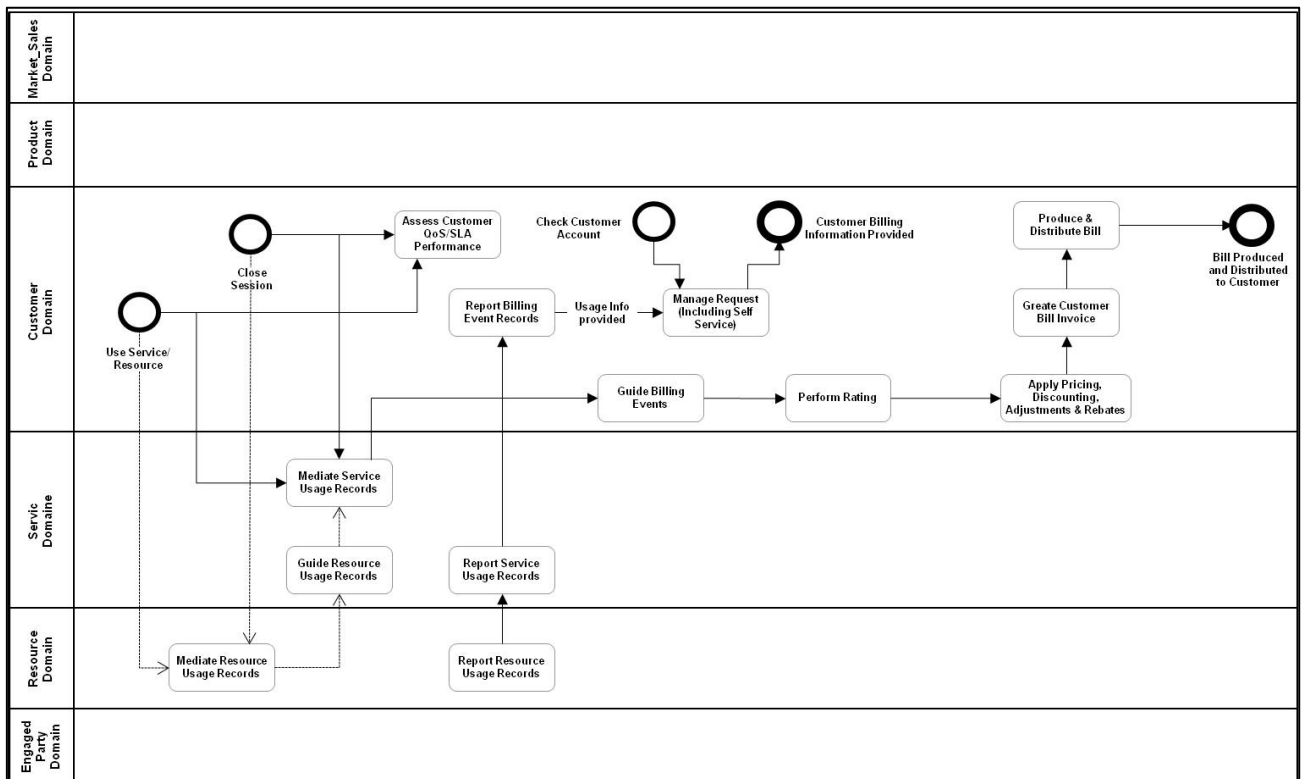


Figure 15. Usage-to-Payment – Process-Flow (Level 3) BPMN

2.3.5. Detailed Information

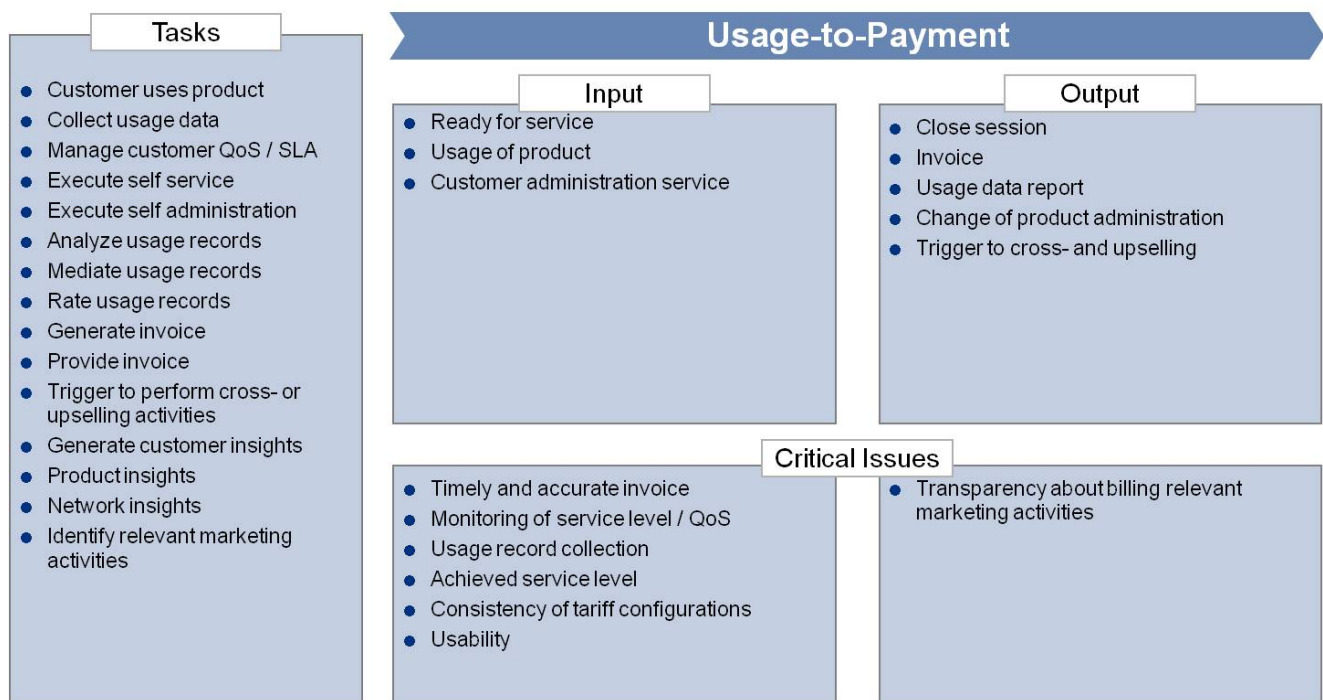


Figure 16. Usage-to-Payment – Detailed Information

2.4. Process Flows for Request-to-Change

2.4.1. Overview

- This process deals with all activities which convert the customer's change request into a 'Ready for use' product.
- This process involves capturing customer order information, triggering the relevant provisioning process and handing over the order to the Service layer.
- Once the product is successfully provisioned, the customer order is closed, and the customer satisfaction is validated.

2.4.2. Assumptions:

- This scenario is relevant for products offered to the mass customer base.
- Pre-order feasibility check verifies whether the requested change can be implemented for the customer based on the service / resource availability, the product portfolio, and the customer's configuration.
- Pre-order feasibility check would not generally involve the reservation of resources prior to issuance of customer order.
- However, there might be a reservation of some critical resources, depending on the operator's policies and procedures.

2.4.3. eTOM Mapping

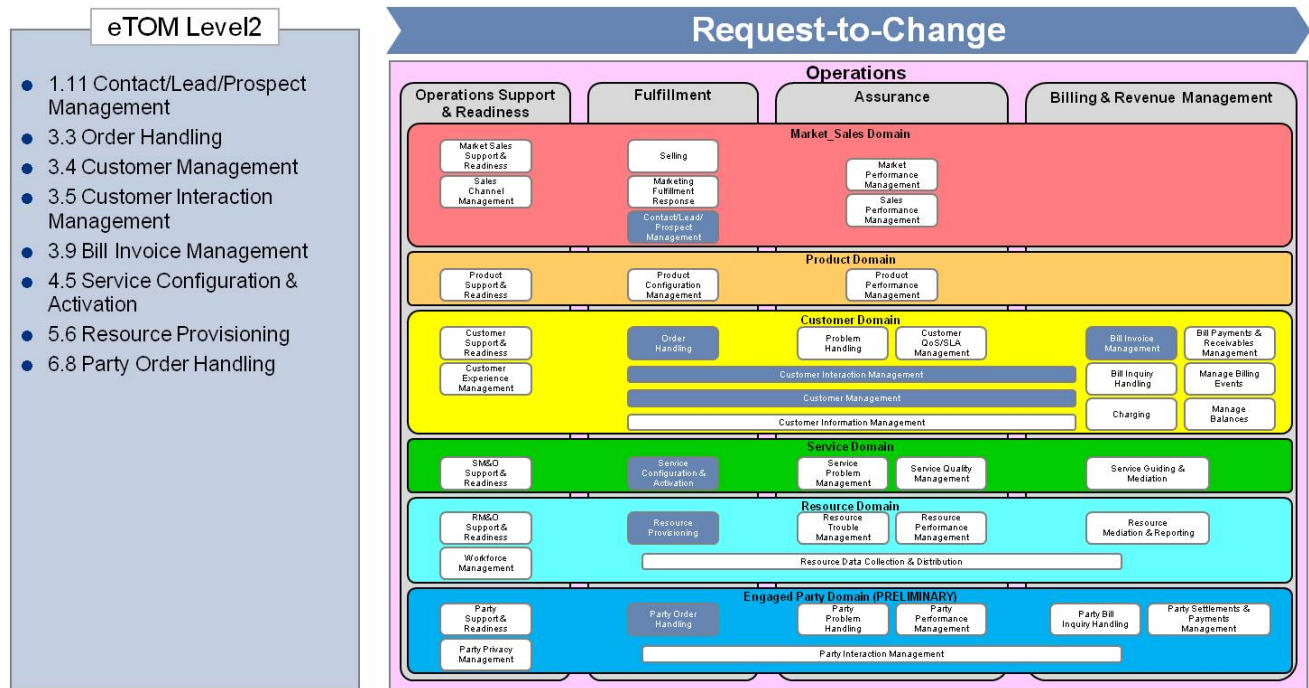


Figure 17. Request-to-Change - Mapping on eTOM Level 2

2.4.4. Detailed Process-Flow (Level 3)

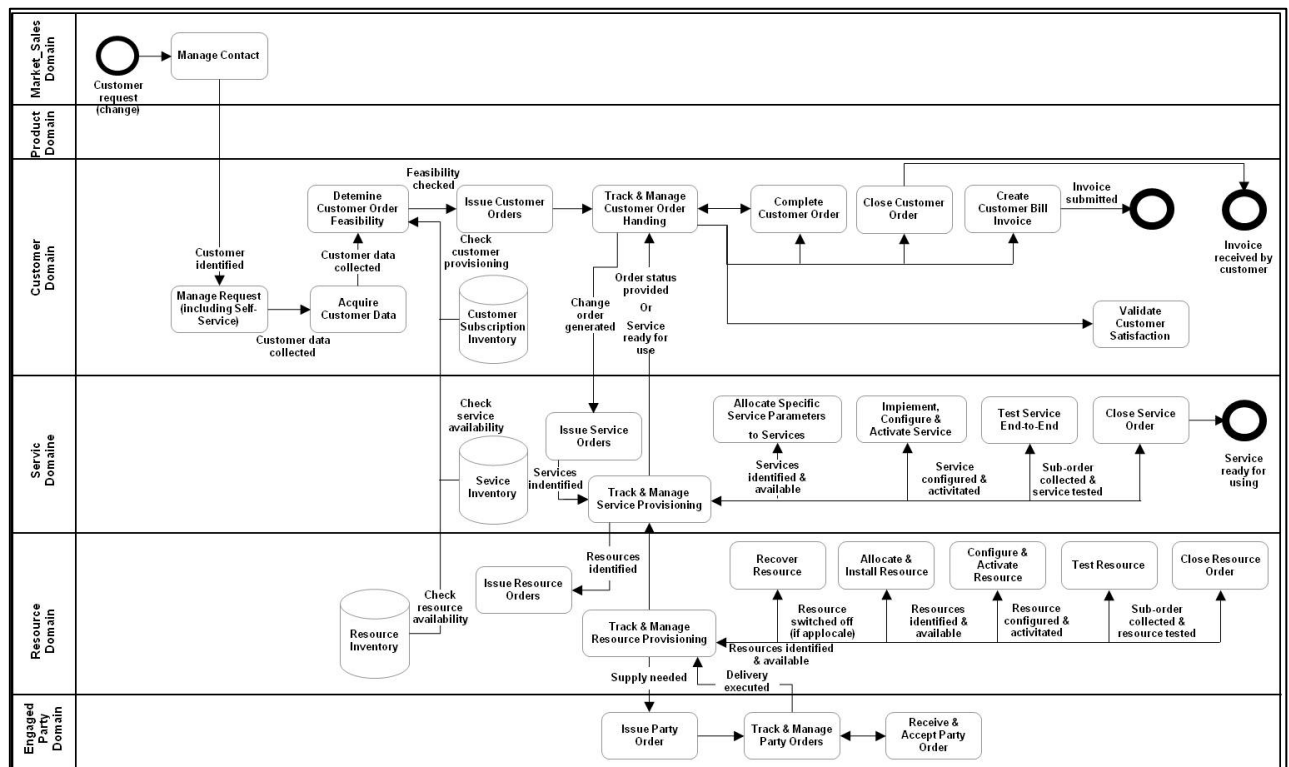


Figure 18. Request-to-Change Process-Flow (Level 3) BPNM

2.4.5. Detailed Information

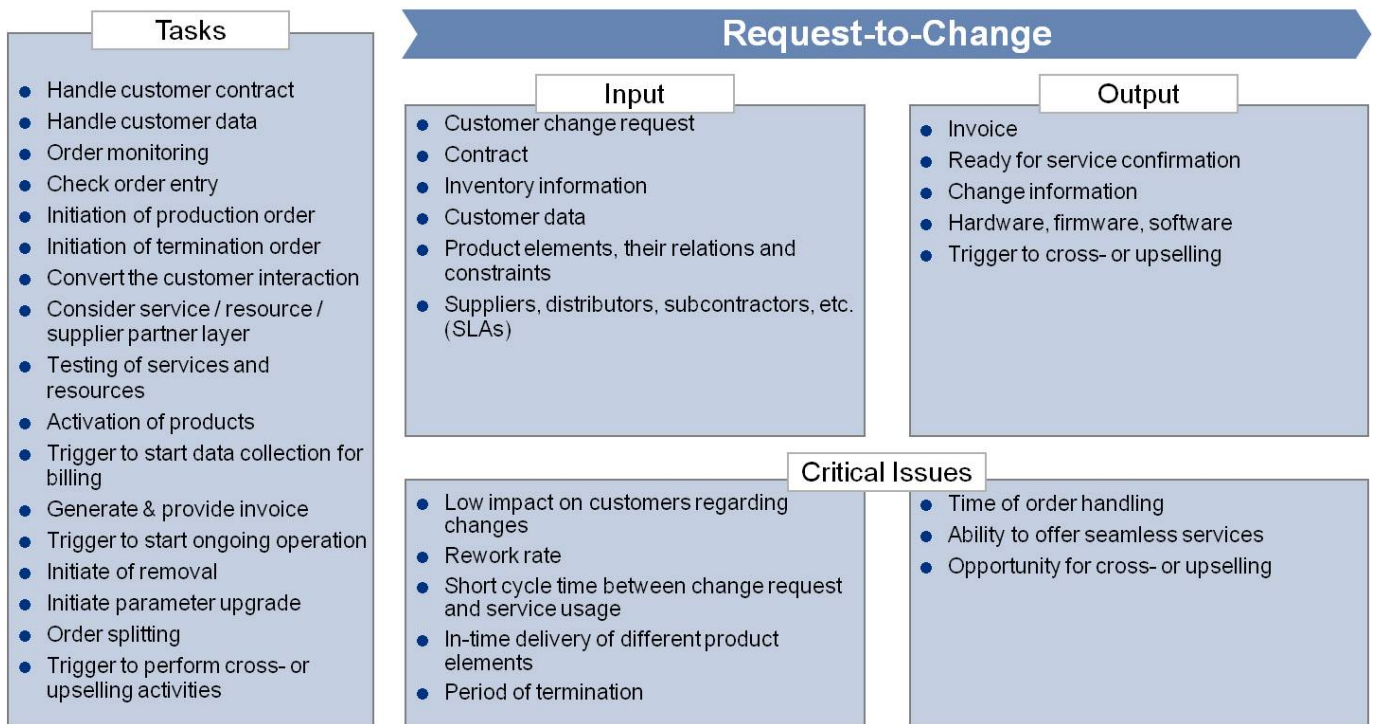


Figure 19. Request-to-Change – Detailed Information

2.5. Process Flows for Termination-to-Confirmation

2.5.1. Overview

- This process deals with all activities related to the execution of customer's termination request.
- This process involves retention activities, capturing customer order information, triggering the relevant provisioning process and handing over the order to the Service layer.
- Once the product is successfully terminated, the customer order is closed, and the customer satisfaction is validated.

2.5.2. Assumptions:

- This scenario is relevant for products offered to the mass customer base.
- This process includes the termination of a customer order or a product.
- Pre-order feasibility check verifies whether the requested termination can be executed based on the customer's current configuration.

2.5.3. eTOM Mapping

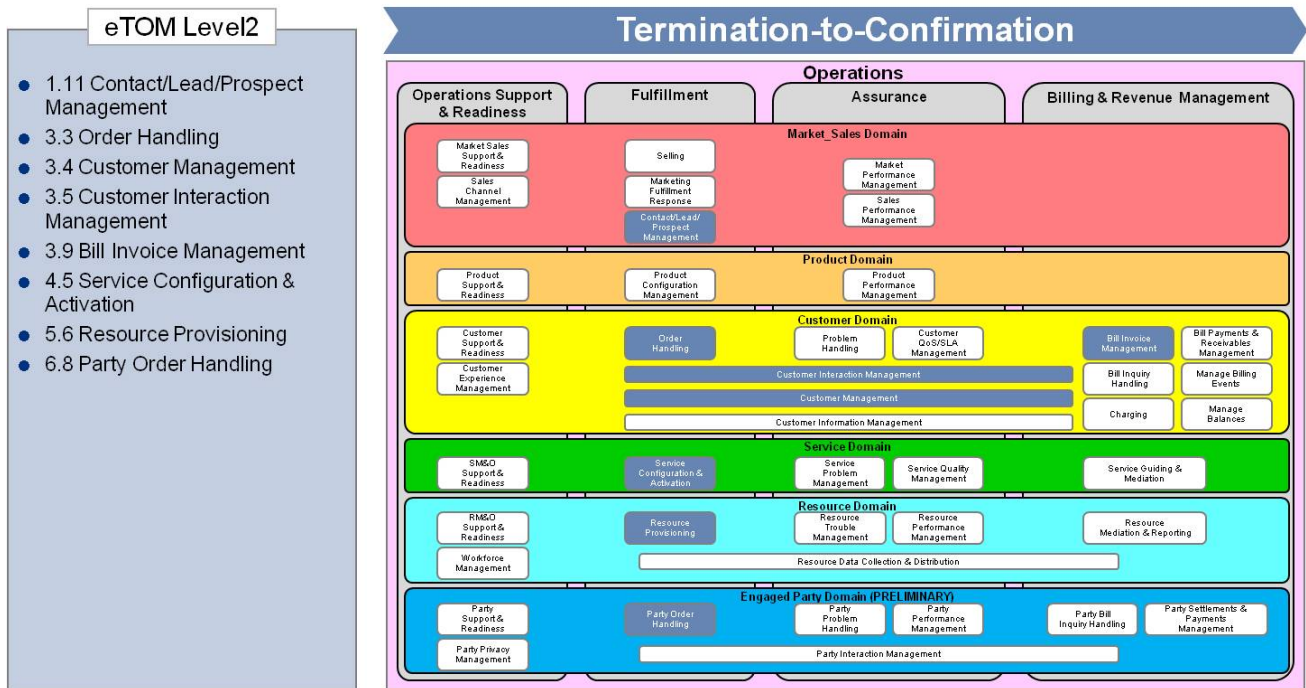


Figure 20. Termination-to-Confirmation - Mapping on eTOM Level 2

2.5.4. Detailed Process-Flow (Level 3)

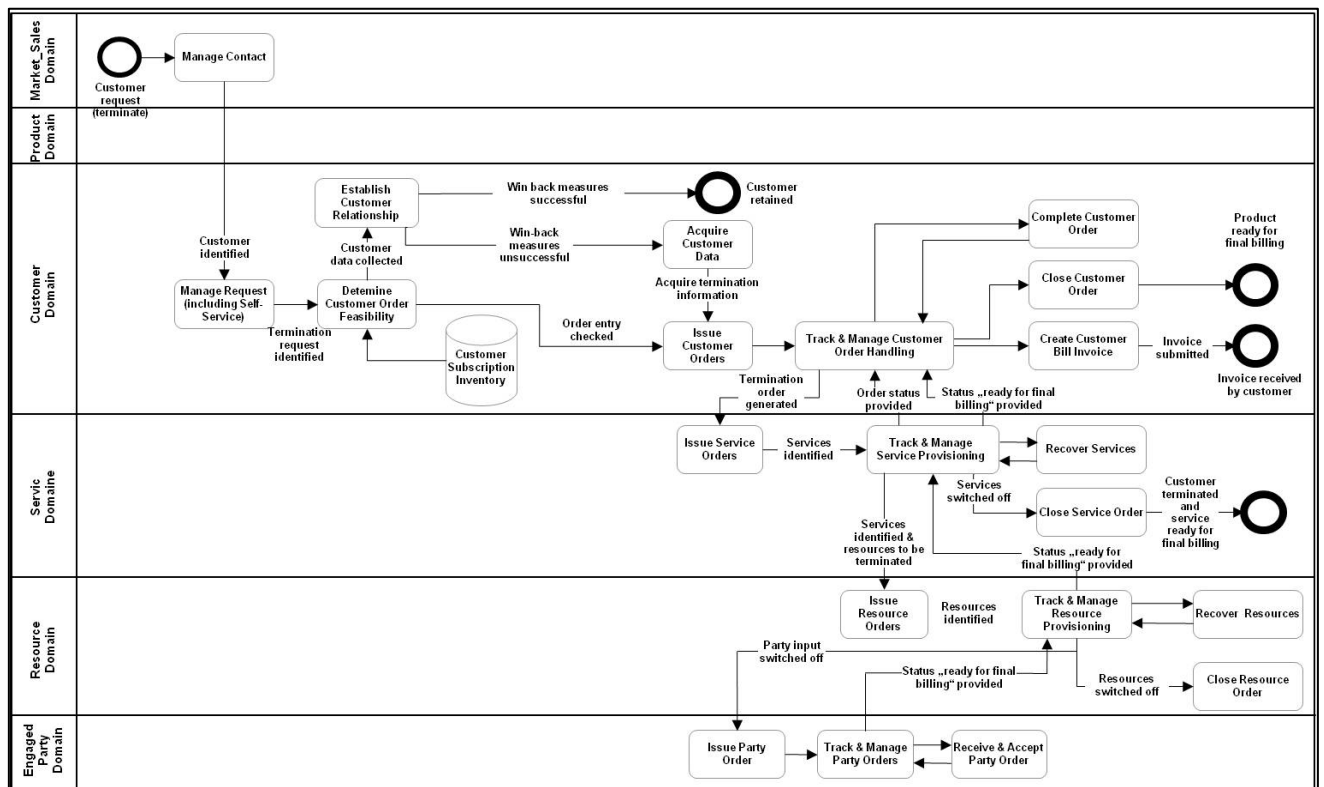


Figure 21. Termination-to-Confirmation – Process-Flow (Level 3) BPNM

2.5.5. Detailed Information

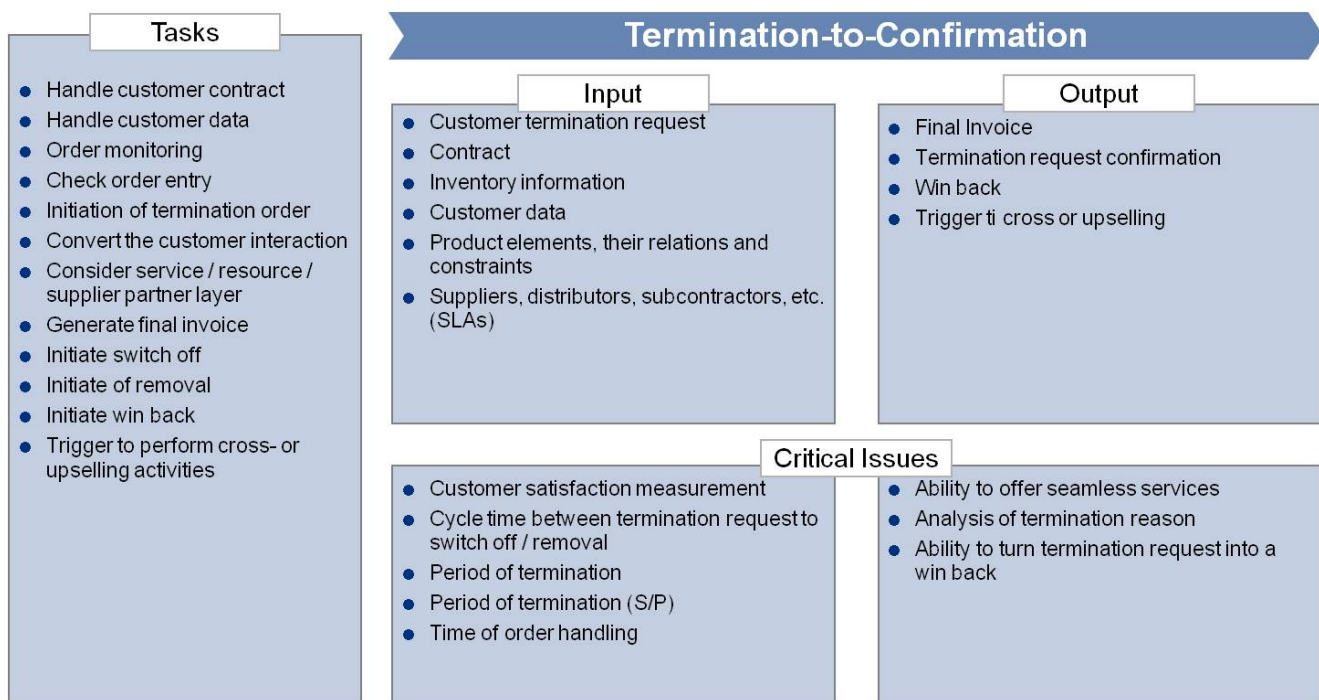


Figure 22. Termination-to-Confirmation – Detailed Information

2.6. Process Flows for Problem-to-Solution

2.6.1. Overview

- This process deals with a technical complaint (problem) initiated by the customer, analyses it to identify the source of the issue, initiates resolution, monitors progress and closes the trouble ticket.
- The basis for a problem is an unplanned interruption to a product / service or reduction in the quality of a product/service. (In comparison, the process “complaint-to-solution” deals with customer inquiries in which the customer is not pleased with a product or handling speed of an inquiry etc.).

2.6.2. Assumptions:

- This scenario is relevant only for technical complaints which are termed as ‘problems’.
- Problems such as no outgoing call or sms and faulty handset, may be resolved at first level support or re-directed to other layers
- Non-technical complaints will be dealt by another scenario ‘Complaint-to-Solution’.

2.6.3. eTOM Mapping

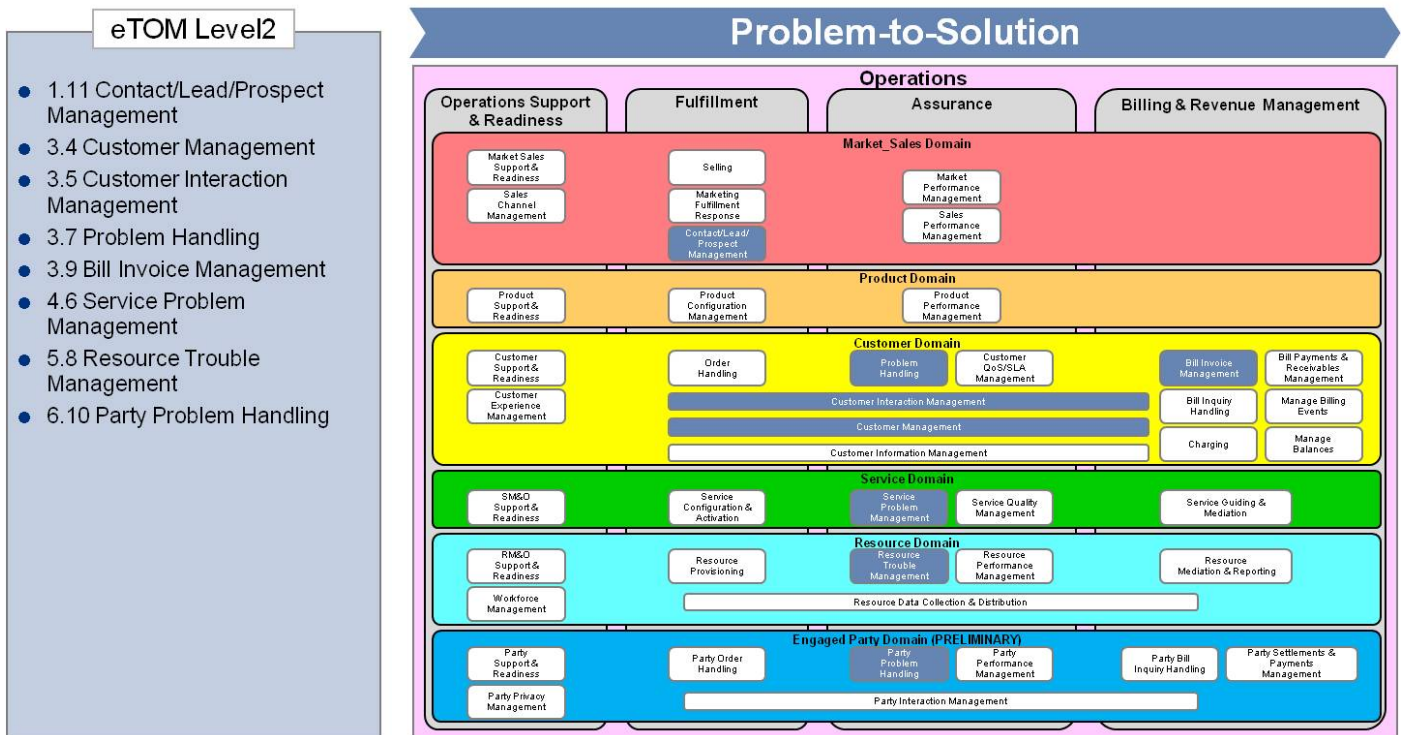


Figure 23. Problem-to-Solution - Mapping on eTOM Level 2

2.6.4. Detailed Process-Flow (Level 3)

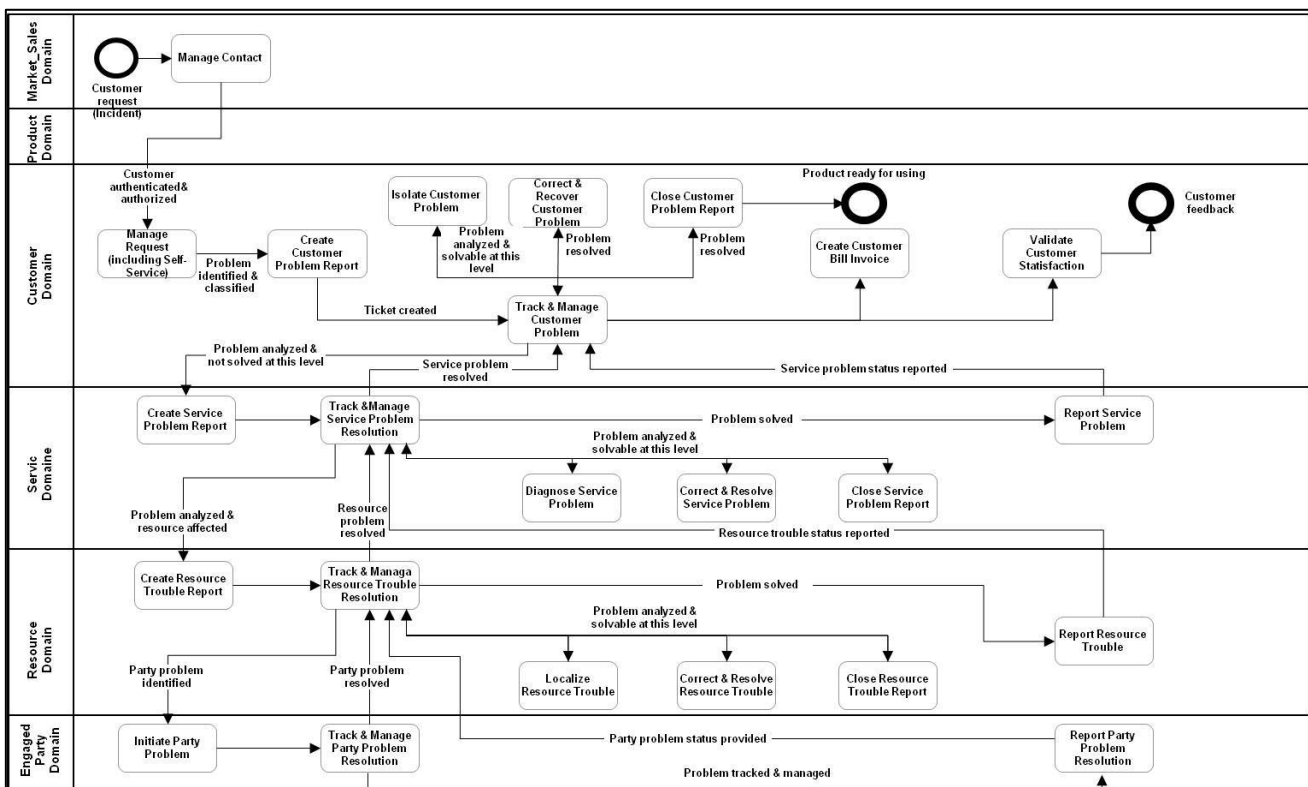


Figure 24. Problem-to-Solution – Process-Flow (Level 3) BPNM

2.6.5. Detailed Information

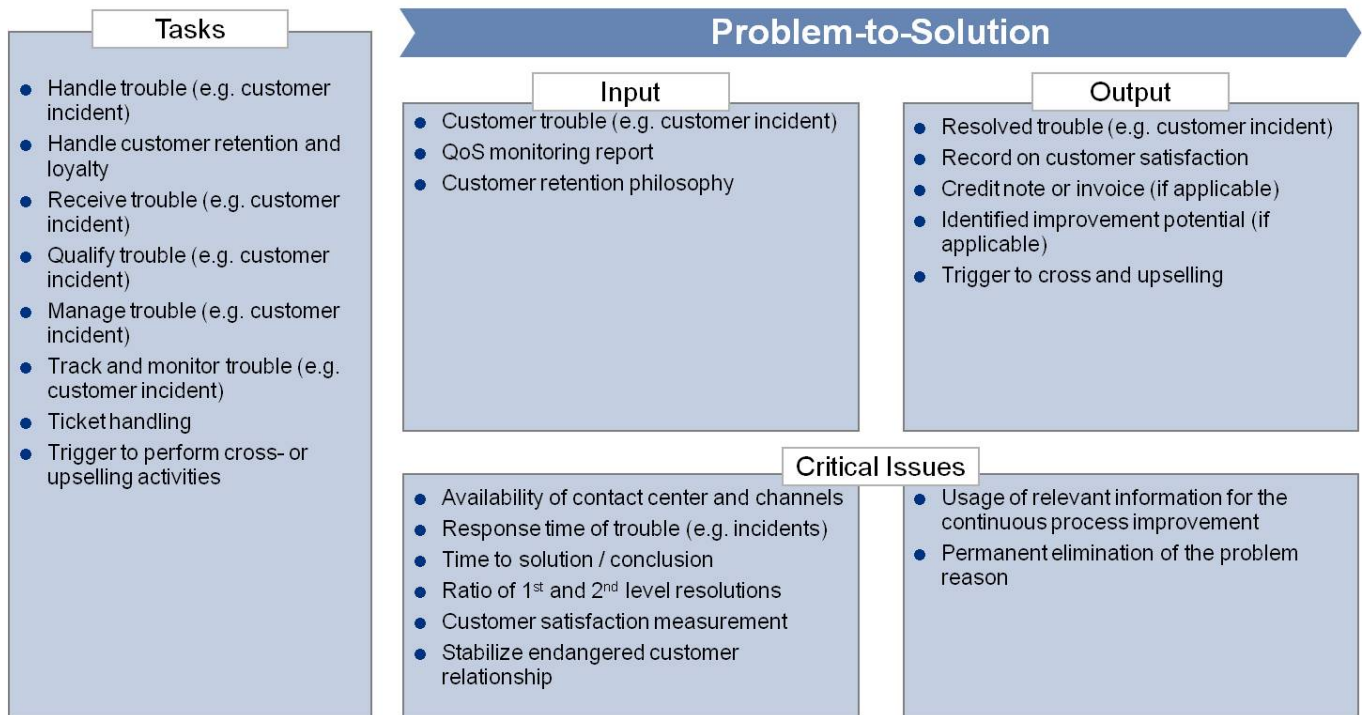


Figure 25. Problem-to-Solution – Detailed Information

2.7. Process Flows for Complaint-to-Solution

2.7.1. Overview

- This process deals with a complaint (problem) initiated by the customer, analyses it to identify the source of the issue, initiates resolution, monitors progress and closes the trouble ticket.
- A complaint is a customer inquiry in which the customer is not pleased with a product or the handling speed of an inquiry etc.

2.7.2. Assumptions:

- This scenario is relevant only for non-technical complaints which are termed as 'complaints'.
- Complaints such as over-due provisioning, quality of customer service and bill adjustments will fall under the scope of this process.

2.7.3. eTOM Mapping

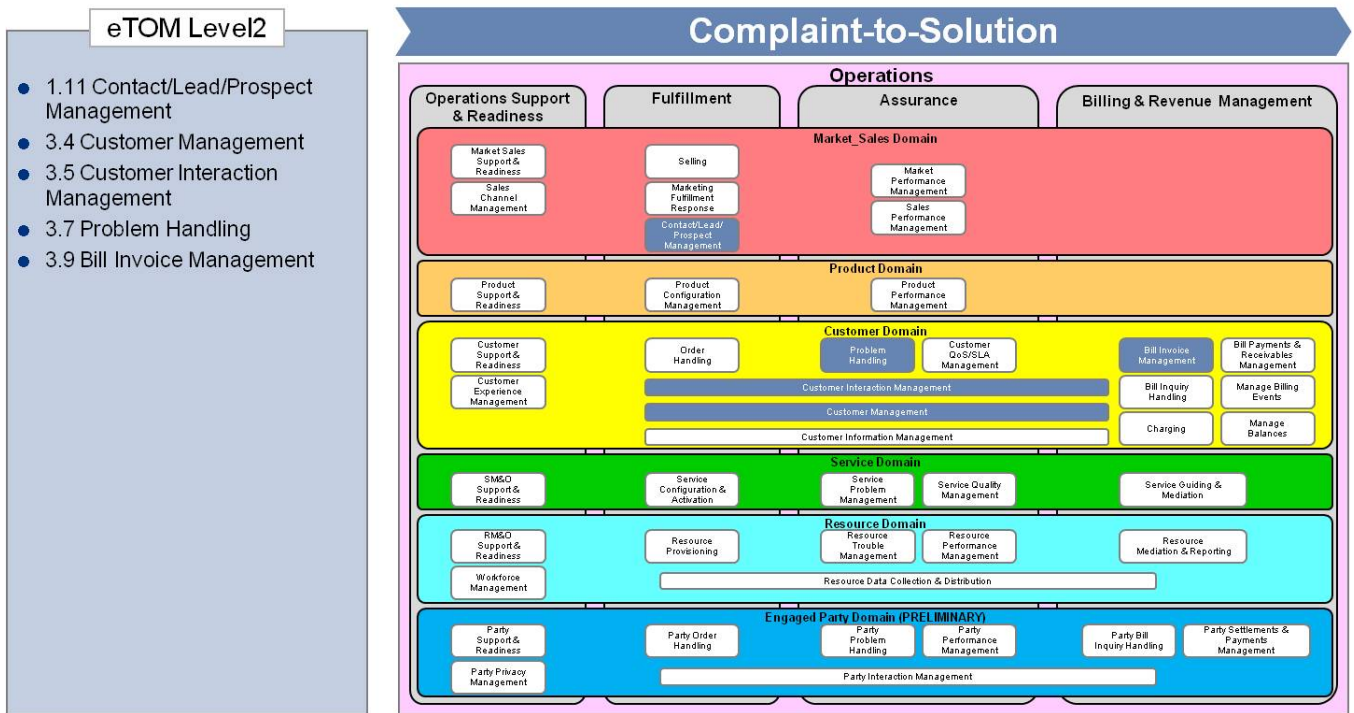


Figure 26. Complaint-to-Solution - Mapping on eTOM Level 2

2.7.4. Detailed Process-Flow (Level 3)

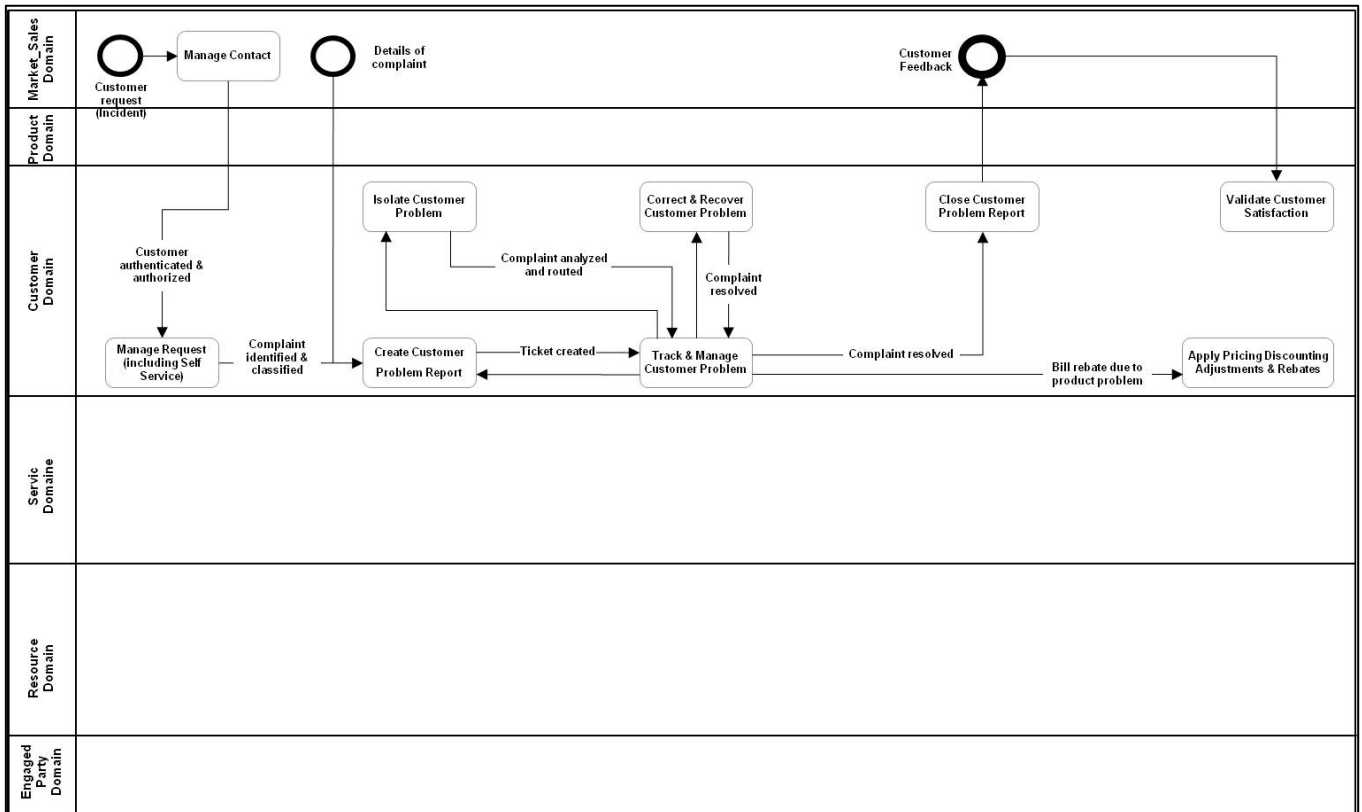
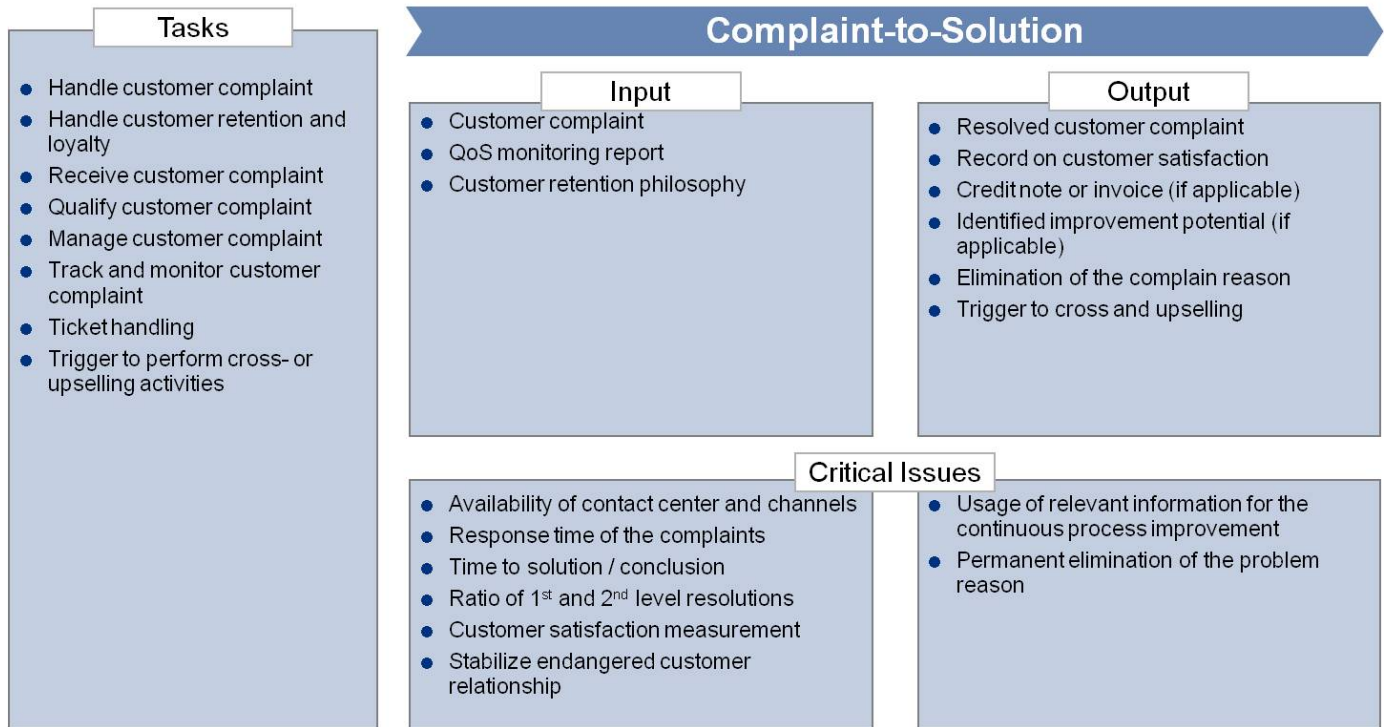


Figure 27. Complaint-to-Solution – Process-Flow (Level 3) BPNM

2.7.5. Detailed Information**Figure 28. Complaint-to-Solution – Detailed Information**

3. Network Centric Process Flows

This section includes sample process-flows (eTOM level 3) relevant to the Network Centric focus. The process-flows for the six network end-to-end processes show how to apply the Operations, Strategy and Enterprise domains of eTOM to a service provider.

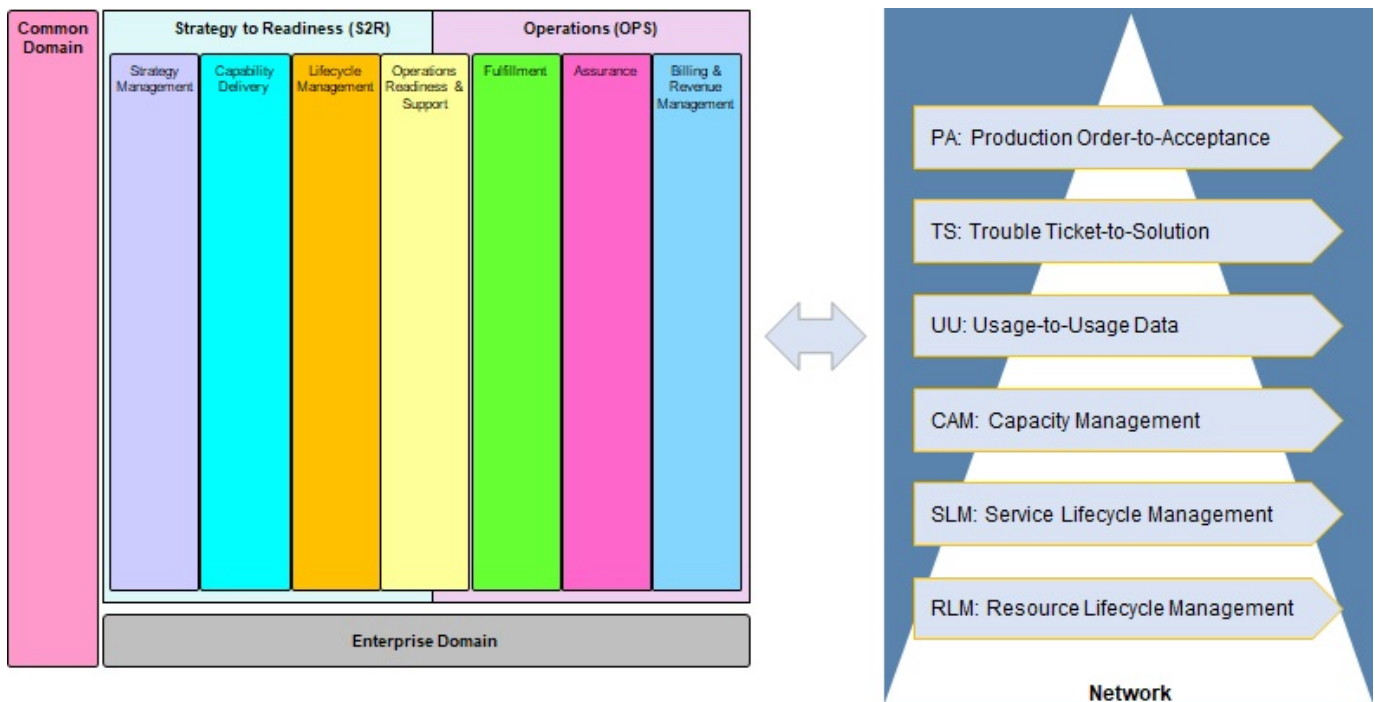


Figure 29. eTOM and Network-Centric processes

3.1. Production Order-to-Acceptance

3.1.1. Overview

- This scenario manages the provisioning and termination process, starting from the feasibility check and ending with the activation of services and resources.
- The order processing includes the verification of service specifications, the availability check for the services and resources, and the identification of critical and uncritical resources.
- The reservation or cancellation of critical resources then takes place, followed by the completion of the suborder with relevant parameters.
- In the end, an overall acceptance test is executed.

3.1.2. Assumptions:

- This scenario is relevant for products offered to the mass customer base.
- This process can be fully automated, but in some cases manual activities are also in scope.

- The processes capacity management, service- and resource lifecycle management manage the requirements of this process.
- The feasibility check considers service-, resource- and workforce capacities.
- The result of the feasibility check is binding for the offer against the customer.
- An order cancellation is not possible at all times. After a certain time period has elapsed, the cancellation of the order cannot be executed.

3.1.3. eTOM Mapping

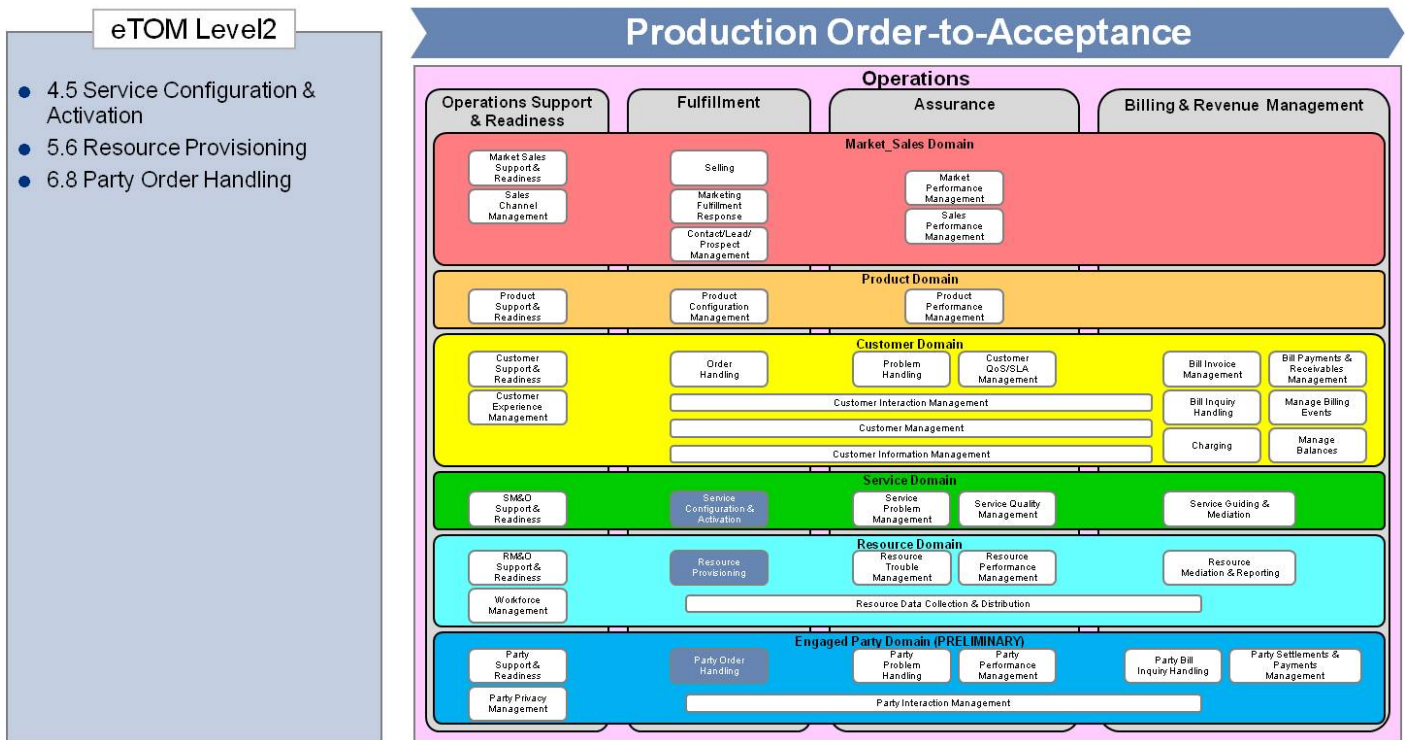


Figure 30. Production Order-to-Acceptance - Mapping on eTOM Level 2

3.1.4. Detailed Process-Flow (Level 3)

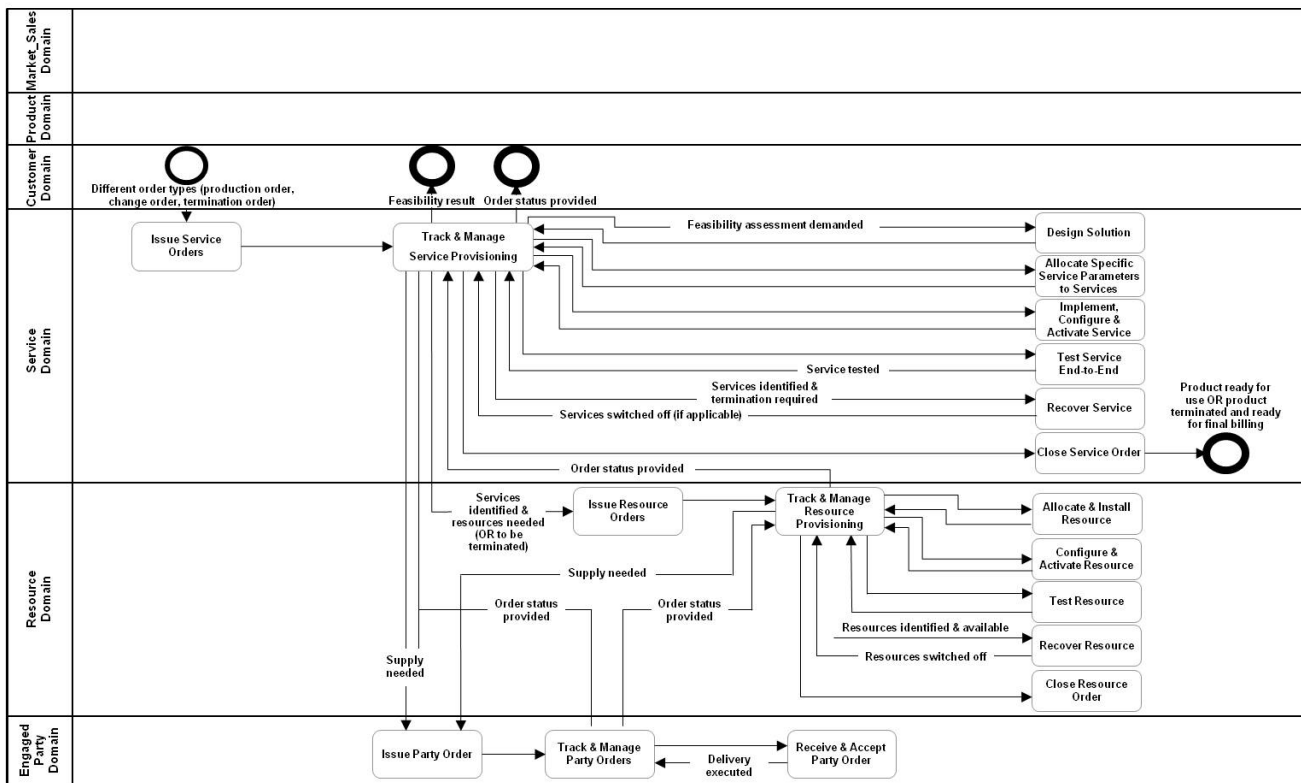


Figure 31. Production Order-to-Acceptance – Process-Flow (Level 3) BPMN

3.1.5. Detailed Information

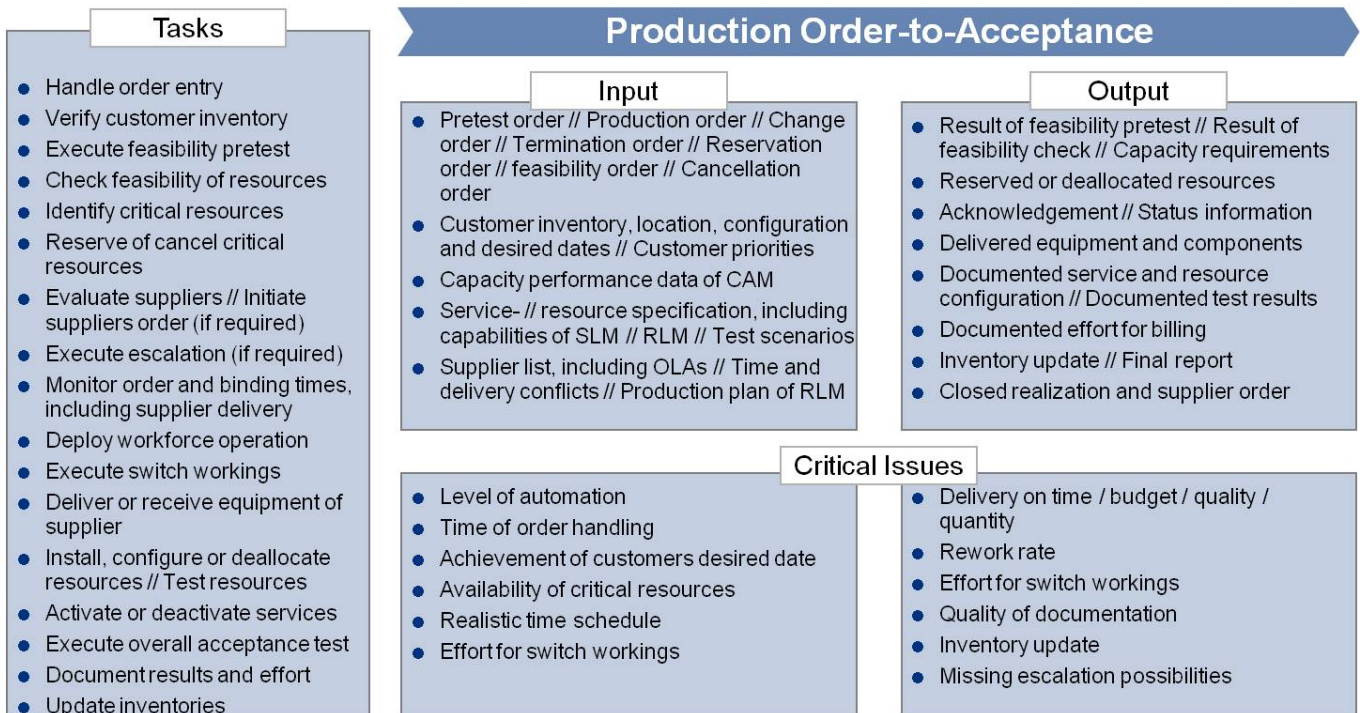


Figure 32. Production Order-to-Acceptance – Detailed Information

3.2. Process Flows for Trouble Ticket-to-Solution

3.2.1. Overview

- This process is either triggered internally through a service or resource alarm, or externally, through a trouble ticket generated based on a complaint of a customer.
- The process covers the qualification, classification and the elimination of the service or resource trouble.
- The analysis of trouble includes the verification of trouble ticket or request, and a delta analysis based on reference and actual values.
- Once the root cause is identified, the trouble eliminated and the acceptance test completed, the trouble ticket is closed.

3.2.2. Assumptions:

- This scenario is relevant for products offered to the mass customer base.
- This process can be fully automated, but in some cases manual activities are also in scope.
- This scenario considers the trouble ticket handling triggered by the customer and the event-based handling triggered by the network.
- The source which opens a ticket is also responsible for closing it.
- An error database is established and can be enhanced quickly and easily.
- The assurance and fulfillment processes work with the same data and Information.

3.2.3. eTOM Mapping

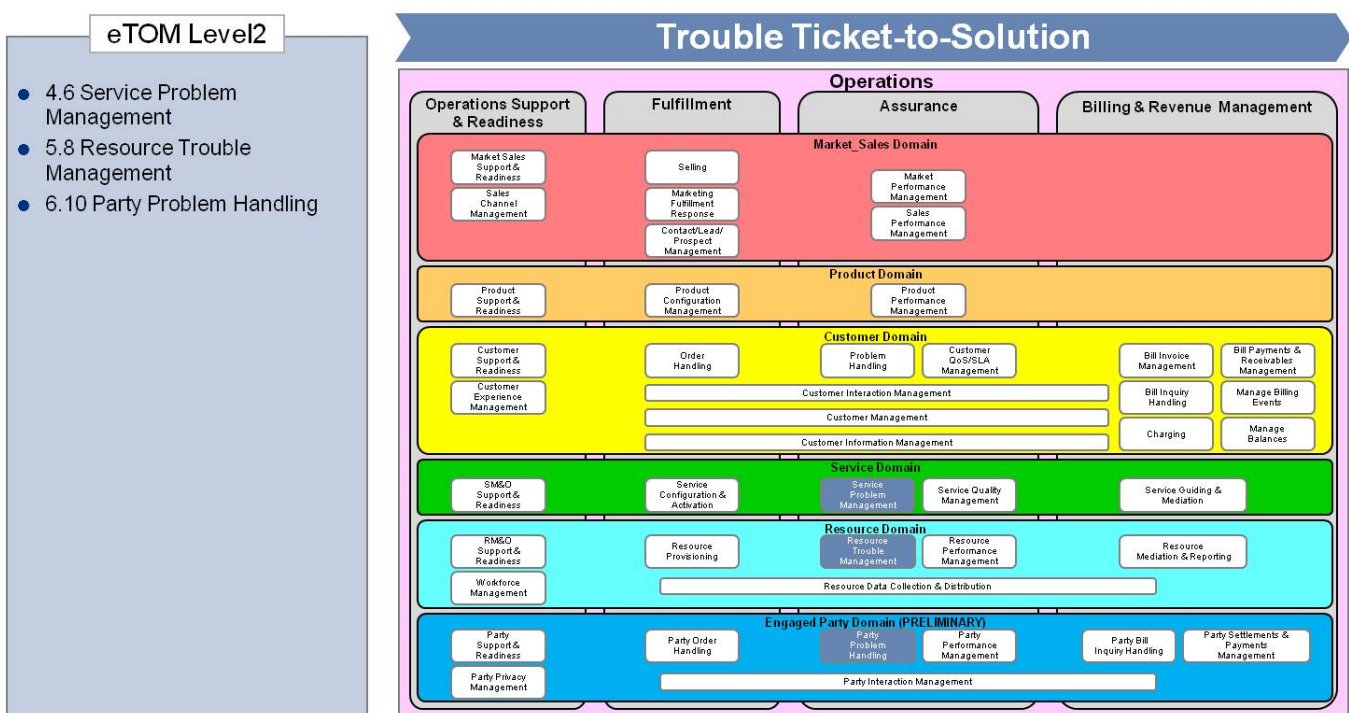


Figure 33. Trouble Ticket-to-Solution - Mapping on eTOM Level 2

3.2.4. Detailed Process-Flow (Level 3)

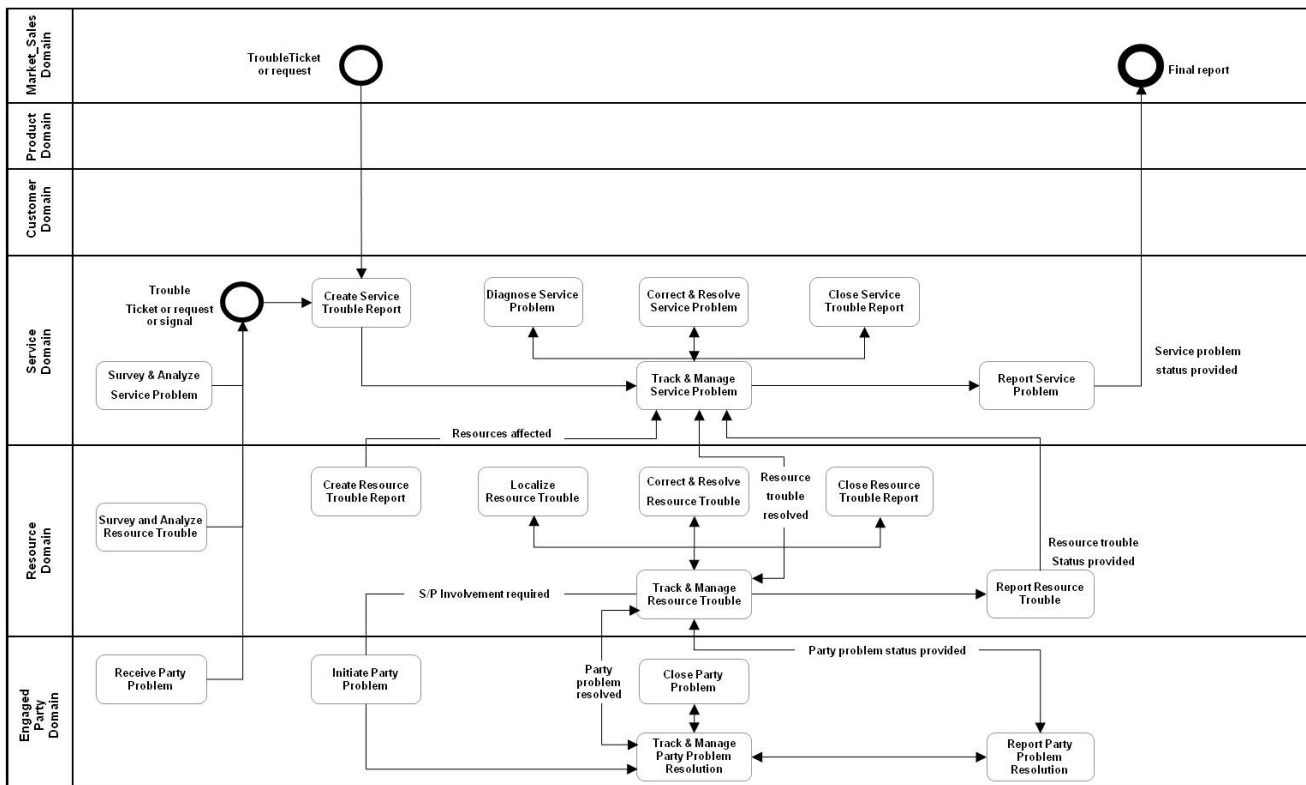


Figure 34. Trouble Ticket-to-Solution – Process-Flow (Level 3) BPNM

3.2.5. Detailed Information

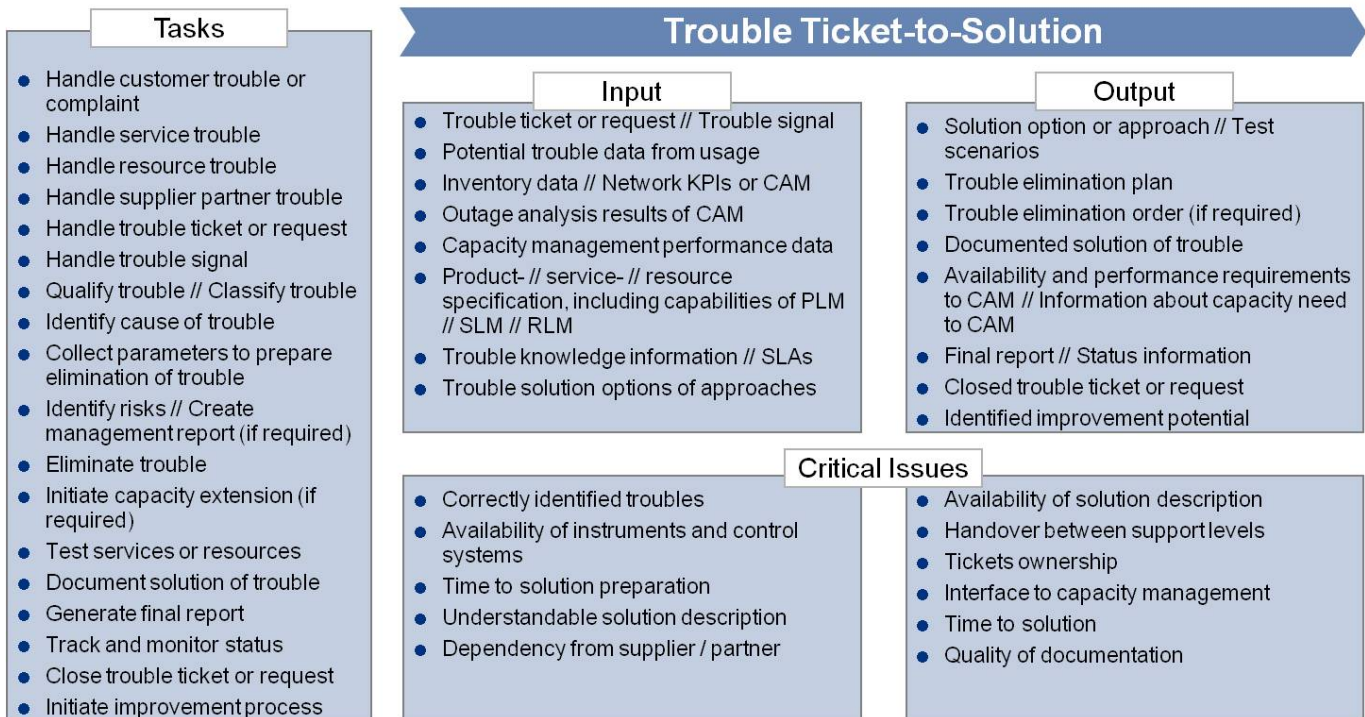


Figure 35. Trouble Ticket-to-Solution – Detailed Information

3.3. Process Flows for Usage-to-Usage Data

3.3.1. Overview

- This process focuses on the enabling of usage, the collection of usage records and the monitoring of performance criteria.
- It takes into consideration the authentication and authorization of the user, including the initiation of subsequently following steps and the providing of self-service.
- During the usage, the collection of usage data records starts.
- After the records pass through the mediation, matching and rating steps, the bill is produced and distributed to the customer.

3.3.2. Assumptions:

- This scenario is relevant for products offered to the mass customer base.
- After the authentication and authorization of the user, the usage is enabled when no connection restrictions exist.
- During the usage, the customer is provided with the option for self-service or other activities.
- The process elements 'Mediate Resource Usage Records' and 'Guide Resource Usage Records' are not always applicable.
- 'Perform Rating' element is usually under the Customer layer for customized products, and under the Service layer for standardized products.

3.3.3. eTOM Mapping

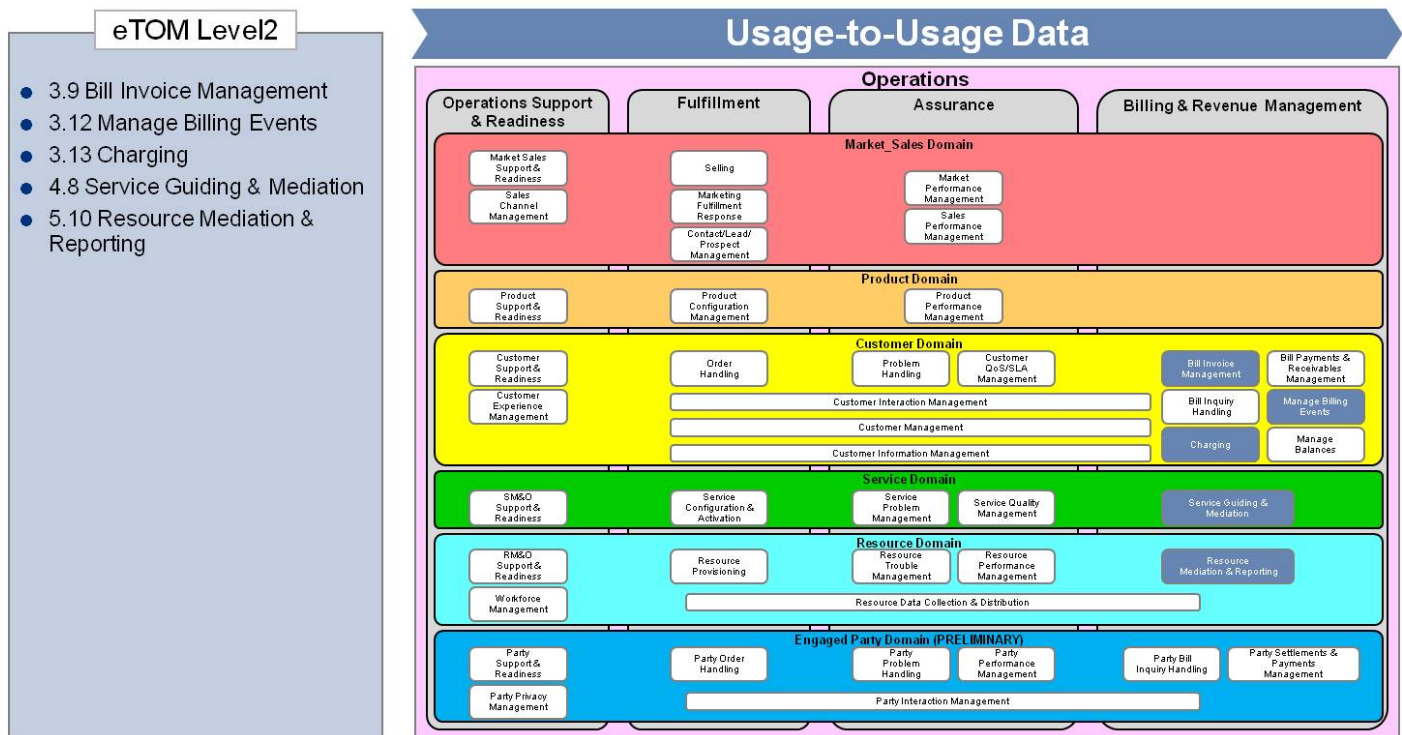


Figure 36. Usage-to-Usage Data - Mapping on eTOM Level 2

3.3.4. Detailed Process-Flow (Level 3)

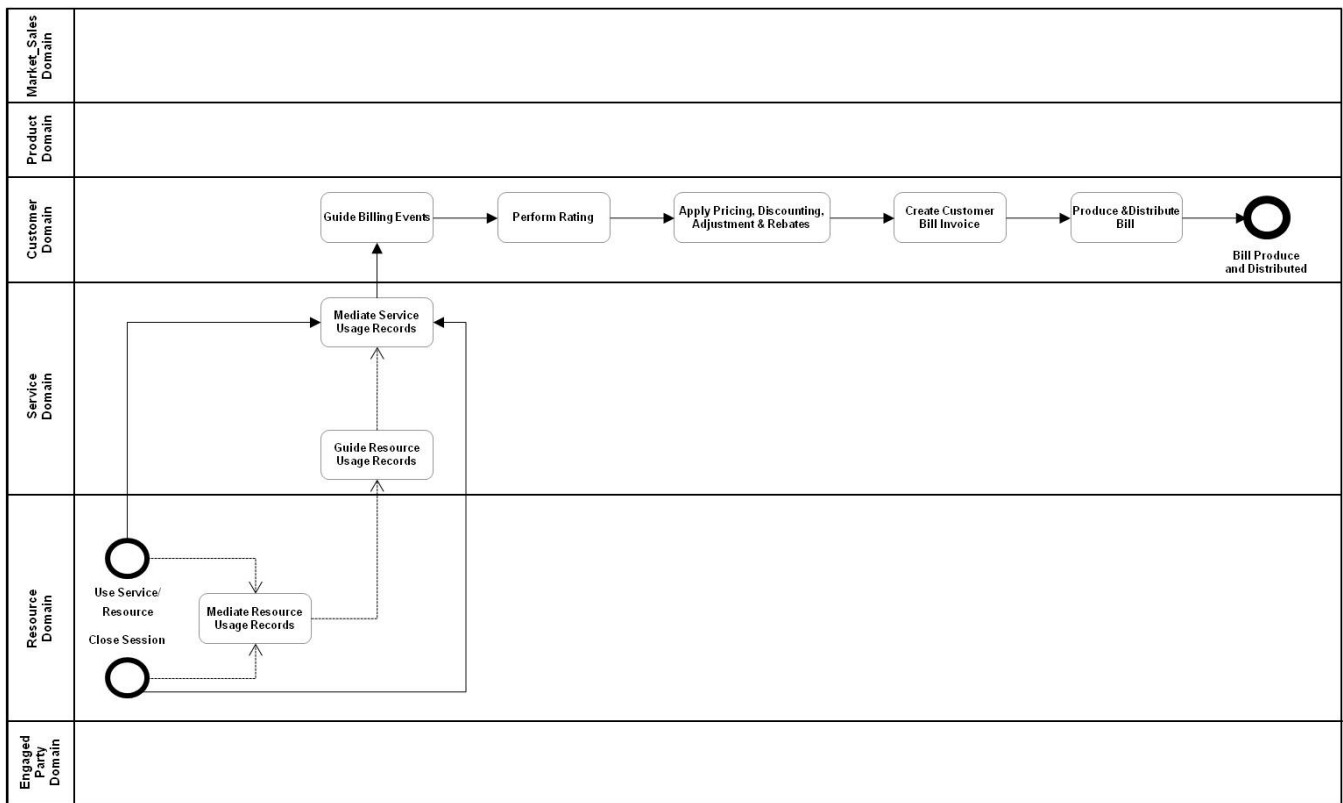


Figure 37. Usage-to-Usage Data – Process-Flow (Level 3) BPNM

3.3.5. Detailed Information

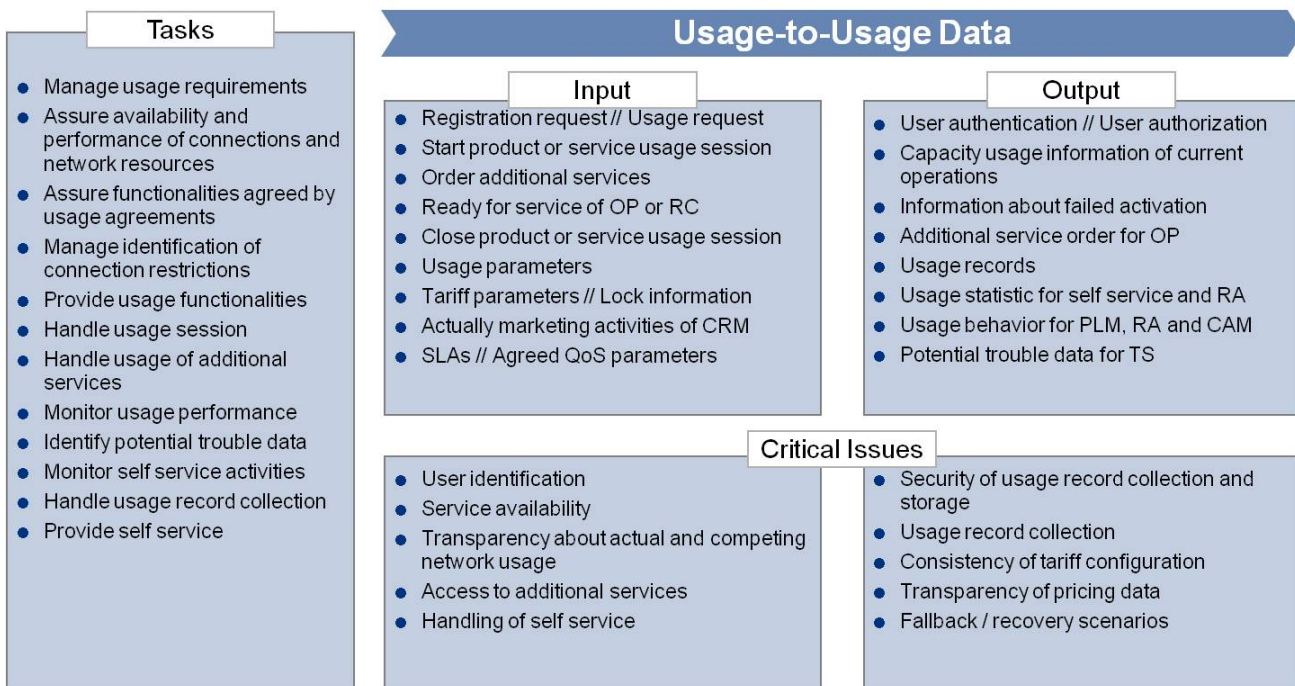


Figure 38. Usage-to-Usage Data – Detailed Information

3.4. Process Flows for Capacity Management

3.4.1. Overview

- This process aims to ensure the timely and cost-efficient provisioning of the accurate capacity of services and components.
- There is a collection and qualification of capacity information and the analysis of capacity needs to guarantee adequate levels of capacity for support of operations.
- This process can be nearly automated, but in some cases manual activities are also in scope. Data collection and analysis can leverage big data analytics and machine learning algorithms to improve efficiency and accuracy. It may require collecting information from other domains' processes such as Market, Sales domain or Product domain processes.
- This process may require aligning with other domains such as Market, Sales domain, Product domain, Service domain, etc. through "Domain Orchestration" process to provide more precise capacity planning.
- Future forecast of business demands regarding capacity may leverage AI to provide more accurate capacity demand.
- The scope of capacity management is specified, including the definition of KPIs and measurement methods.
- After the planning and design phase, the implementation plan is developed and executed.
- Lastly, the results of capacity management are handed over to operations and performance is continuously monitored.

3.4.2. Assumptions:

- KPIs and performance measurement features are designed and established.
- Type of capacity which required to be managed and capacity period are pre-defined.
- The resource, sales & marketing, product and service relevant data is available and usable.
- This process scenario is integrated in the network expansion planning process and aims to support rolling expansion planning (quarterly, monthly, etc.) requirements.
- The process scenario manages the requirements of the fulfillment process.
- This scenario is not technically independent.
- This scenario takes into consideration the capacity and availability management as per ITIL V3.

3.4.3. eTOM Mapping

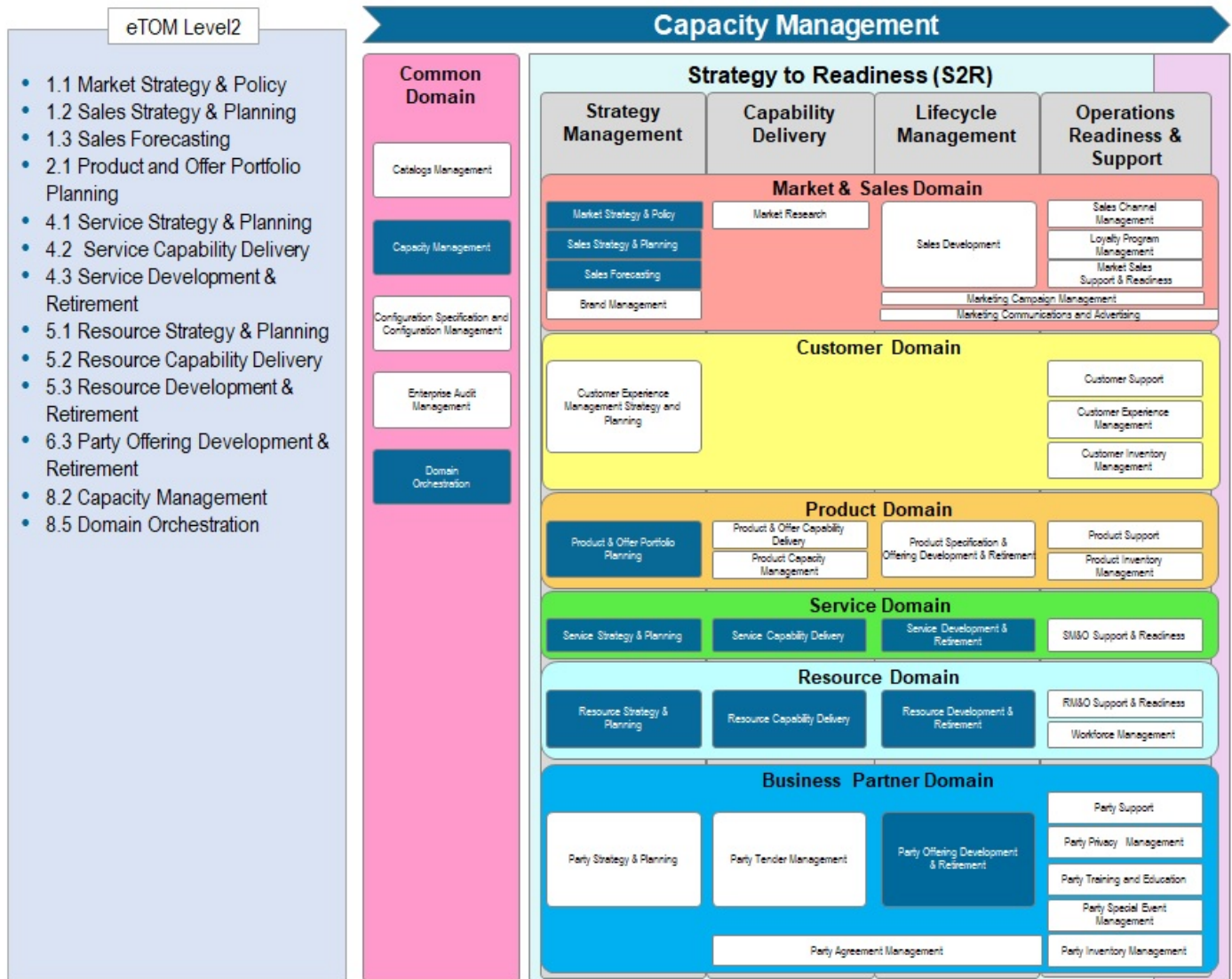


Figure 39. Capacity Management - Mapping on eTOM Level 2 – Strategy to Readiness

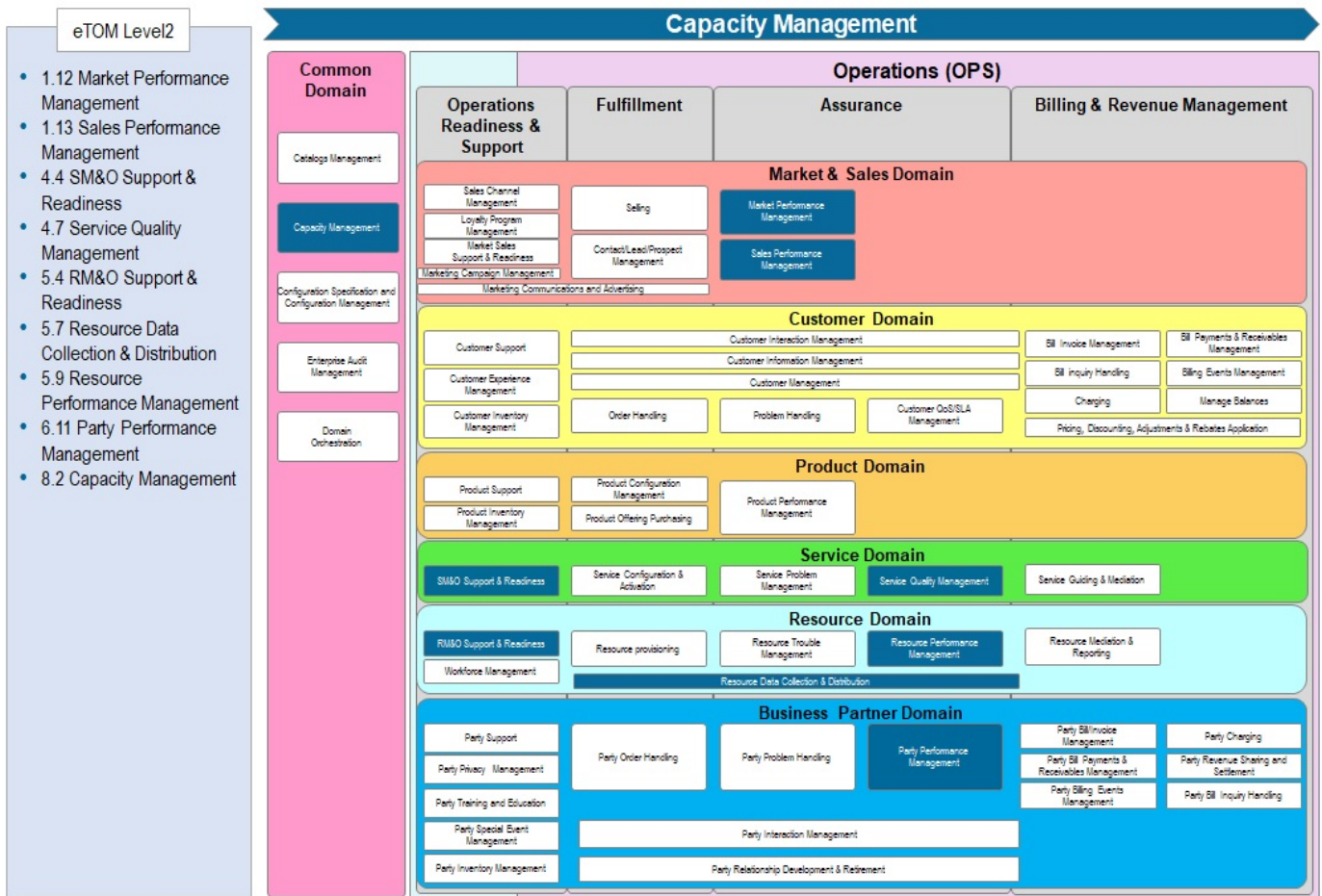


Figure 40. Capacity Management - Mapping on eTOM Level 2 – Operations

3.4.4. Detailed Process-Flow (Level 3)

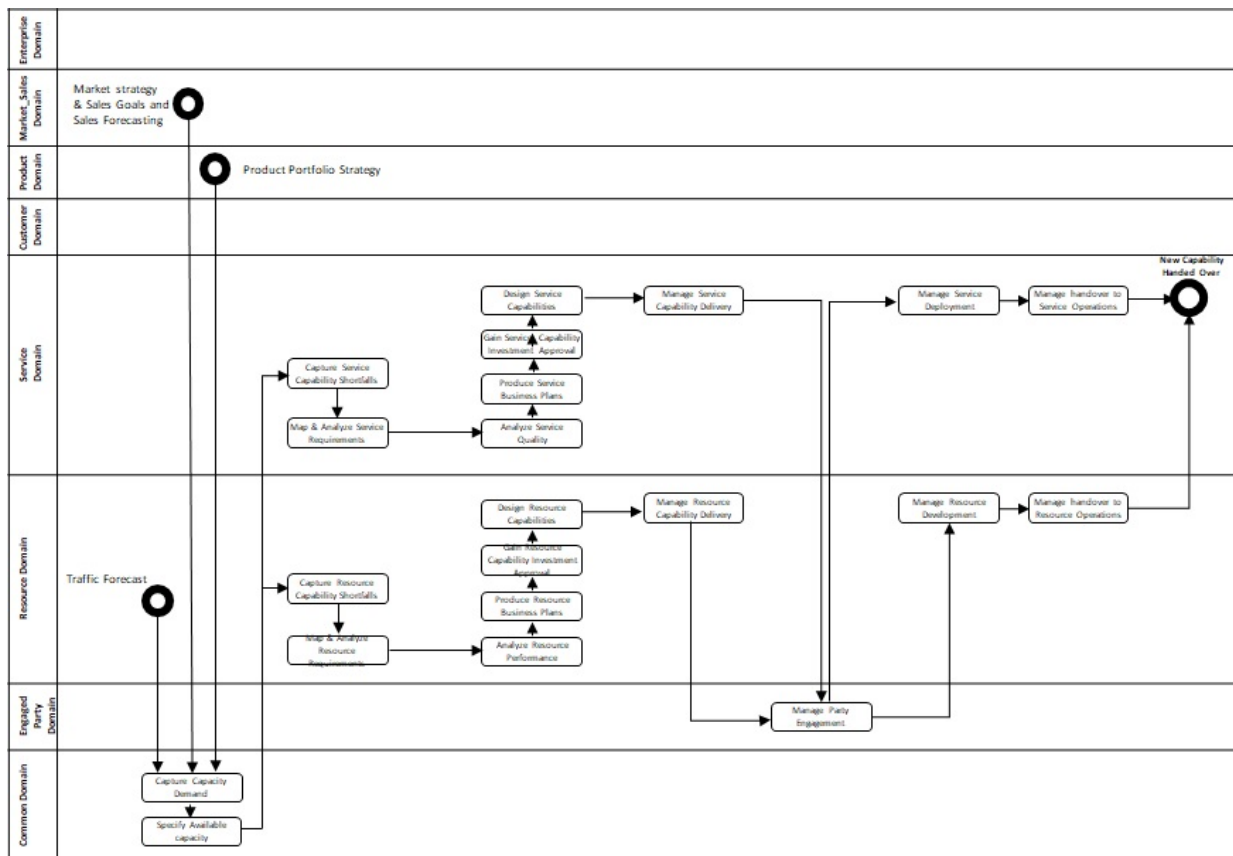


Figure 41. Capacity Management – Process-Flow (Level 3) BPNM

3.4.5. Detailed Information

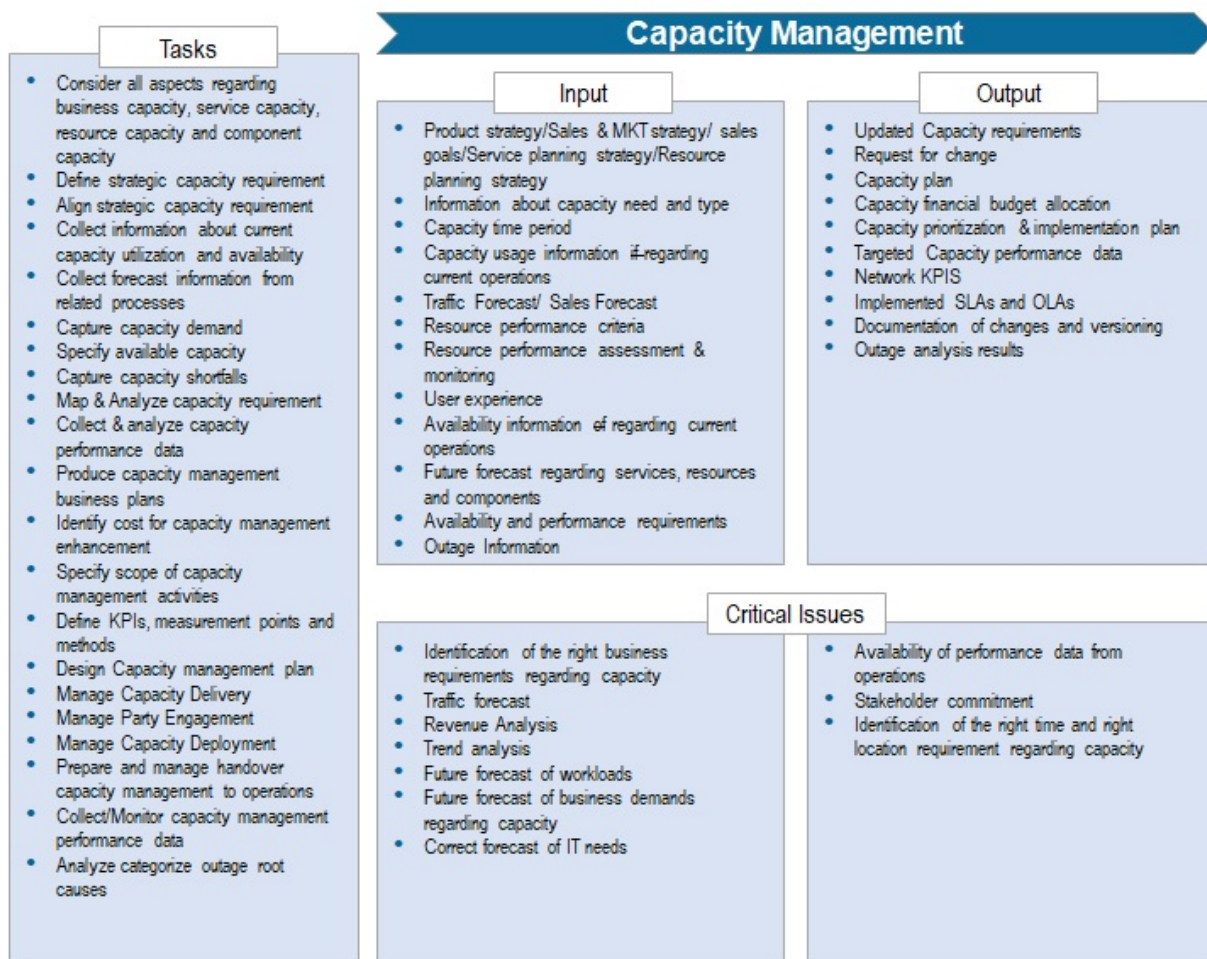


Figure 42. Capacity Management – Detailed Information

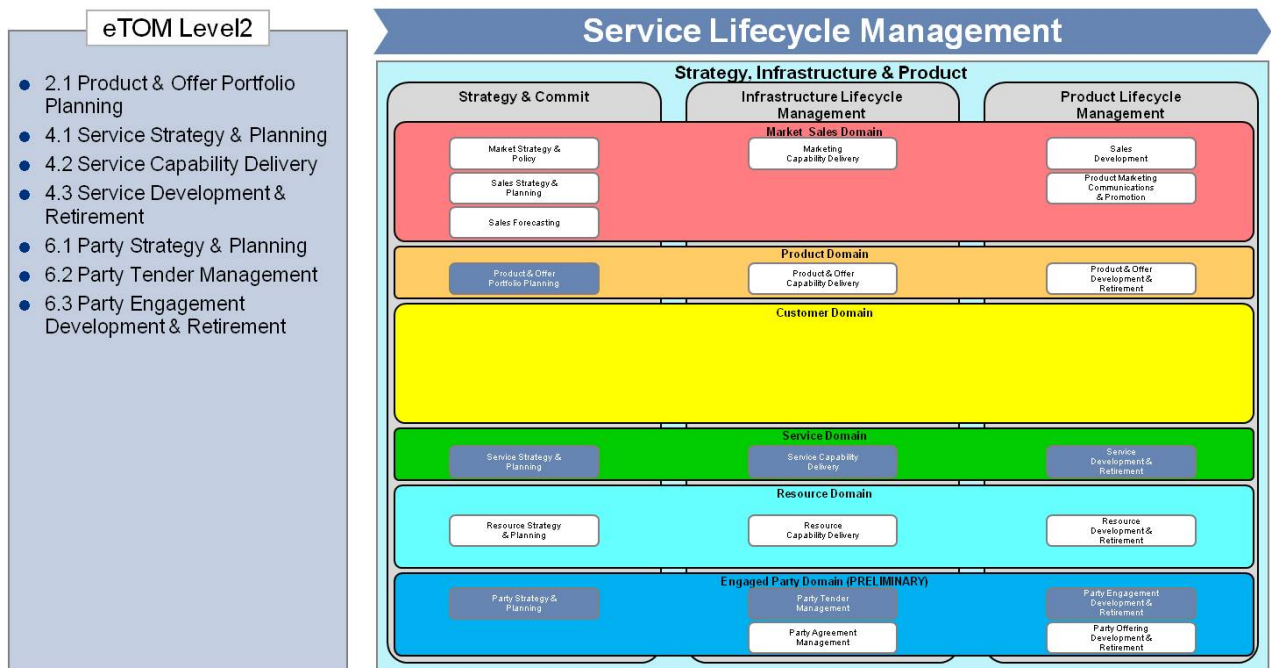
3.5. Process Flows for Service Lifecycle Management

3.5.1. Overview

- This process defines, plans, designs and implements all services in order to support the introduction, operations and retirement of market products.
- The process starts with the evaluation of market trends, service specific needs and the definition of the service development and exit strategy.
- This is followed by the creation of the business plan and definition of the service architecture and other strategic documents.
- The design and development of service solutions are in line with the strategic goals and required service capabilities.
- After the implementation and a successful testing phase, the service is handed over to operations.

3.5.2. Assumptions:

- The product portfolio requirements are defined as per the input from Product Marketing.
- A result of the Service Lifecycle Process may act as an input for the fulfillment process
- Further results of this scenario are requirements for the capacity management process.
- Direct links between products and services are modeled and stored in the product catalog.
- This scenario does not take into consideration the retirement phase.
- In some cases, migration scenarios are needed.

3.5.3. eTOM Mapping**Figure 43. Service Lifecycle Management- Mapping on eTOM Level 2 – Strategy**

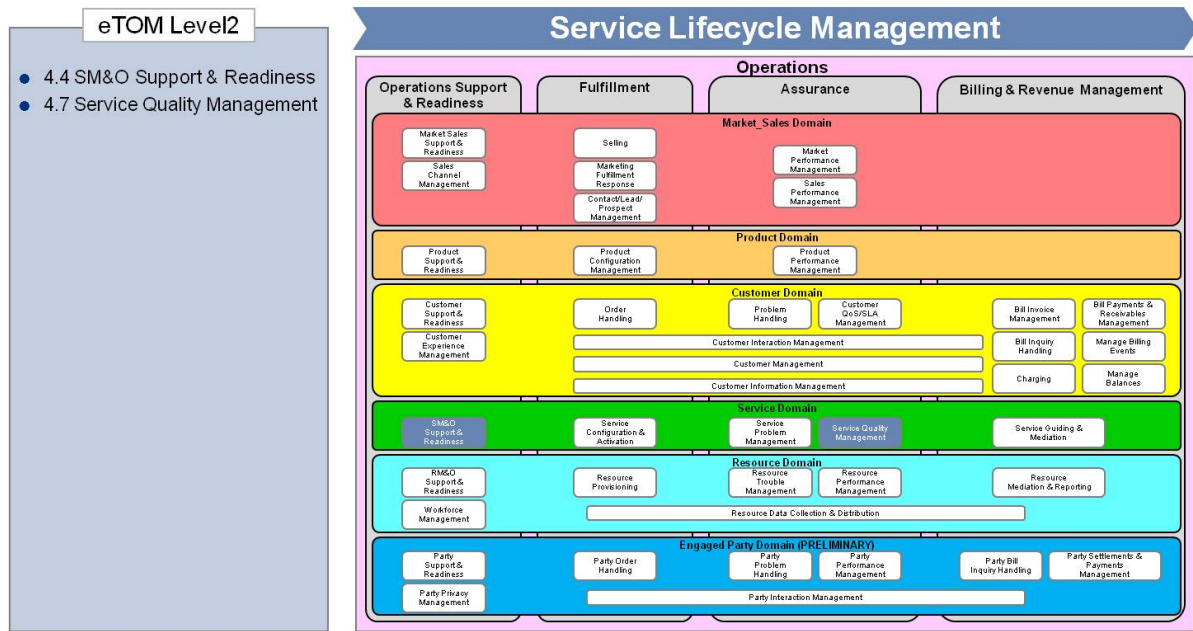


Figure 44. Service Lifecycle Management - Mapping on eTOM Level 2 – Operations

3.5.4. Detailed Process-Flow (Level 3)

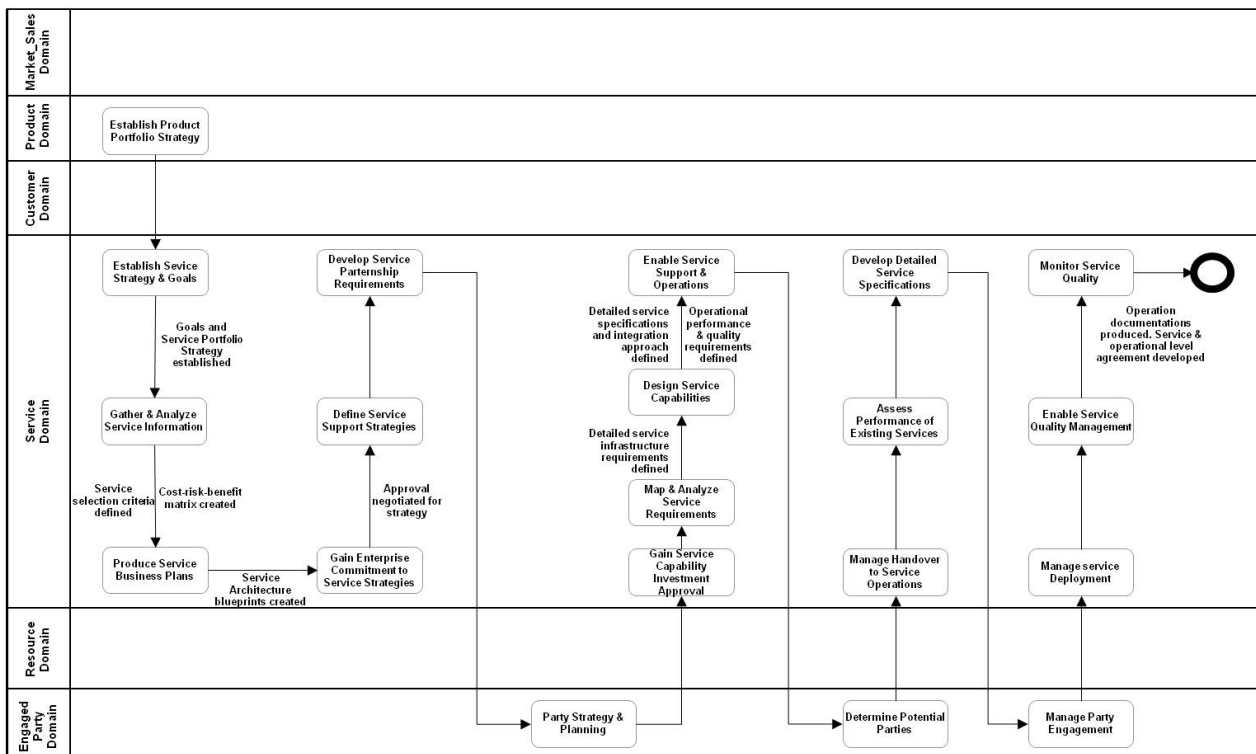


Figure 45. Service Lifecycle Management – Process-Flow (Level 3) BPNM

3.5.5. Detailed Information

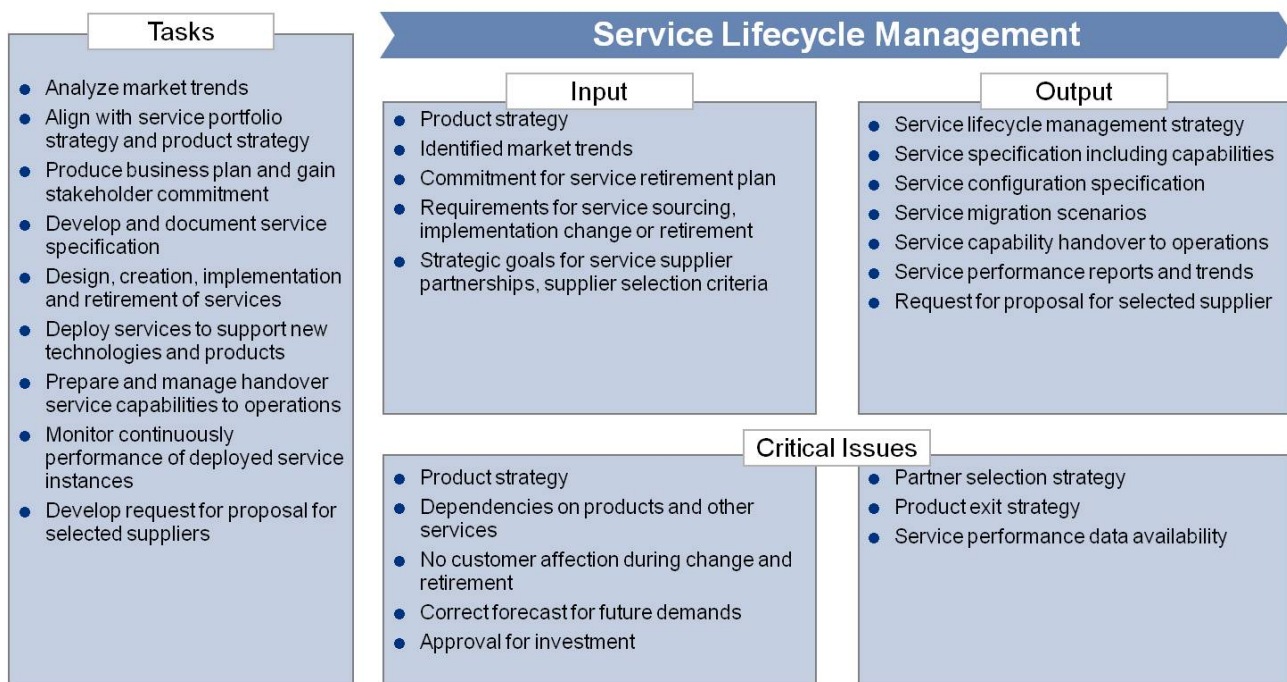


Figure 46. Service Lifecycle Management – Detailed Information

3.6. Process Flows for Resource Lifecycle Management

3.6.1. Overview

- This process defines, plans, designs and implements all resources in order to support the introduction, operations and retirement of market products.
- The process starts with the evaluation of market trends, resource specific needs and the definition of the resource development and exit strategy.
- This is followed by the creation of the business plan and definition of the resource architecture and other strategic documents.
- The design and development of resource solutions are in line with the strategic goals and required resource capabilities.
- After the implementation and a successful testing phase, the resource is handed over to operations.

3.6.2. Assumptions:

- The product portfolio requirements are defined as per the input from Product Marketing.
- A result of the Resource Lifecycle Process may act as an input for the fulfillment process
- Further results of this scenario are requirements for the capacity management process.
- Direct links between products, services and resources are modeled and stored in the product catalog.
- This scenario does not take into consideration the retirement phase.
- In some cases, migration scenarios are needed.

3.6.3. eTOM Mapping

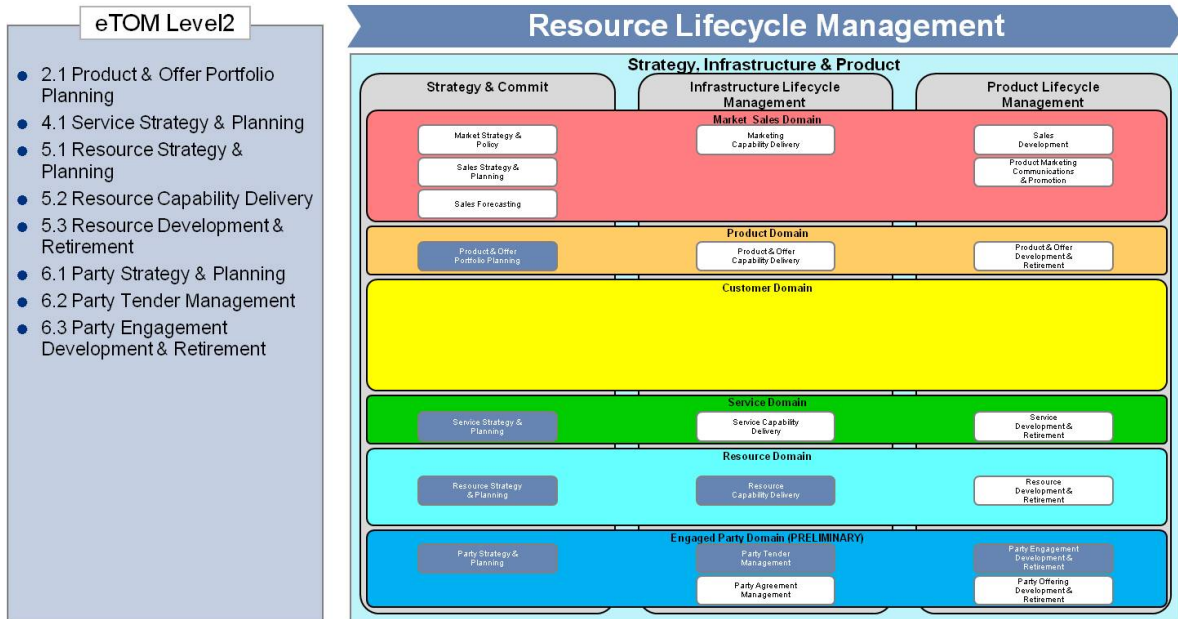


Figure 47. Resource Lifecycle Management- Mapping on eTOM Level 2 – Strategy

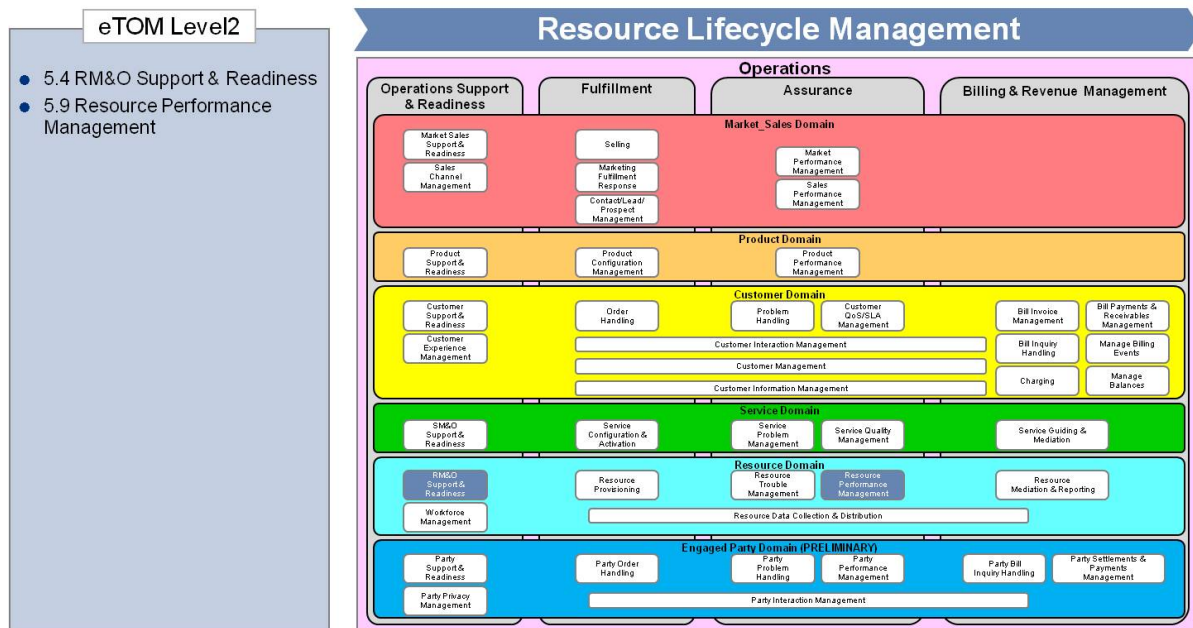


Figure 48. Resource Lifecycle Management- Mapping on eTOM Level 2 – Operations

3.6.4. Detailed Process-Flow (Level 3)

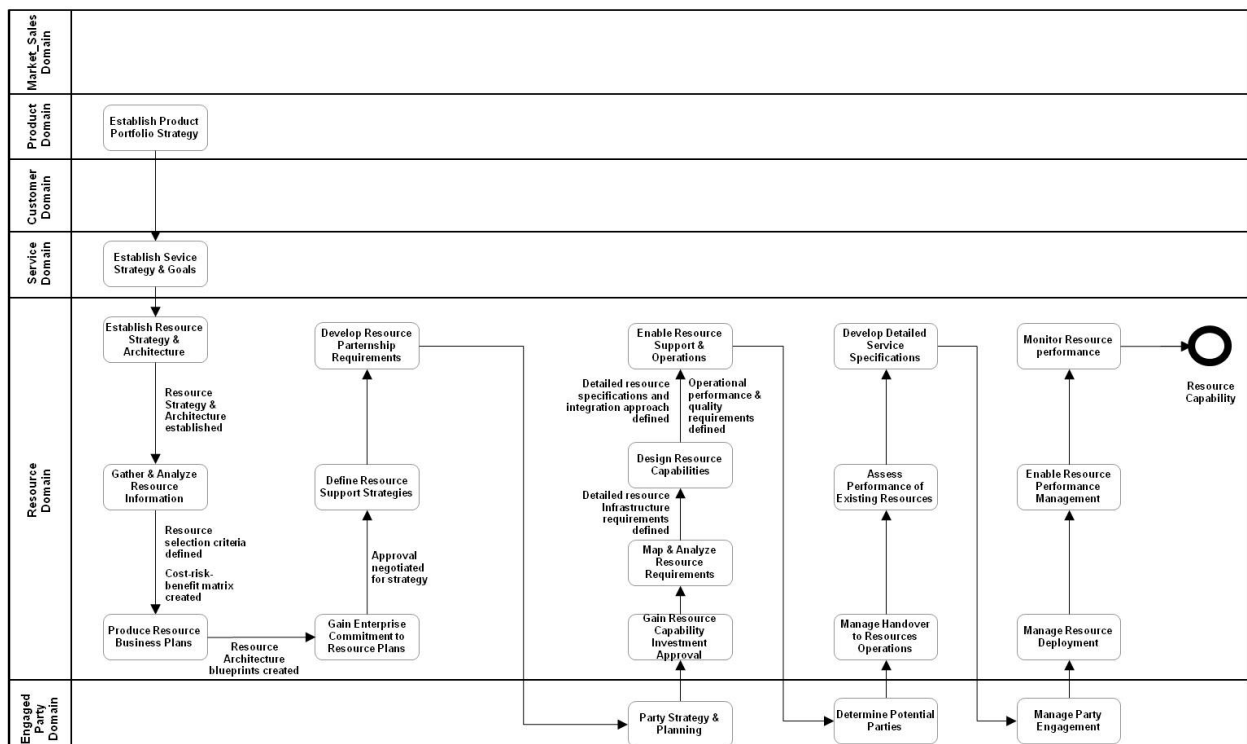


Figure 49. Resource Lifecycle Management – Process-Flow (Level 3) BPMN

3.6.5. Detailed Information

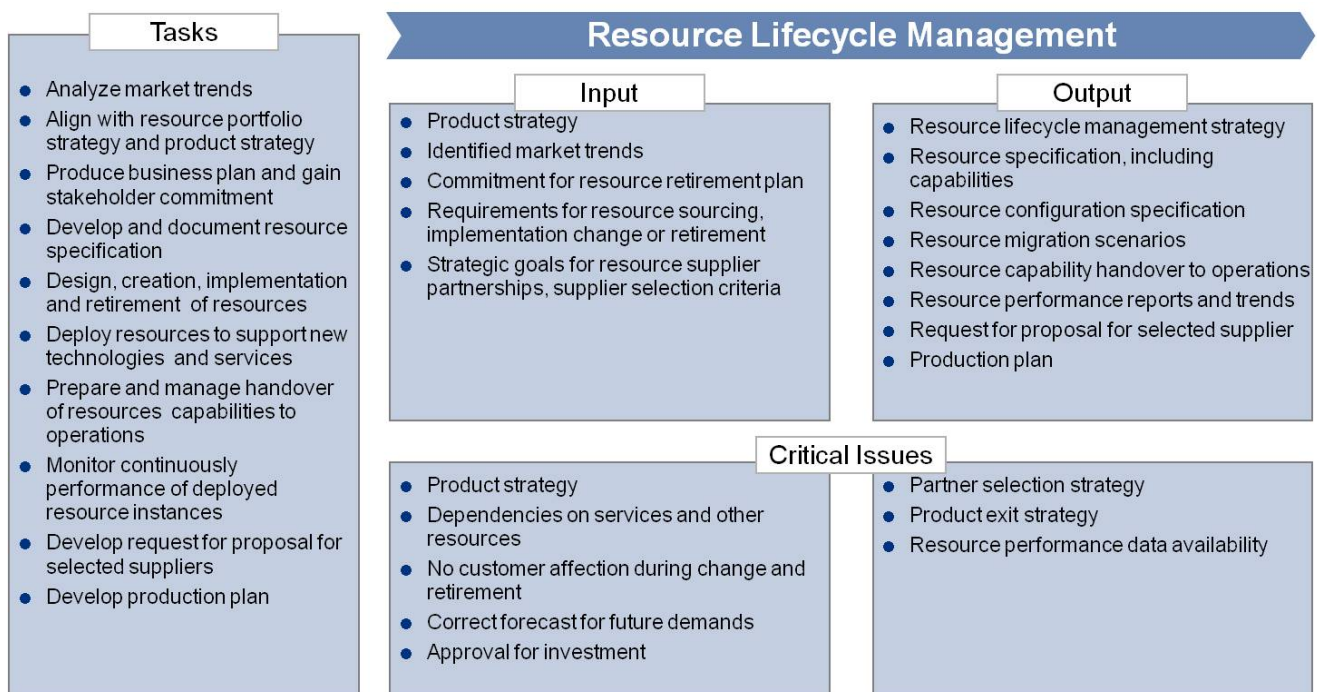


Figure 50. Resource Lifecycle Management – Detailed Information

4. End-to-End Product Flow

This section includes sample process-flows (eTOM level 3) relevant to the Product Centric focus. The process flows for the six Product Centric end-to-end processes recommend how to apply the Strategy, Infrastructure and Product domain of eTOM to a service provider.

The described examples of end-to-end processes from the product domain are based on Detecon project experience and telco best practices from various operators, and focuses on the following processes:

- Idea-to-Business Plan
- Idea-to-Business Proposal
- Business Proposal-to-Launch
- Assessment-to-Relaunch
- Assessment-to-Retirement
- Market Strategy-to-Campaign

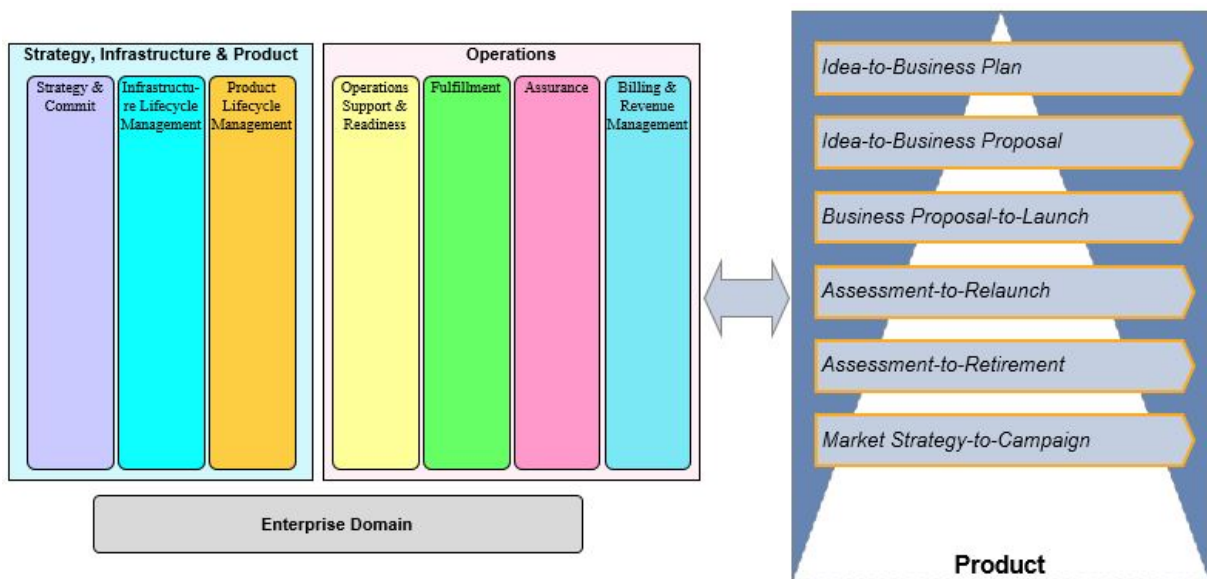


Figure 51. eTOM and Product-Centric processes

4.1. Process Flows for Idea-to-Business Plan

4.1.1. Overview

- This process shows the strategic part of portfolio management, starting with monitoring of market trends (external perspective) and gathering product information (internal perspective) and ending with an approved business plan.

- A high-level analysis of Customer Experience Management goals, service requirements and partnership requirements deliver inputs to the product portfolio strategy.
- An approval of investment in enhancement of available service capabilities may also be part of the process flow
- A product portfolio strategy and related business plans are outputs of this process and are the framework for the following management of product offerings

4.1.2. eTOM Mapping

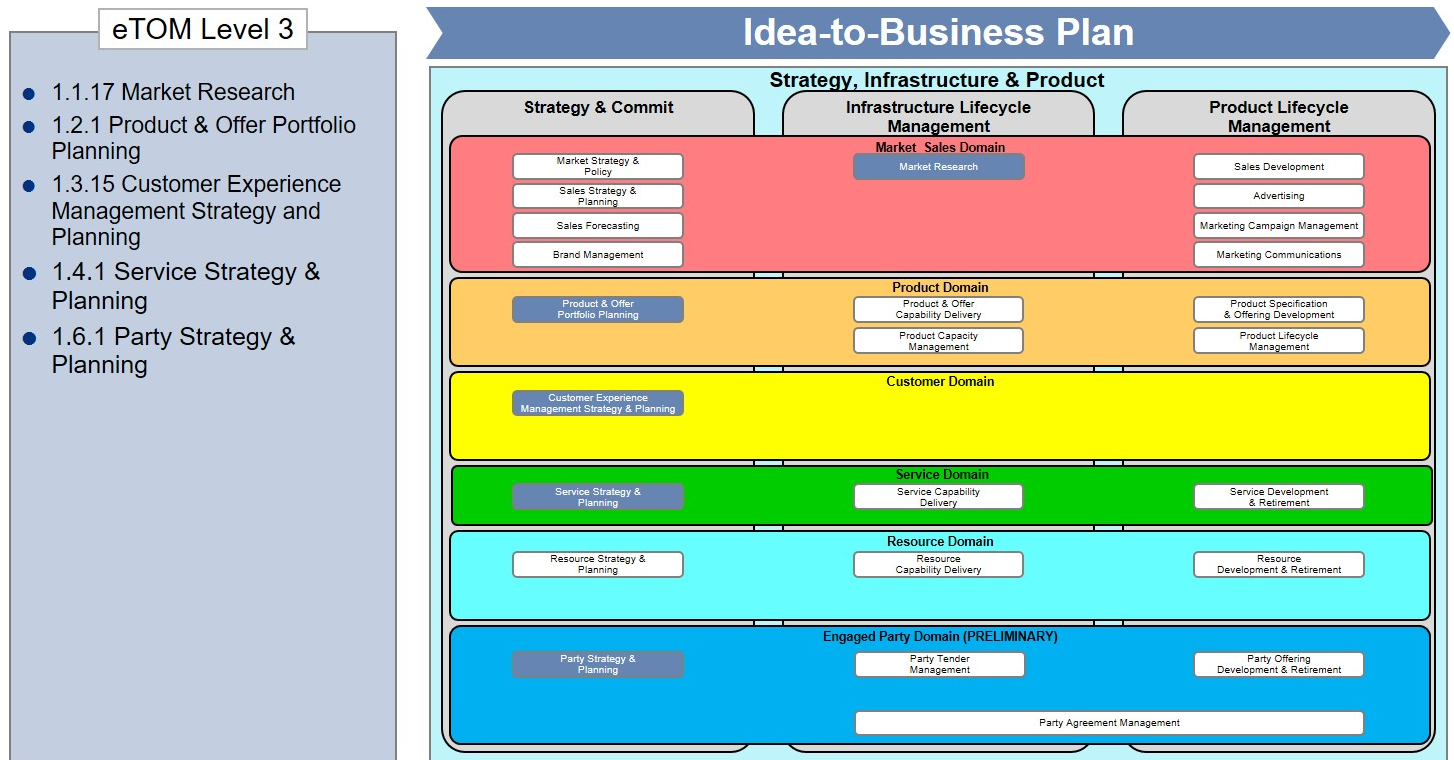


Figure 52. eTOM Mapping for Idea-to-Business Plan

4.1.3. Detailed Process-Flow (Level 3)

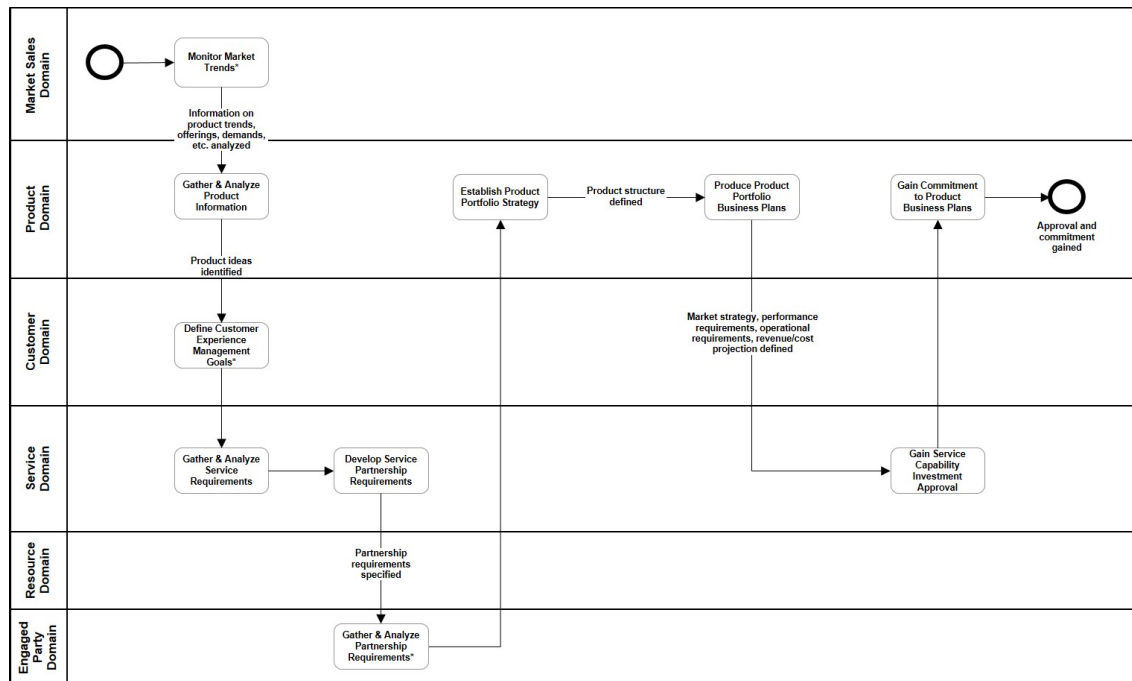


Figure 53. Workflow for Idea-to-Business Plan

4.1.4. Detailed Information

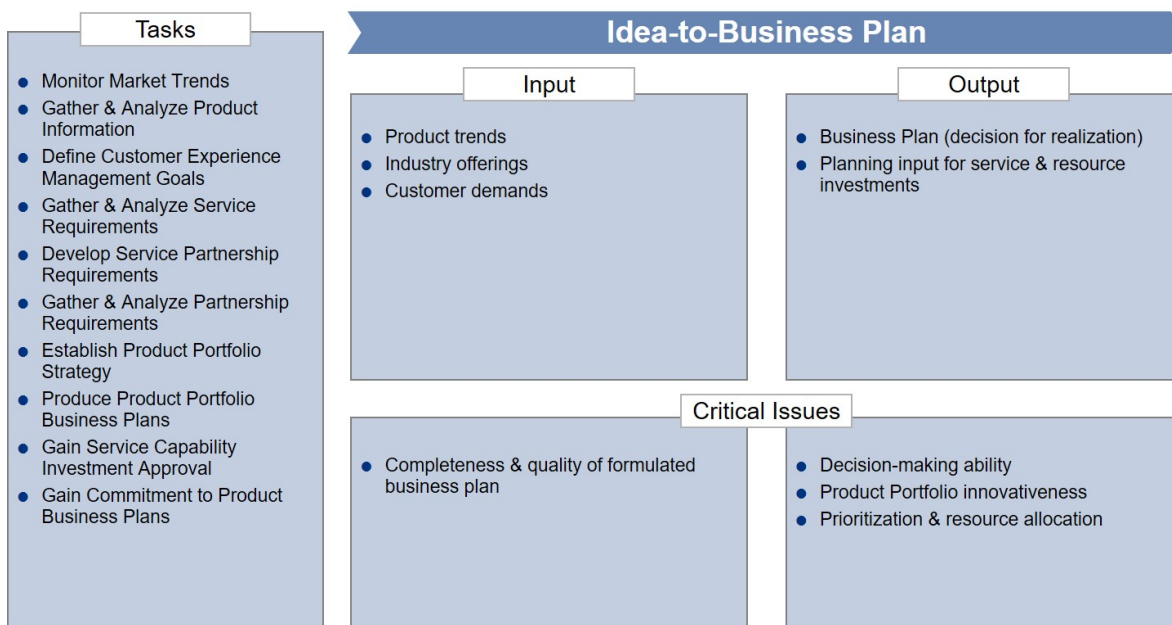


Figure 54. Details for Idea-to-Business Plan

4.2. Process Flows for Idea-to-Business Proposal

4.2.1. Overview

- The process describes the activities related to the management of single product offerings, starting with idea generation and with an approved business proposal (incl. product blueprint, forecasts & financial figures)
- The main purpose of this process is to develop (high-level) product specifications and assess the feasibility of launching the product
- At the end of this process a product committee (or a similar decision making body) takes a Go/No-Go decision for commercial product launch

4.2.2. eTOM Mapping

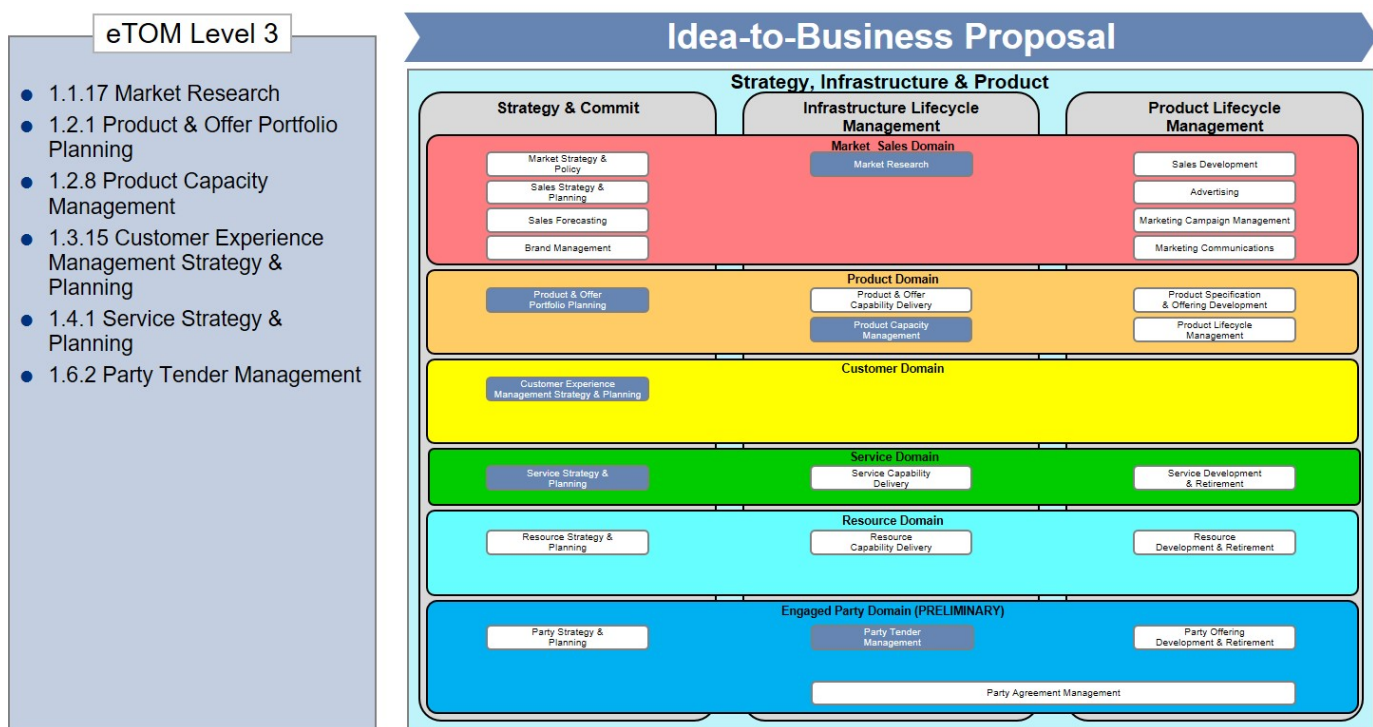


Figure 55. eTOM Mapping for Idea-to-Business Proposal

4.2.3. Detailed Process-Flow (Level 3)

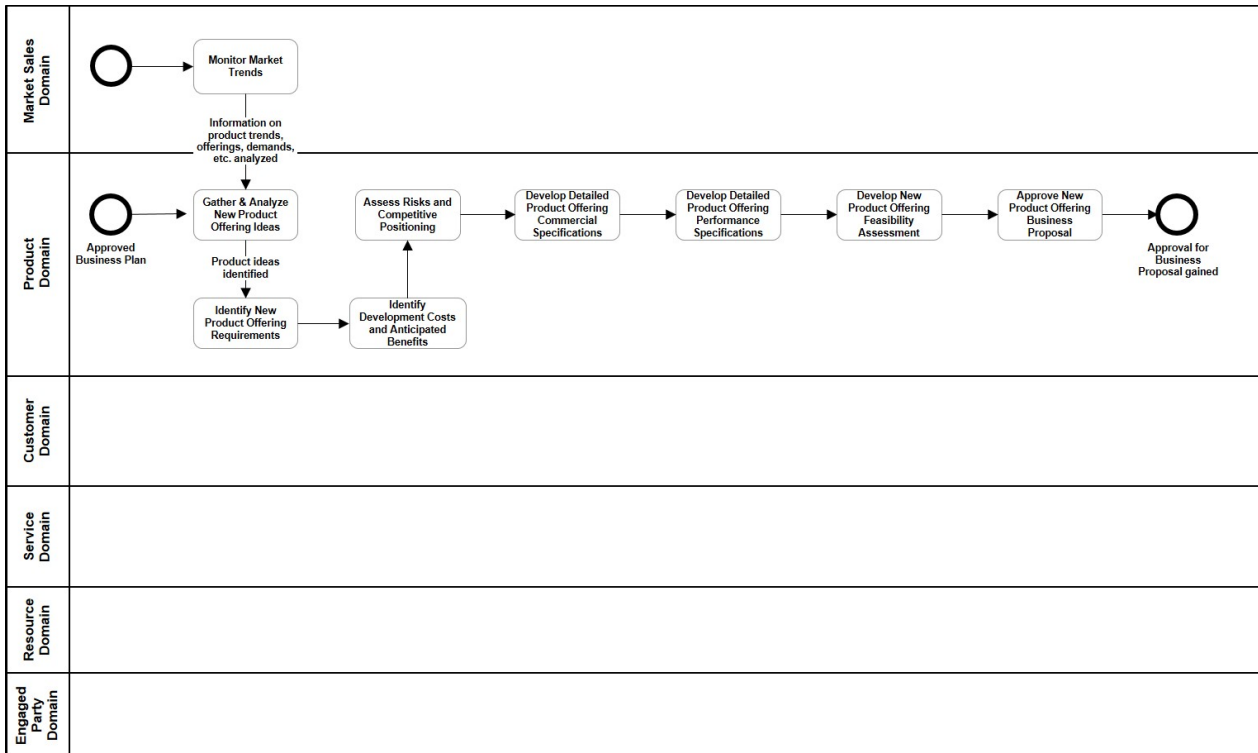


Figure 56. Workflow for Idea-to-Business Proposal

4.2.4. Detailed Information

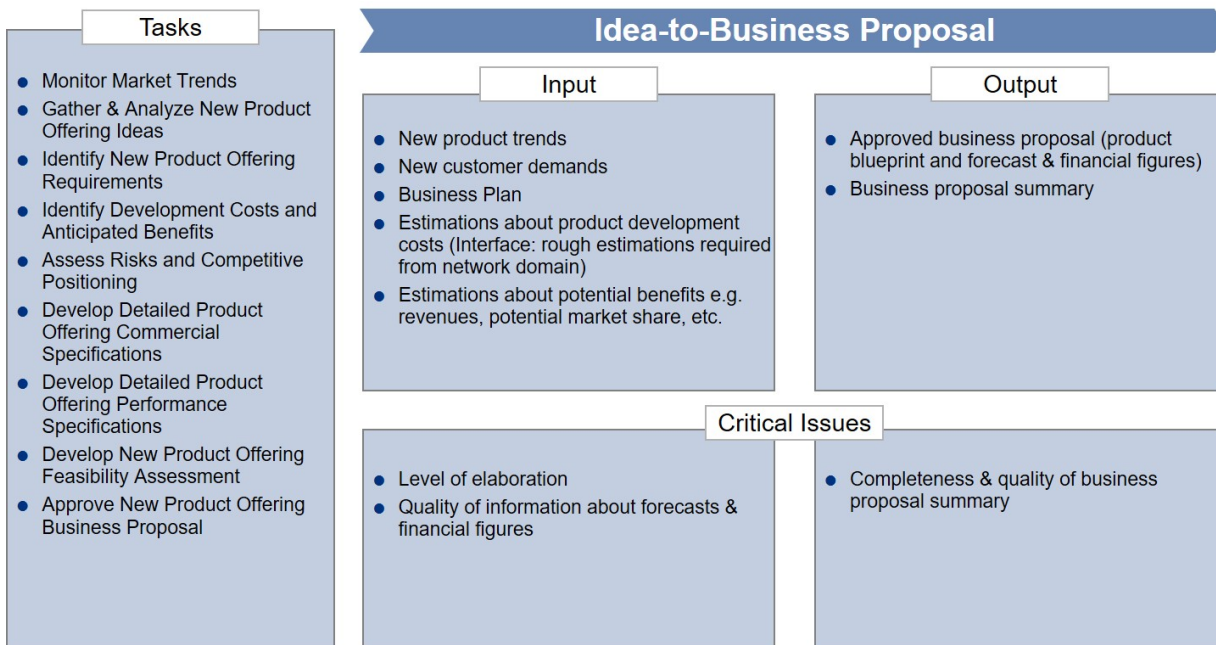


Figure 57. Details for Idea-to-Business Proposal

4.3. Process Flows for Business Proposal-to-Launch

4.3.1. Overview

- The process describes the activities related to the preparation of product launch, starting with the Go-decision and ending with the commercial launch.
- The definition of related service requirements, resources and, if involved, third party services is an essential part of this process.
- Service Lifecycle Management and Resource Lifecycle Management are very important interfaces to this process and need to be synchronized carefully

4.3.2. eTOM Mapping

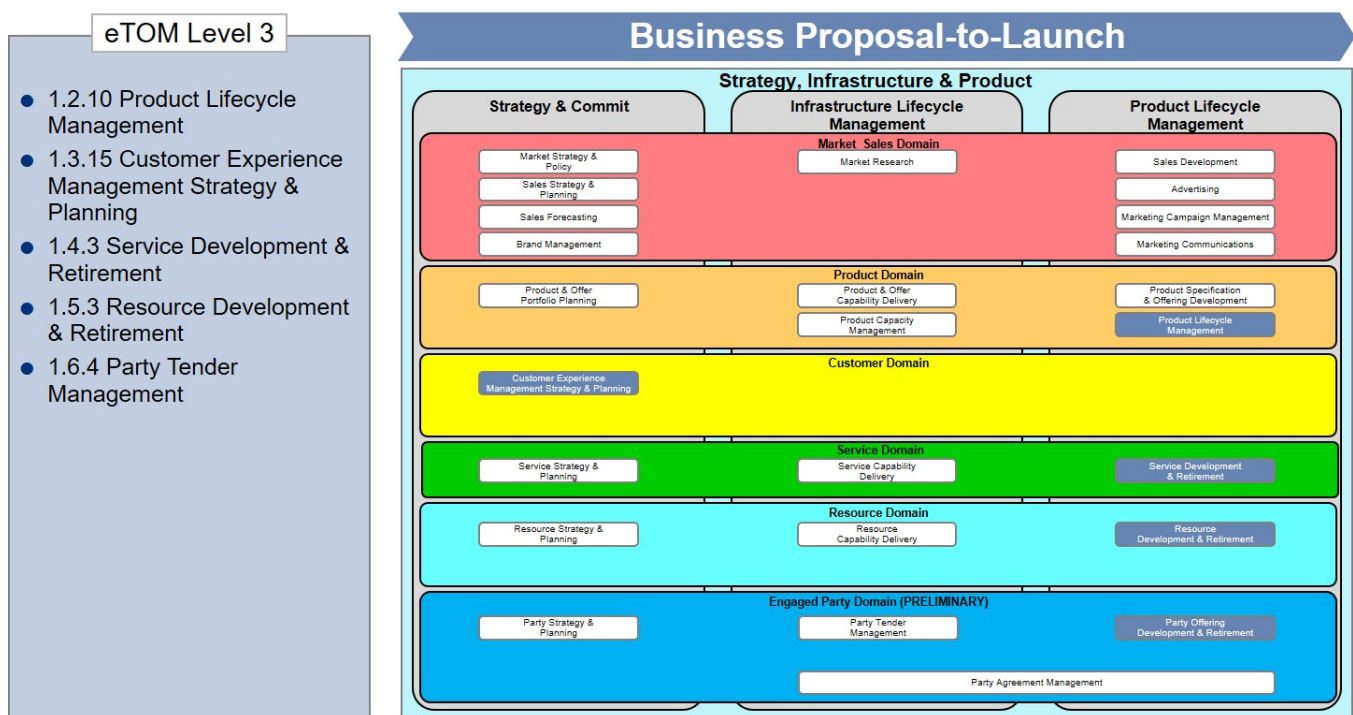


Figure 58. eTOM Mapping for Business Proposal-to-Launch

4.3.3. Detailed Process-Flow (Level 3)

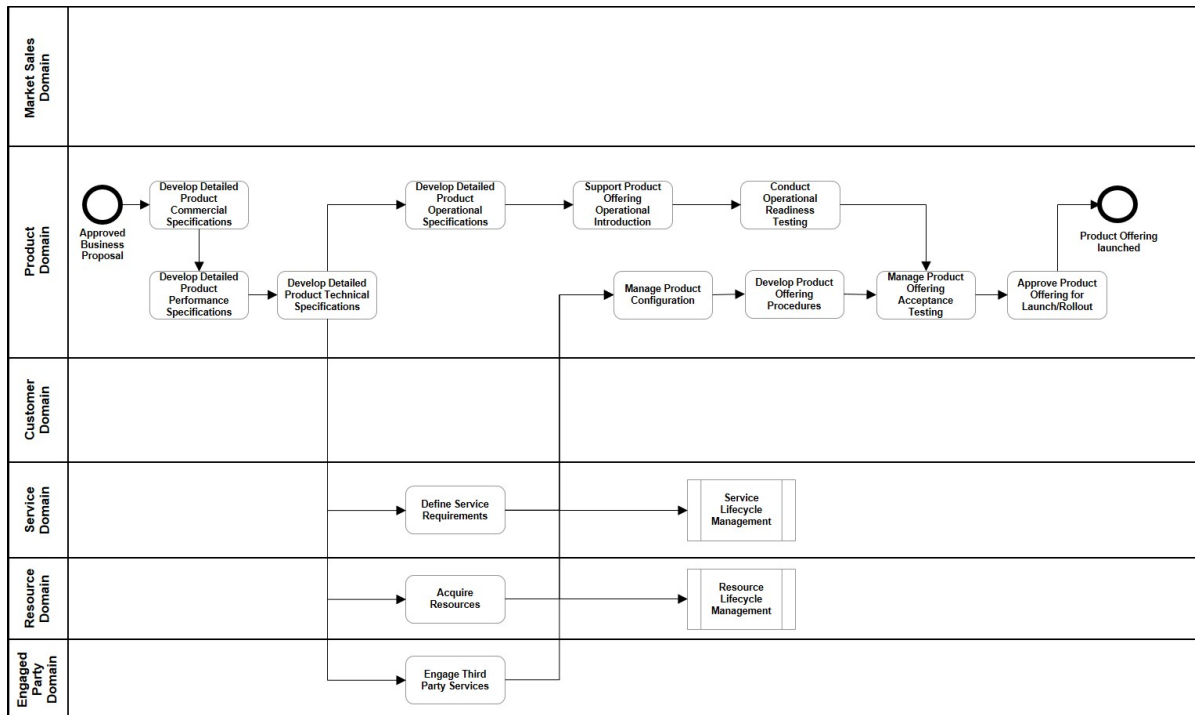


Figure 59. Workflow for Business Proposal-to-Launch

4.3.4. Detailed Information

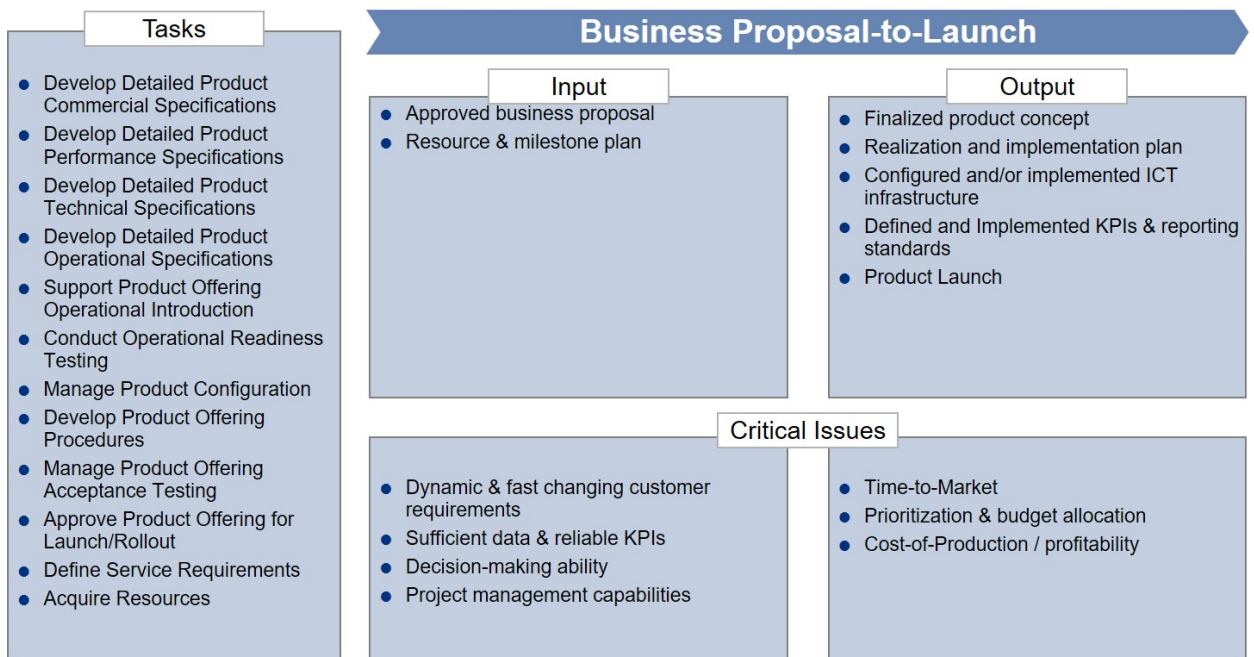


Figure 60. Details for Business Proposal-to-Launch

4.4. Process Flows for Assessment-to-Relaunch

4.4.1. Overview

- The goal of this process is the continuous improvement and enhancement of the existing product offerings as a reaction to customer needs or internal commercial, technical or financial decisions
- The activities include assessment of product offering performance, identification of issues and updating required aspects like product specification, configuration, etc.
- The outcome of the process is a relaunched product offering with enhanced specifications

4.4.2. eTOM Mapping

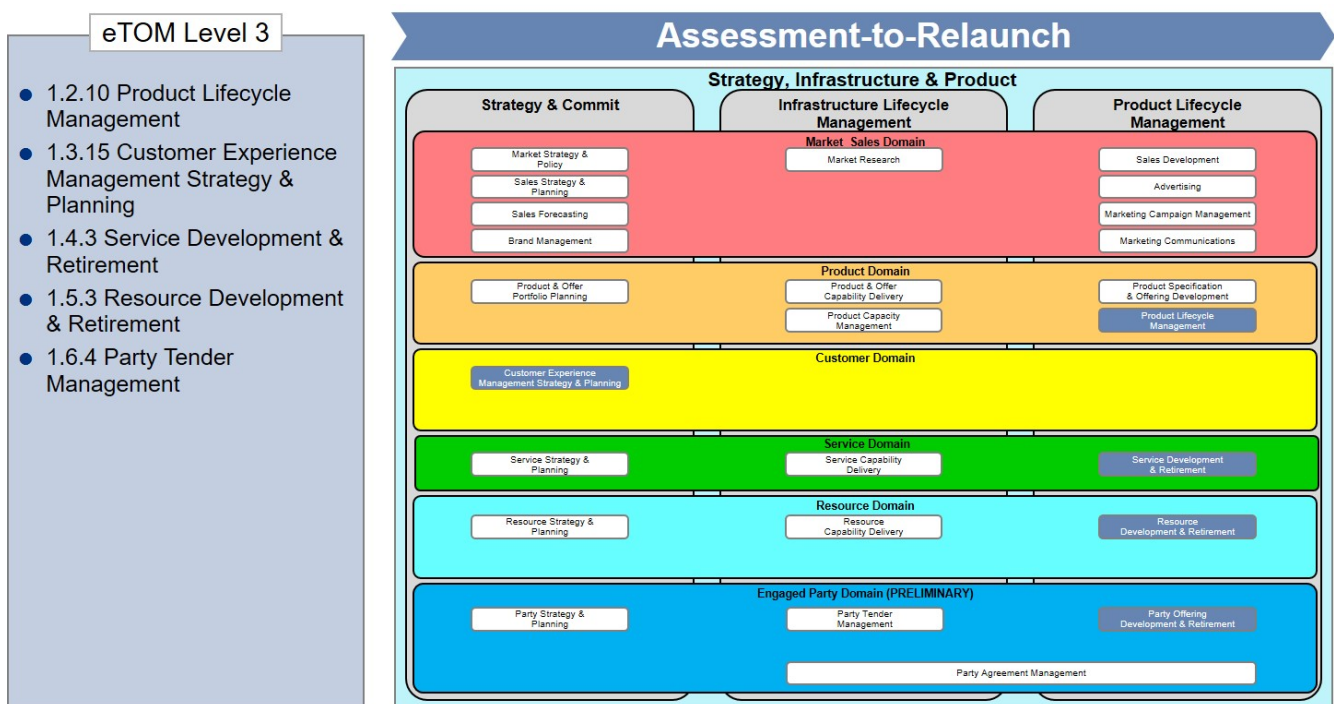


Figure 61. eTOM Mapping for Assessment-to-Relaunch

4.4.3. Detailed Process-Flow (Level 3)

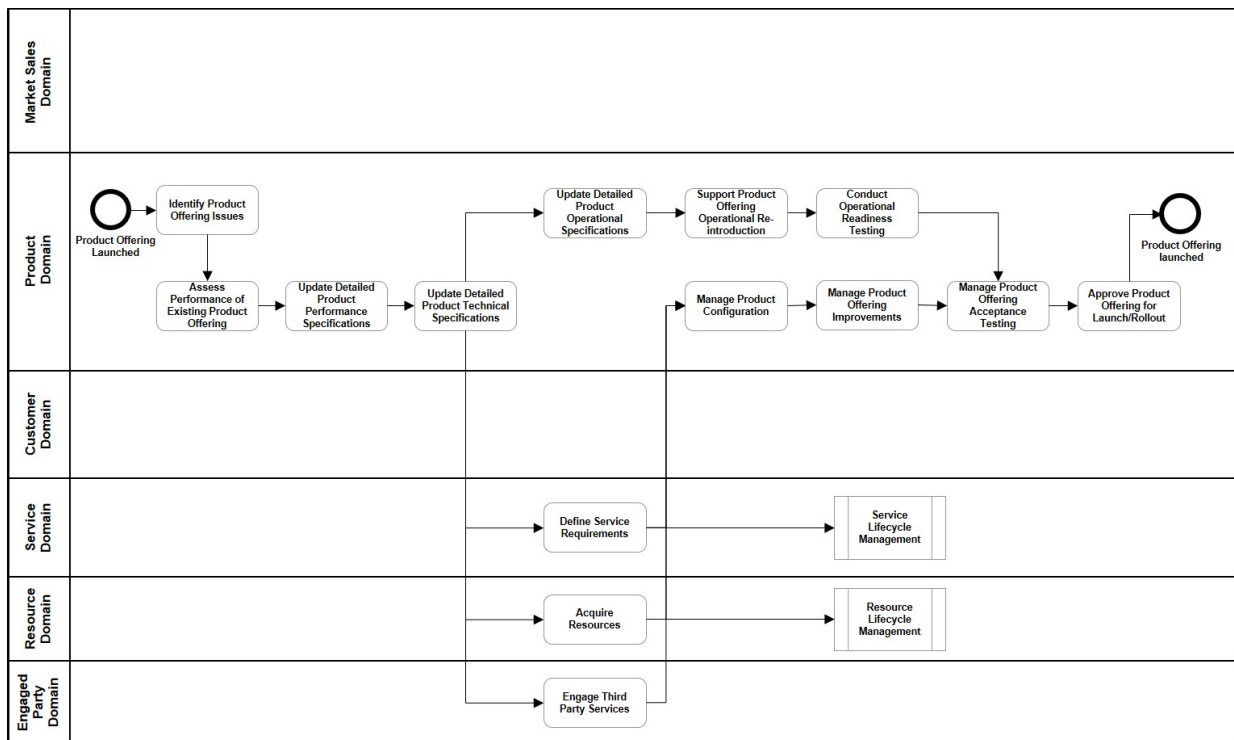


Figure 62. Workflow for Assessment-to-Relaunch

4.4.4. Detailed Information

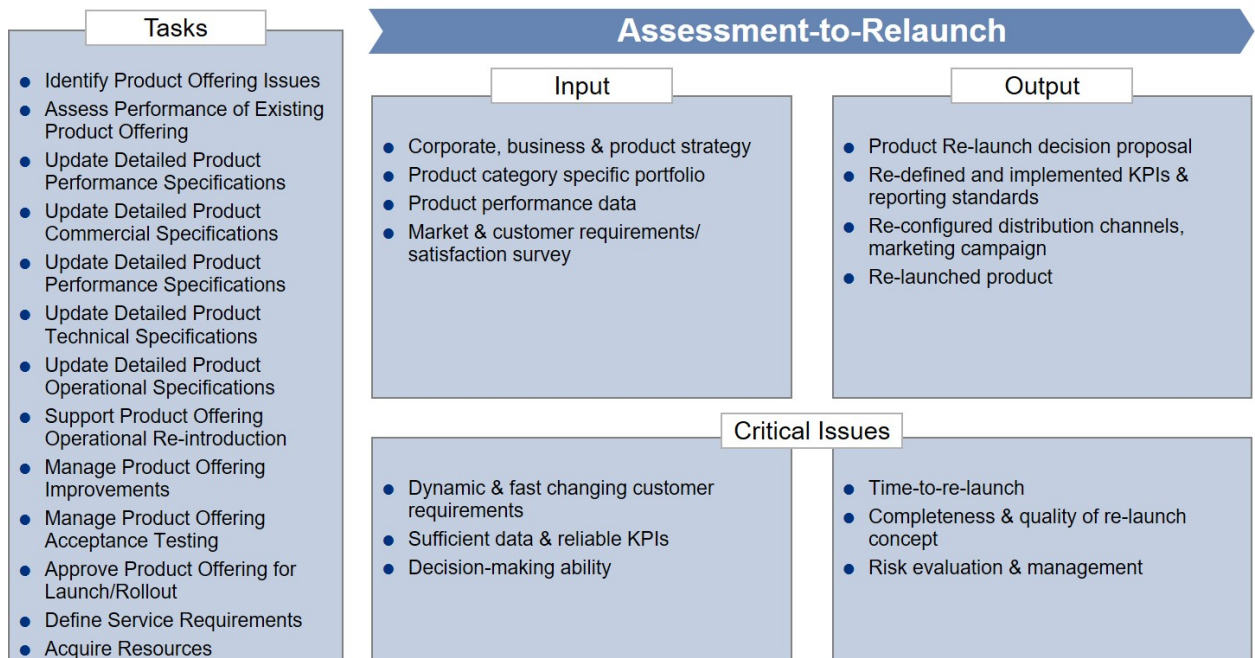


Figure 63. Details for Assessment-to-Relaunch

4.5. Process Flows for Assessment-to-Retirement

4.5.1. Overview

- This process describes the activities related to the retirement of product offerings, which have been identified as unviable.
- The correct identification of the impact caused by the product exit and the right exit strategy are the most important issues.
- Definition of released service and resource capacity is also part of the process; the management of these capacities is executed in Service Lifecycle Management and Resource Lifecycle Management processes

4.5.2. eTOM Mapping

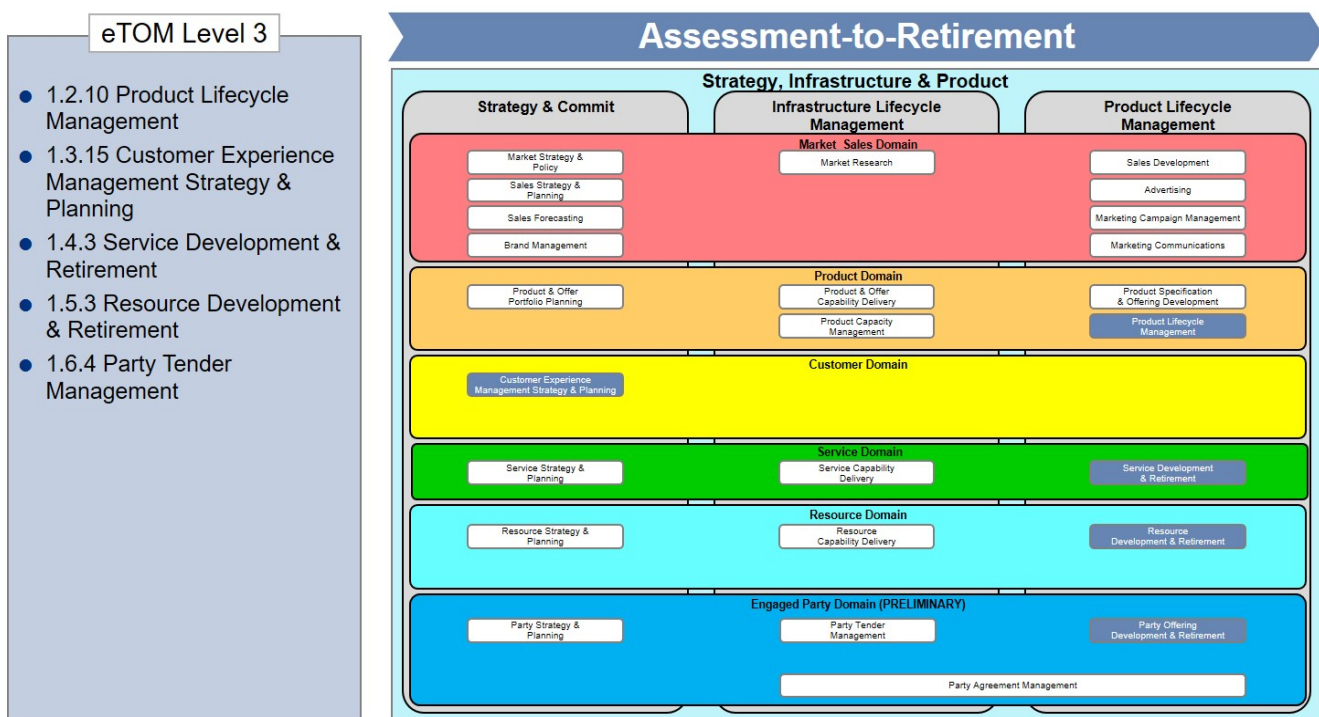


Figure 64. eTOM Mapping for Assessment-to-Retirement

4.5.3. Detailed Process-Flow (Level 3)

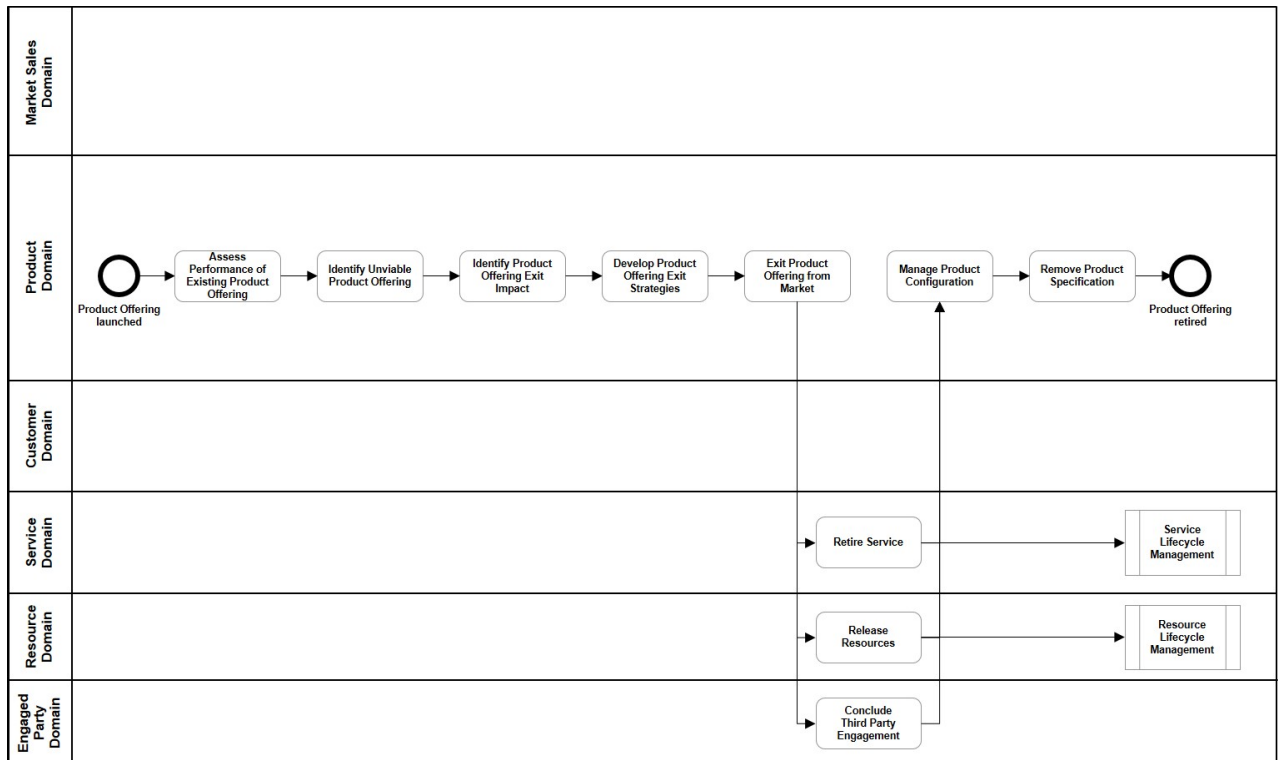


Figure 65. Workflow for Assessment-to-Retirement

4.5.4. Detailed Information

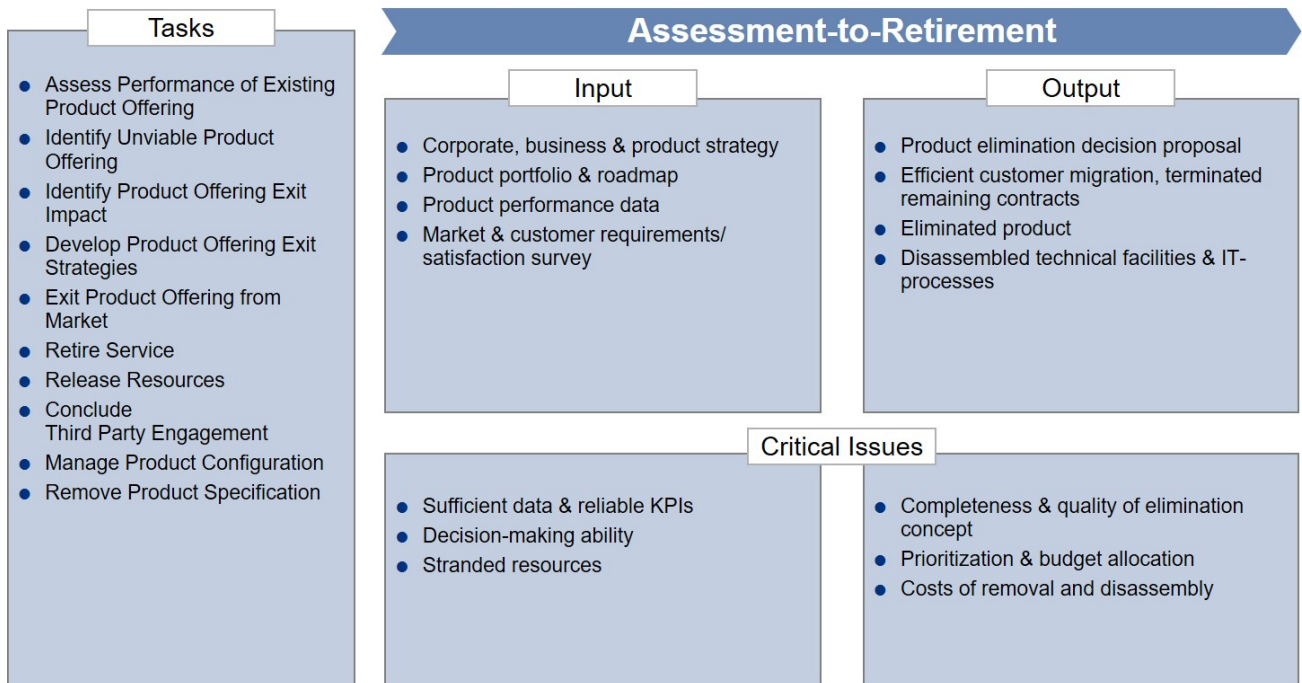


Figure 66. Details for Assessment-to-Retirement

4.6. Process Flows for Market Strategy-to-Campaign

4.6.1. Overview

- The process flow includes activities related to the establishing market strategy and execution of single marketing campaigns.
- Portfolio strategy, product delivery feedback and customer feedback are the most critical inputs to the process.

4.6.2. eTOM Mapping

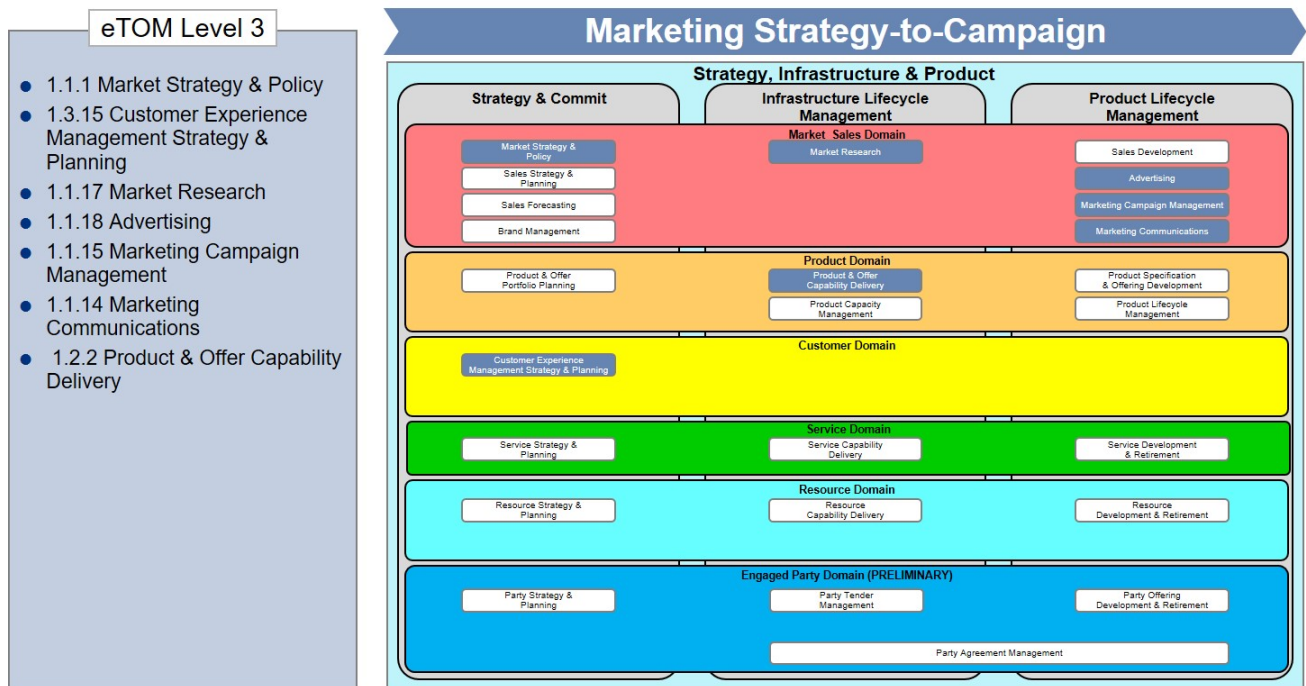


Figure 67. eTOM Mapping for Marketing Strategy-to-Campaign

4.6.3. Detailed Process-Flow (Level 3)

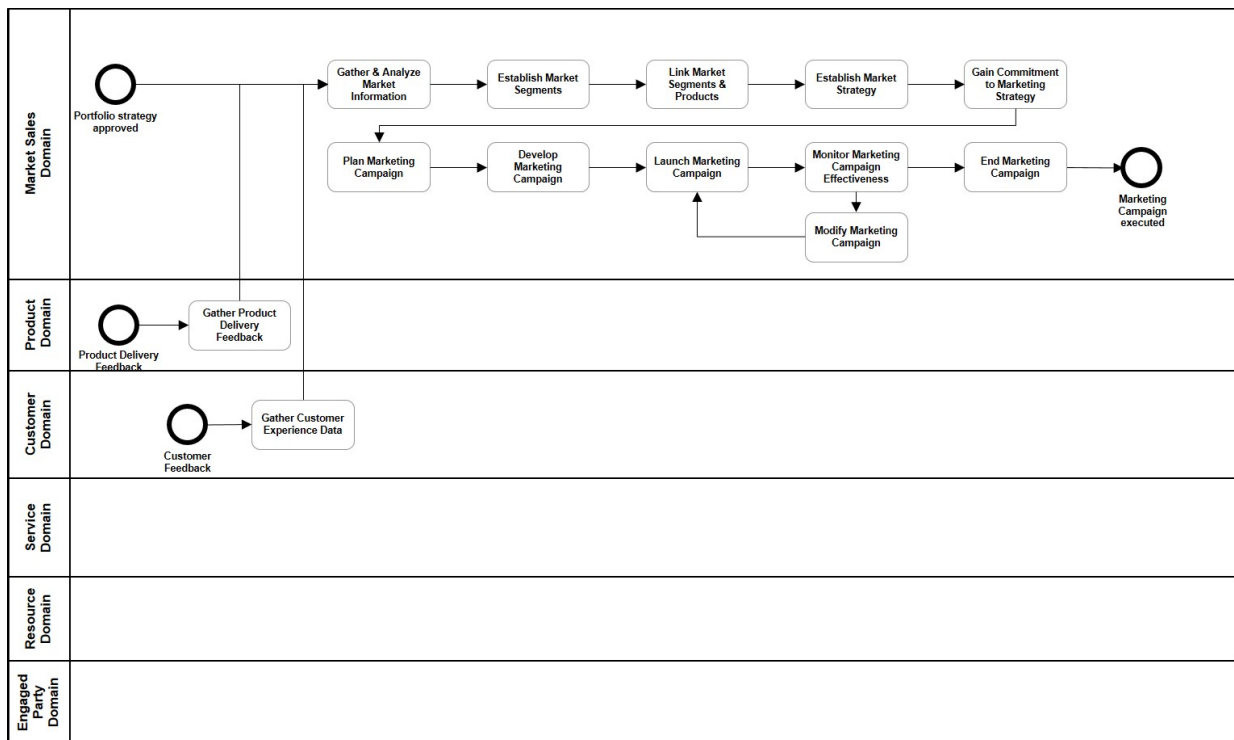


Figure 68. Workflow for Marketing Strategy-to-Campaign

4.6.4. Detailed Information

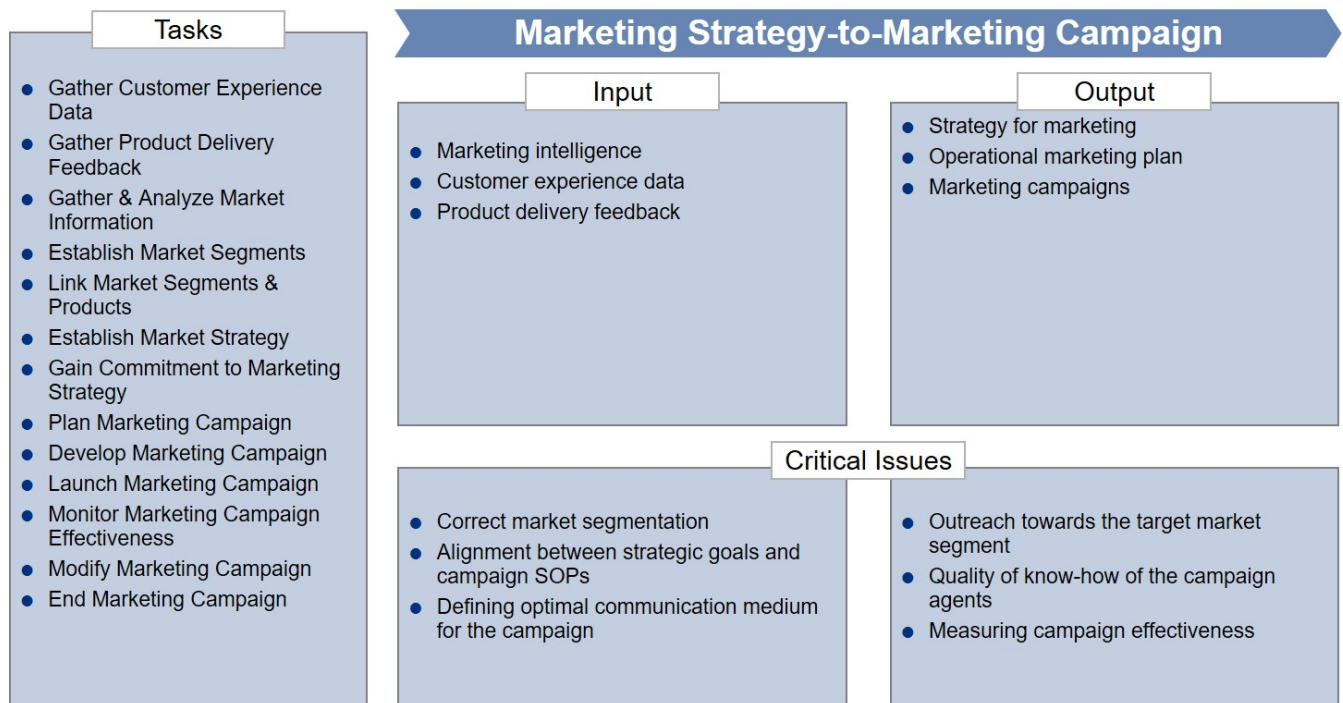


Figure 69. Details for Marketing Strategy-to-Campaign

5. Engaged Party Flows

This section includes sample process-flows (eTOM level 3) relevant to the Engaged Party Flows. The process-flows show how to apply the Engaged Party Domain, a part of eTOM to six phases of procurement process.

We look at addressing the “**Effective management of procurement process using Business Process Framework (eTOM)**”. There are various processes in industry to carry out procurement, but doing it in a standard, well known, flexible approach will have a good value addition throughout procurement lifecycle.

Hence, we choose to decompose and identify standard business process from Engaged Party of Business Process Framework which can be easily mapped to various phases of procurement process. We may not be able to get one-to-one mapping of procurement phase against eTOM processes, however we can identify the applicable group of processes.

5.1. Process Flows for Procurement using Engaged Party

5.1.1. Overview

- Procurement process is a key activity for any organization which gets complex and challenging with growing need for value addition. Likewise, there is always parallel need of standard, easy to adopt, industry-neutral process definition for procurement. The example defined in this document tries to address this and provide best practice that can be derived from business process framework.
- The idea behind this case study is to provide a road map and best practices organizations can use during procurement process using standards defined in Framework.
- As mentioned, this section decomposes/ identifies process involved till Level 3 of business process framework that can be mapped to various phases of procurement process.

5.1.2. Assumptions

- In the scope of this example, process involving performance and problem reporting of external parties is not considered. This can be included as part of next versions of document. This case study was originally based on Framework 14.5 Model and was updated according to the latest 16.0 Model. Engaged party domain is considered throughout the case study (since the Supply partner domain was replaced with Engaged party domain). This process can be fully automated, but in some cases manual activities are also in scope.

5.1.3. Phases of procurement:

Below flow depicts procurement process at high level and the idea is to make this model as domain/industry neutral as possible.



Figure 70. Phases of Procurement Process

- **Phase 1: Define Scope of work/Definition of work:** Define and review the definition of work, Intellectual property considerations, vendor selection method etc.
- **Phase 2: Verify Vendors and Issue RFP** – Collect information about the vendors list and issue RFP stating details. If required, conduct a session on RFP briefing.
- **Phase 3: Validate RFP response** – verify RFP response from vendors including commercials, SOW and technical requirements and other criteria.
- **Phase 4: Purchase Agreement/ Formal award of contract** – Publish contract award notice to selected vendor. Explain the terms and conditions of contract.
- **Phase 5: Contract Management** – Set up a kick-off meeting, conduct regular follow up on progress of work.
- **Phase 6: Post Contract** – review the overall progress and milestones accordingly.

5.1.4. eTOM Mapping

We can now choose which processes from the Business Process Framework are needed, based on the definition of the business problem set out originally, and based on the various phases of procurement .We inspect the Business Process Framework and identify where the main focus of process activity is for the case concerned .

Procurement – Phase 1 to 4:

In this section let us try to map eTOM processes to phase 1,2,3 and 4 of procurement process defined in section 4.1.5. Phase 4 of procurement is spanned across two parts in this document for the ease of understanding and continuity.

Identification of L1 process:

As we start with Mapping of business process framework with various phases of procurement, it can be observed that L1– Engaged Party Domain is involved. We may not be able to identify one-to-one mapping of procurement phase against eTOM processes; however, we can identify the applicable group of processes. L1 process identified is common for all phases of procurement.

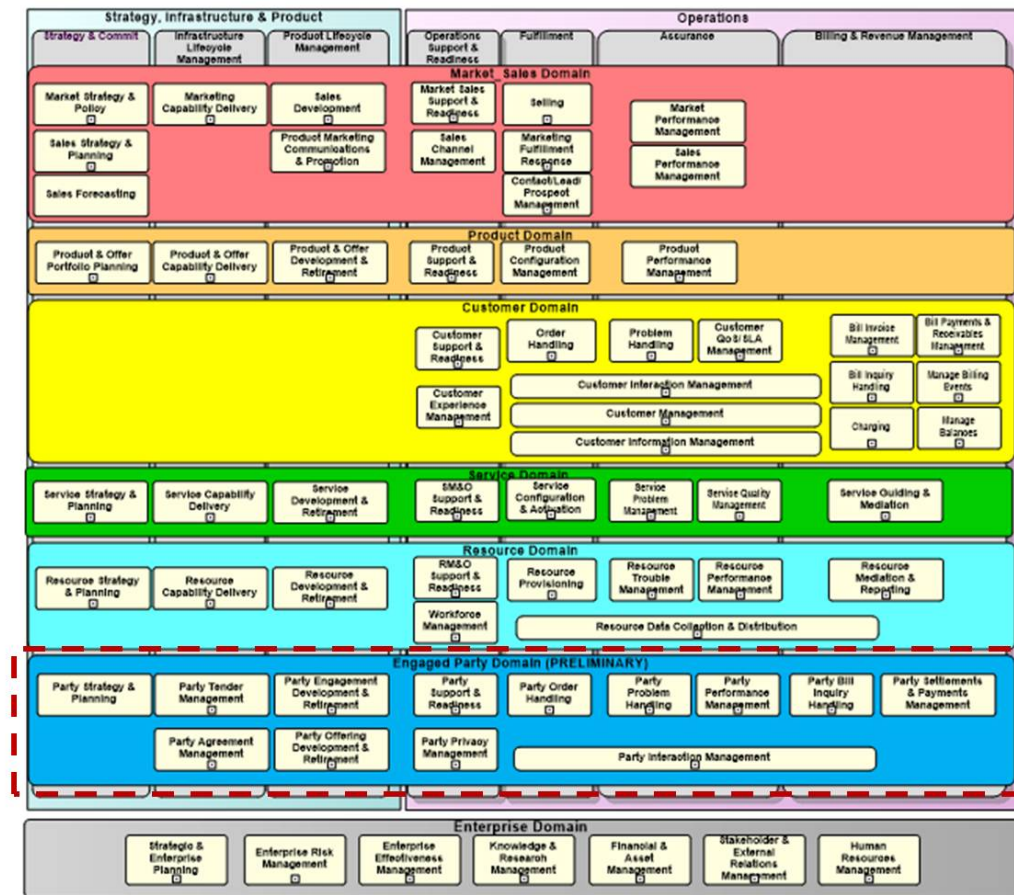


Figure 71. Level 1 Process Framework mapping for procurement process

We observe that each of Level1 process identified can further be detailed to L2, L3 and L4 process depending on complexity involved and granularity needed at various stages. Let us identify L2 process from identified L1 process.

Identification of L2 process:

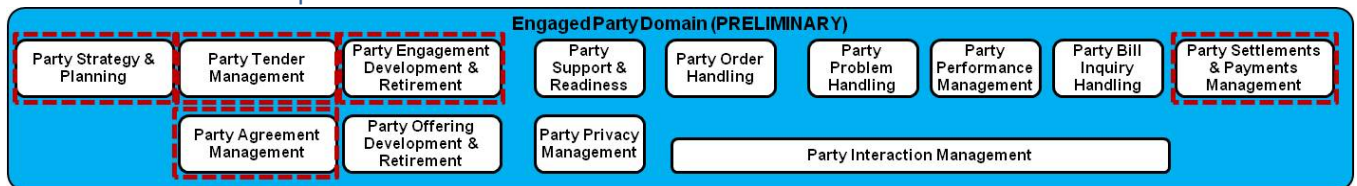


Figure 72. Identification of Level 2 process in "Engaged Party Domain (L1)"

Now we have a set of Level 2 process elements that we can use to address each phase of procurement process. **These processes can be mapped to phase 1, 2, 3 and 4 of procurement.**

Level 2 processes identified are:

1. Party Strategy and Planning, Party Tender Management, Party Engagement Development and Retirement, Party Agreement Management, Party Settlements & Payments Management from L1- Engaged Party Domain

For example: - scoping and planning of outsourcing IT transformation project to a software service provider/vendor.

Now, let us identify L3 process from identified L2 processes.

Identification of L3 process:

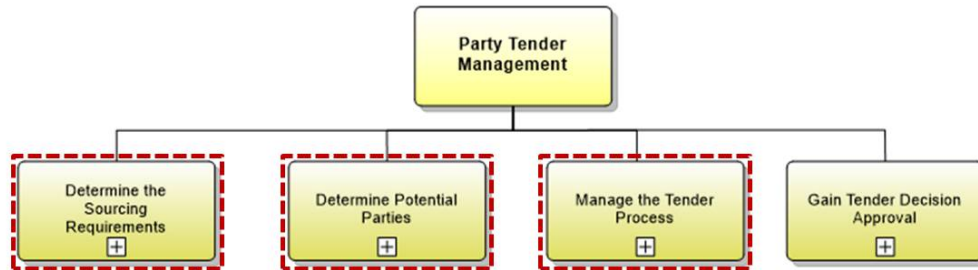


Figure 73. Identification of Level 3 process in “Party Tender Management (L2)”

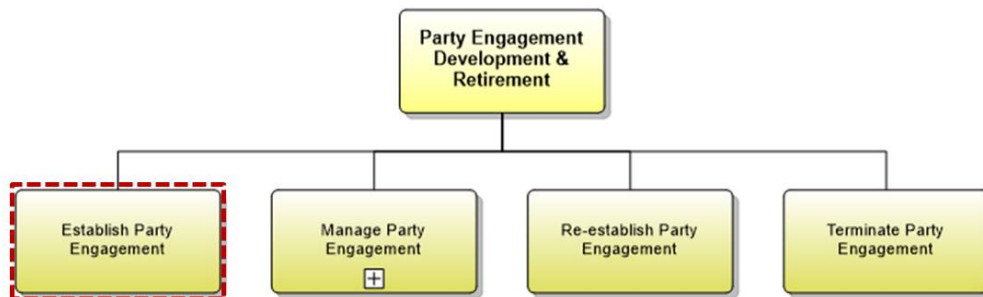


Figure 74. Identification of Level 3 process in “Party Engagement Development and Retirement (L2)”

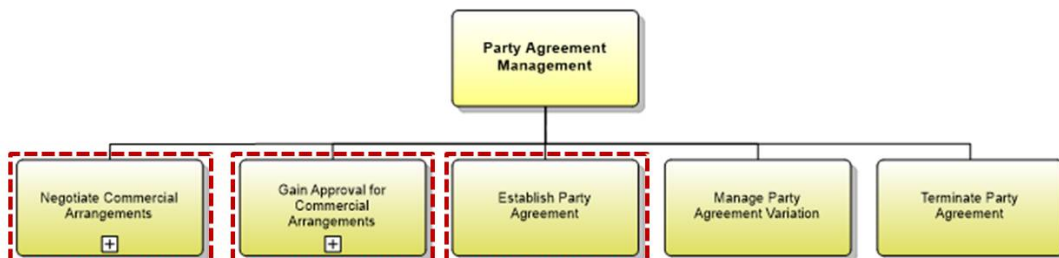


Figure 75. Identification of Level 3 process in “Party Agreement Management (L2)”

Figure above depicts the processes that can be mapped to L3 process under L2 process. This process defines policies, principles and performance standards of the organization which need to be adhere by vendor/external service provider.

Example: a process defining SLA for a software system.

Now we have a set of Level 3 process elements that we can use to address each phase of procurement process. **These processes can be mapped to phase 1, 2, 3 and 4 of procurement.**

Level 3 processes identified are

1. Determine Sourcing Requirements, Determine Potential Parties, Manage Tender Process from L2 Party Tender Management.

2. Establish Party Engagement from L2: Party Engagement, Development & Retirement.
3. Negotiate Commercial Arrangements, Gain Approval of Commercial Arrangements, and Establish Party Agreement from L2 Party Agreement Management.

Procurement – phase 4, 5 and 6:

Processes decomposed in this section are applicable to procurement phase 5 and 6 (defined in section 4.1.5 of this document) - awarding contract ,contract management and post contract phase. It is to be observed that phase 4 has spanned across two sections as process related to supplier identification /award/maintaining spans across multiple processes. Now, let us identify Level 2 process from identified L1 process.

Identification of L2 process:

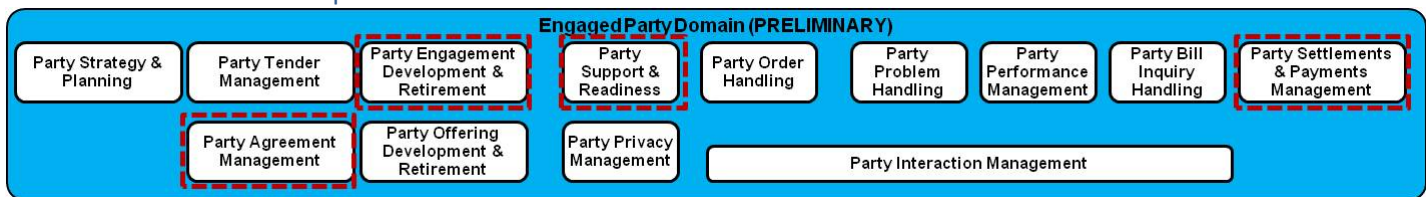


Figure 76. Identification of Level 2 process in “Engaged party domain (L1)”

Now we have a set of Level 2 process elements that we can use to address each phase of procurement process. **These processes can be mapped to phase 4, 5 and 6 of procurement.**

Level 2 processes identified are:

1. Party Engagement Development & Retirement, Party Support & Readiness, Party Agreement Management, Party Settlements & Payments Management from L1- Engaged Party Domain.

Now, let us identify Level 3 process from identified L2 process.

Identification of L3 process:

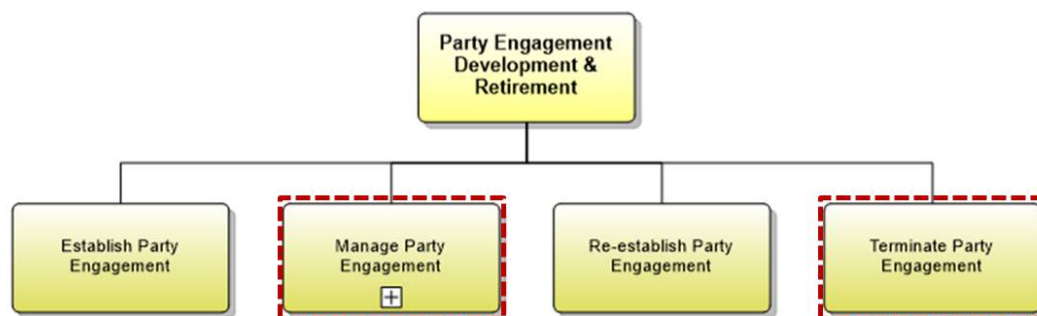


Figure 77. Identification of Level 3 process in “Party Engagement Development & Retirement (L2)”

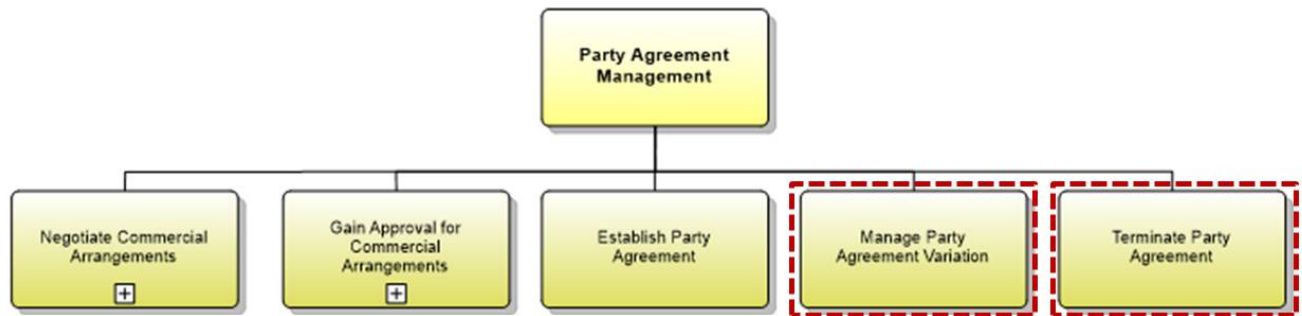


Figure 78. Identification of Level 3 process in “Party Agreement Management (L2)”

Now we have a set of Level 3 process elements that we can use to address each phase of procurement process further. **These processes can be mapped to phase 4, 5 and 6 of procurement.**

Level 3 processes identified are

1. Manage Party Engagement and Terminate Party Engagement in L2 Party Engagement Development & Retirement.
2. Manage Party Agreement Violation and Terminate Party Agreement from L2 Party Agreement Management.
3. All of the processes in L2: Party Settlements & Payments Management
4. All of the processes in L2: Party Support and Readiness

We observe that **Manage Party Engagement** process addresses various activities involved post-awarding contract. Some of the activities can be arranging a kickoff meeting, monitoring and tracking the progress, escalations etc. We can also make use of any other process than ones identified, depending on the nature of the deal/contract.

6. Administrative Appendix GB921E

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
0.1	15./March /2010	Georg Vitt	first issue of document
9.0	August 2010	Mike Kelly	Minor formatting and document number updated
9.1	August 2010	Alicja Kawecki	Minor cosmetic corrections for web posting and ME
9.2	December 2010	Mike Kelly	Amendments to reflect comments received, to clarify document as Application Note and to explain terminology
9.3	January 2011	Mike Kelly	Amendment to Appendix to make more generic, and addition of Acknowledgments
9.4	April 2011	Alicja Kawecki	Updated to reflect TM Forum Approved status
1.13.0	October 2013	Chin-Gi Hong (+ minor reformatting by Mike Kelly in line with revised document template)	Major update of process flows to align to GB921 Addendum K ("Construction Guidelines for Process Flows").
1.13.1	October 2013	Mike Kelly	Small formatting changes
1.13.2	October 2013	Alicja Kawecki	Updated cover & header, inserted Notice & list of figures
1.13.3	March 2014	Alicja Kawecki	Updated cover, copyright year & Notice reflecting TM Forum Approved status
13.5.0	October 2014	Avi Talmor	Updated Release information and file name
16.5.0	November 2016	Johannes Eigenseer	Converted artifact from Word doc to Confluence- Major updates of process mappings and flows to align with eTOM Release 16.5
16.5.1	1 December 2016	Alicja Kawecki	Updated cover, minor formatting/style edits prior to publication for Fx16.5

Version Number	Date Modified	Modified by:	Description of changes
17.5.0	December 2017	Johannes Eigenseer, Jawahar Sajjad	Developed Product centric process flows; updated document
20.0	29-May-2020	Sepide Sahari, Neil Vincent Lawrence, Huangxiarong	Update Capacity Management Process in Network Centric Process Flows to align with latest eTOM Release 19.5 which includes Common Domain. Capacity Management Process Mappings and flow has been updated to support short to mid-term capacity management scenario such as rolling expansion planning (quarterly, monthly, etc.) requirements.
20.0.1	20-Jul-2020	Adrienne Walcott	Updated to reflect TM Forum Approved Status

6.1.2. Release History

Release Number	Date Modified	Modified by:	Description of changes
9.0	August 2010	Mike Kelly (based on document developed by Georg Vitt)	Minor changes to align with detailed GB921 documents.
9.1	December 2010	Mike Kelly	Amendments to reflect comments received, to clarify document as Application Note and to explain terminology.
13.5	October 2013	Chin-Gi Hong (+ minor reformatting by Mike Kelly in line with revised document template)	Major update of process flows to align to GB921 Addendum K ("Construction Guidelines for Process Flows").
15.0	May 2015	Sreepad Kodandapani	Adding Procurement Engaged Party
16.5.0	September 2016	Johannes Eigenseer	Major update of process mappings and flows to align with eTOM Release 16.5
17.5.0	December 2017	Johannes Eigenseer, Jawahar Sajjad	Introduction of Chapter 5 "Product Centric Process Flows"
Pre-production	29-May-2020	Sepide Sahari, Neil Vincent Lawrence, Huangxiarong	Update Capacity Management Process in Network Centric Process Flows
Production	20-Jul-2020	Adrienne Walcott	Updated to reflect TM Forum Approved Status

6.2. Acknowledgments

A number of individuals and companies have contributed to the development of this document, through discussion and suggestions in a series of collaborative reviews and walkthroughs of the process flows.

The TM Forum and the Business Process Framework Team would like to express special thanks to Detecon Int. GmbH for providing the initial input and thought leadership in the development of GB921E.

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Kathrin Hahne	Detecon International
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Marcus Ras	TMF

Thanks to all here who attended the discussions or provided input to the flow design and development.